

Chapter 4

The Medium Access Control Sublayer

媒体访问控制/介质接入控制

(about 8-9 hours)

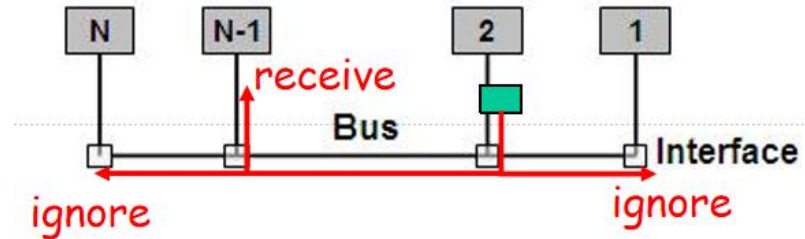
Data Link Types

■ Point to point links(点-点链路)

- ◆ Address may not be necessary
- ◆ Reliable communication, HDLC, PPP...

■ Broadcast links(广播链路)

- ◆ LAN
- ◆ Many senders and potential receivers.
- ◆ Addressing is necessary (physical address)
- ◆ Accessing a shared medium
 - Multiple access problem must be solved



Medium Access Control: a means of controlling access to the medium to promote **orderly** and **efficient** use.

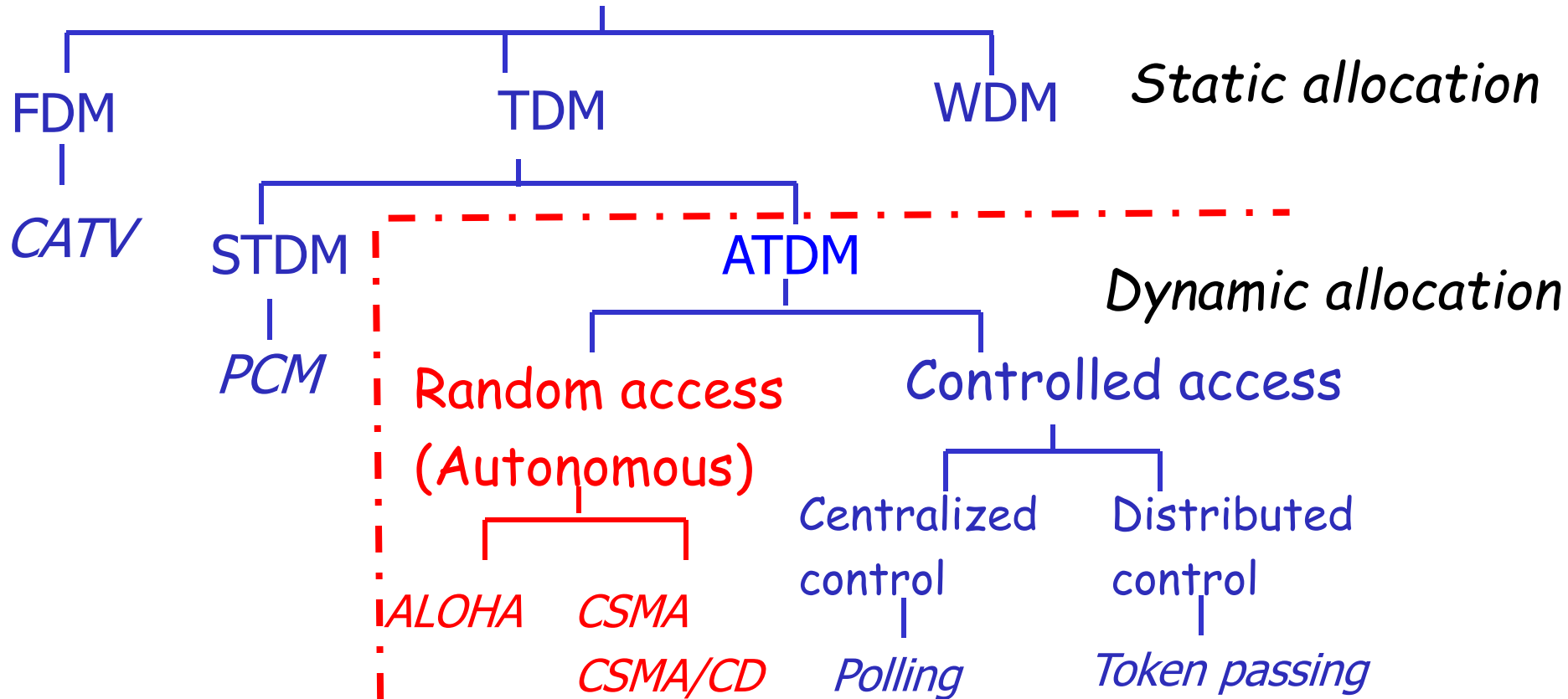
- Sample protocols: **ALOHA**, **CSMA/CD**, **MACA**...

Outline

- 4.1 Channel Allocation Problem
- 4.2 Multiple Access Protocols
- (補)LAN Reference Model: IEEE 802 RM
- 4.3 Ethernet
- 4.4 Wireless LANs
- 4.8 Data Link Layer Switching

4.1 Multiple Access Schemes

Sharing a common channel



STDN: Synchronous TDM 同步时分复用

ATDM: Asynchronous TDM 异步时分复用(统计复用)

4.1.1 Static Channel Allocation (静态信道分配)

- Channel is divided into N static sub-channels

- ◆ Multiple Access Schemes: FDM
- ◆ Telephone networks: user number is small and constant, with heavy load of traffic of each user (simple and efficient)

$$T_{\text{FDM}} = NT$$

WHY?

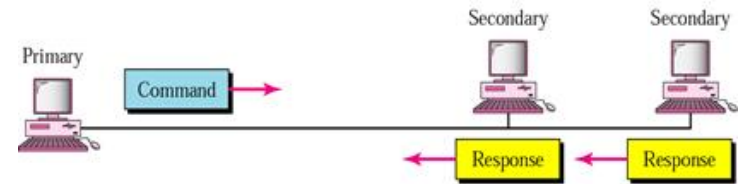
Using M/M/1 model

- ◆ Computer networks: the number of senders is large and varying or the traffic is bursty. (inefficient)
- None of the static channel allocation methods work well with bursty traffic
- Dynamic Channel Allocation are adopted

Classification of Dynamic Channel Allocation

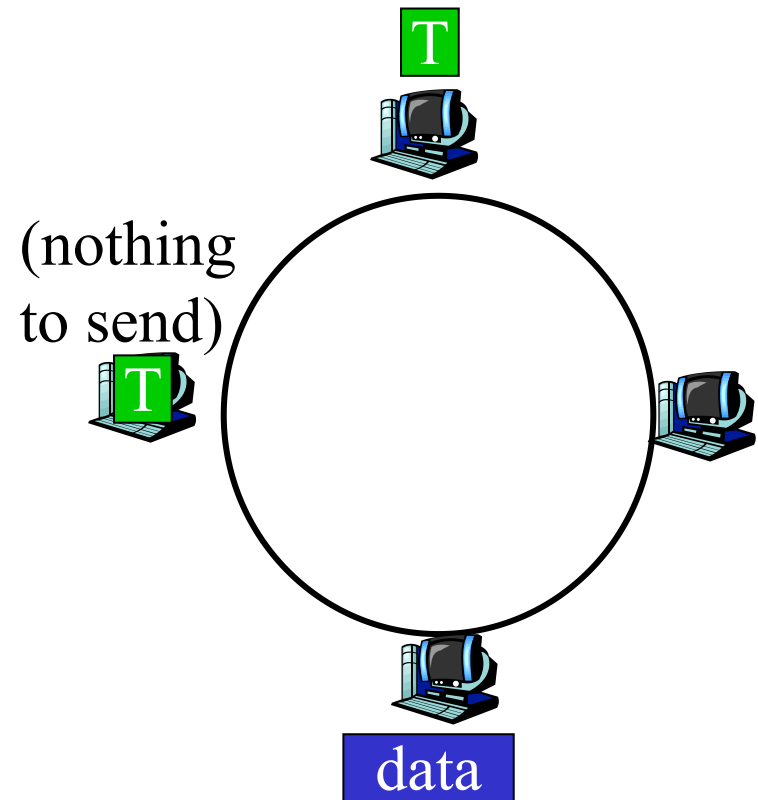
■ Controlled multiple access

- ◆ Centralized: **polling**
- ◆ Decentralized: **token**



■ Random multiple access

- ◆ **ALOHA**
- ◆ **CSMA, CSMA/CD, CSMA/CA**



4.1.2 Assumptions of Dynamic Allocation (动态信道分配)

1. Independent Traffic: independent and single user system
2. Single channel
3. Observable Collisions
4. Time assumption { Continuous time(连续时间)
Slotted time(分时隙)
5. Carrier sense assumption { Carrier sense
No carrier sense
载波监听：发送之前
能否知道信道是否空闲

Outline

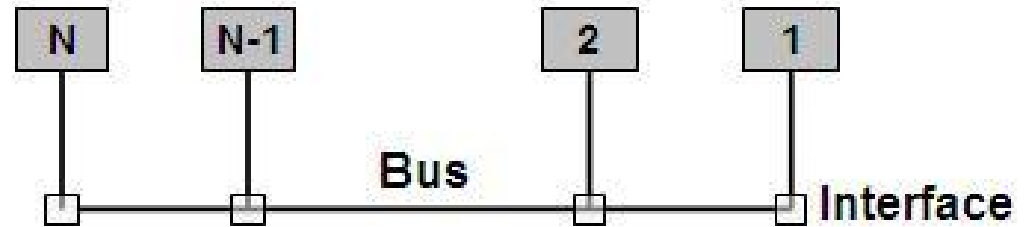
- 4.1 Channel Allocation Problem
- 4.2 Multiple Access Protocols
- (補)IEEE 802 Reference Model
- 4.3 Ethernet
- 4.4 Wireless LANs
- 4.8 Data Link Layer Switching

4.2 Multiple Access Protocols

- 4.2.1 ALOHA
- 4.2.2 Carrier Sense Multiple Access Protocols(CSMA)
- 4.2.5 Wireless LAN Protocols(MACA)

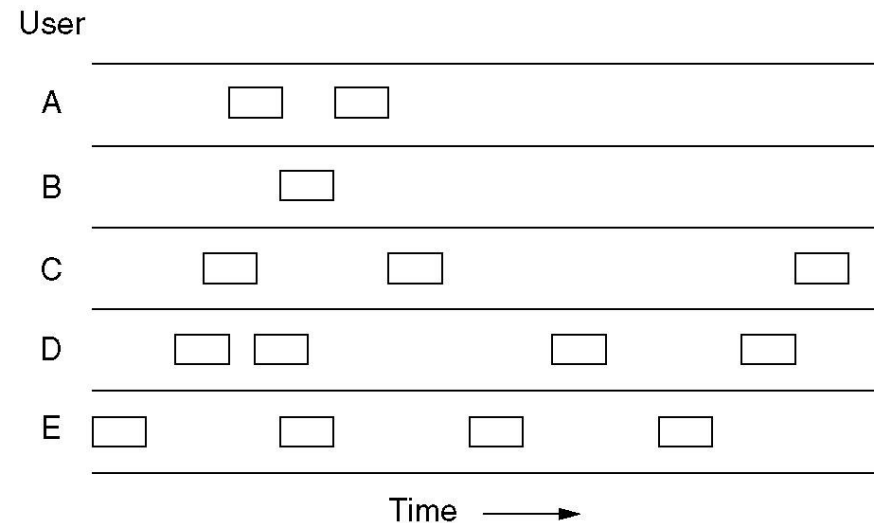
(1) Pure ALOHA

- First protocol for multiple access channels; first analysis of such protocols (Norm Abramson, Univ. of Hawaii, 1970)



- Basic idea

- stations transmit a frame whenever they have one **without coordination** with others
- If the **acknowledge** not received (maybe collision), then do **retransmission**

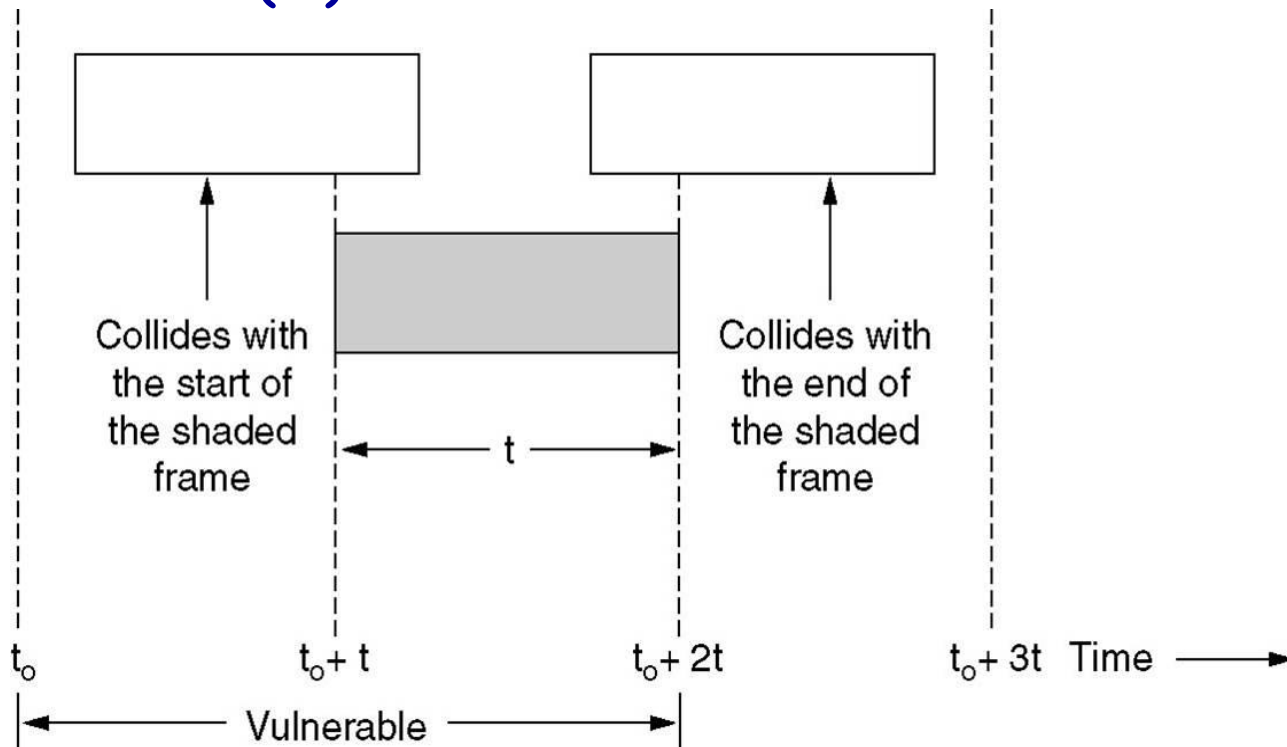


Pure ALOHA: Performance Analysis

■ A frame is successful if

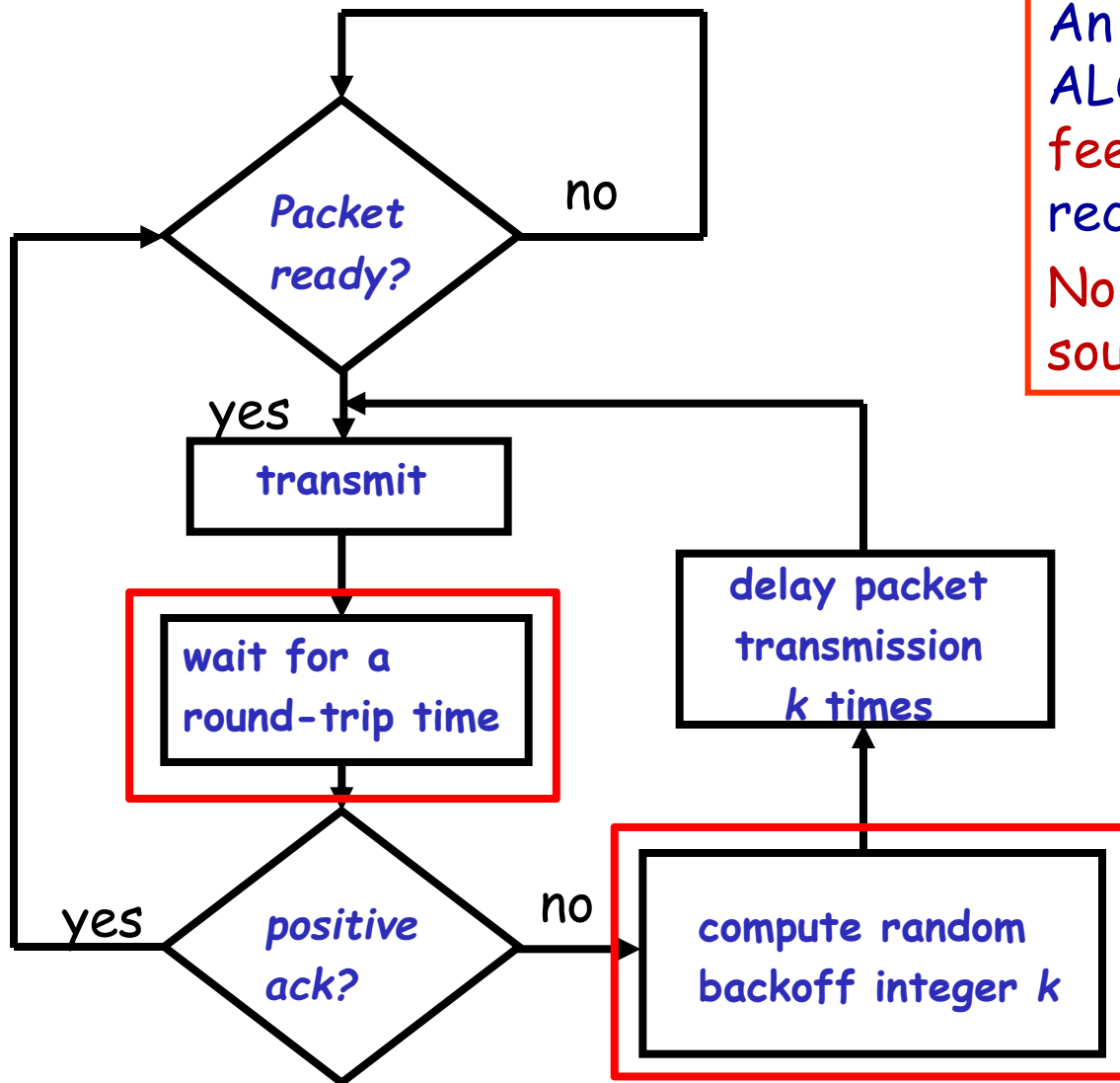
- ◆ no packets arrive within t seconds before it starts or while it is being transmitted

■ Max utilization(S): 18.4%



Vulnerable Period(脆弱期/易冲突期): $2t$ (t is frame time, the transmission delay of a frame)

Flow of Pure ALOHA

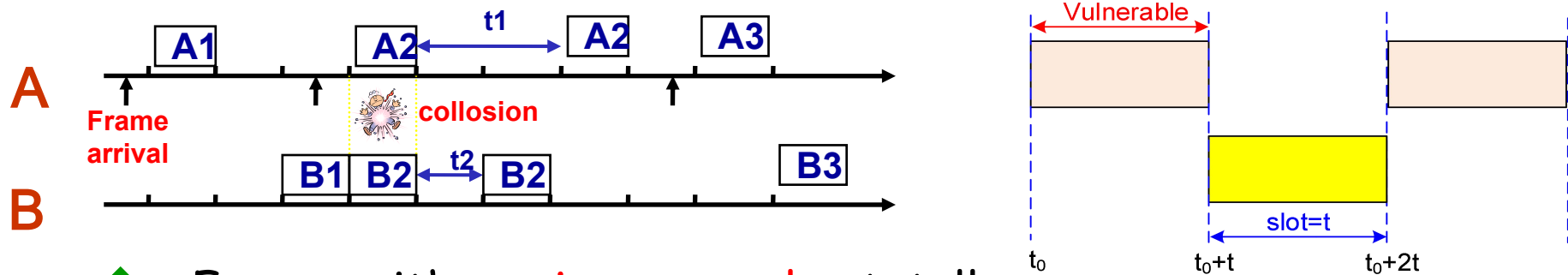


An integral part of the ALOHA protocol is **feedback** from the receiver.

No coordination among sources.

(2) Slotted ALOHA(时隙/分槽ALOHA)

- Time in uniform slots equal T_0
- Need central clock (or other sync mechanism)
- Transmission begins at slot boundary



- Frames either miss or overlap totally
- Vulnerable Period: t , Max utilization(S): 36.8%
- Question : How long is the best duration for slots?
 - Slot $< t$, Vulnerable Period $> t$
 - Slot $> t$, Vulnerable Period = slot
 - slot=frame time

Multiple Access Protocols

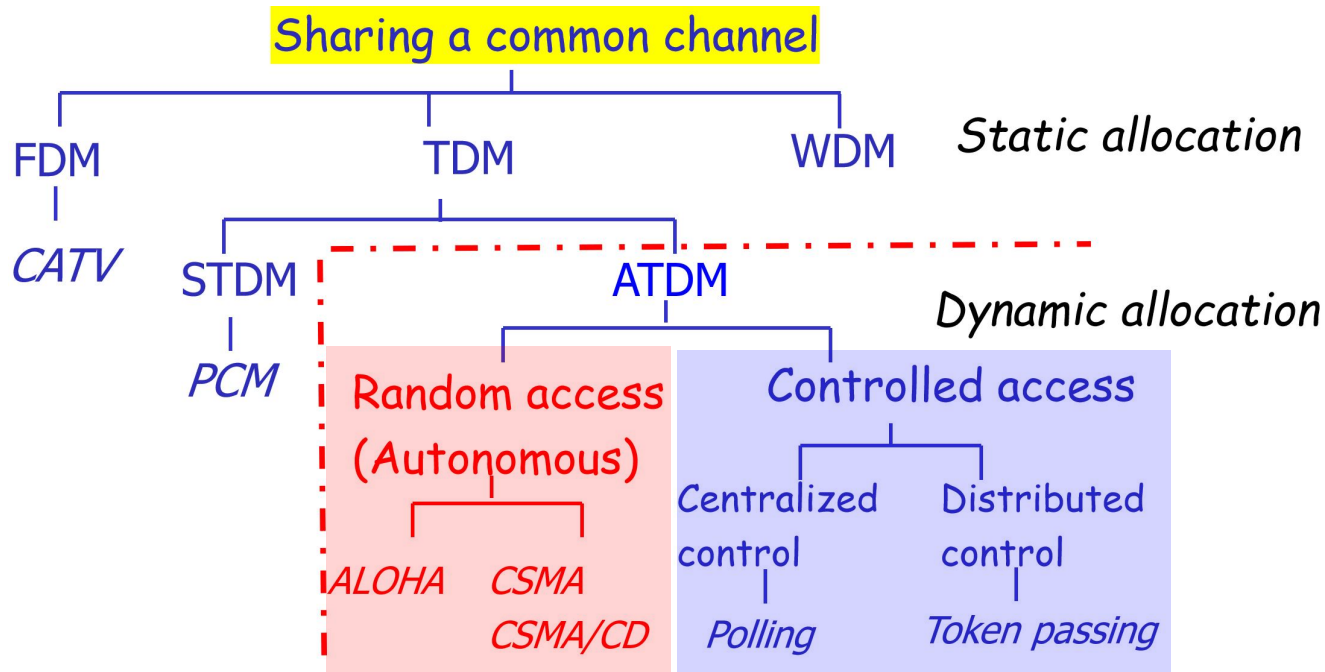
- 4.2.1 ALOHA
- 4.2.2 Carrier Sense Multiple Access Protocols
- 4.2.5 Wireless LAN Protocols

(1) CSMA(载波监听多路访问/多点接入)

- Basic idea: By listening to the channel before sending (*Carrier Sense*), stations try to reduce the Vulnerability Period to *one propagation delay*
 - ◆ Propagation delay is much less than transmission delay
- CSMA
 - ◆ If channel is idle, transmit with a probability (p).
 - ◆ If the channel is busy, the station waits until it becomes idle
 - ◆ If two stations start at the same instant, *collision*
 - ◆ Wait reasonable time (round trip plus ACK contention)
 - ◆ No ACK then *retransmit*

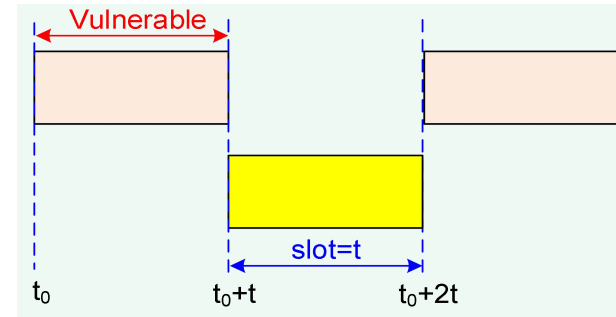
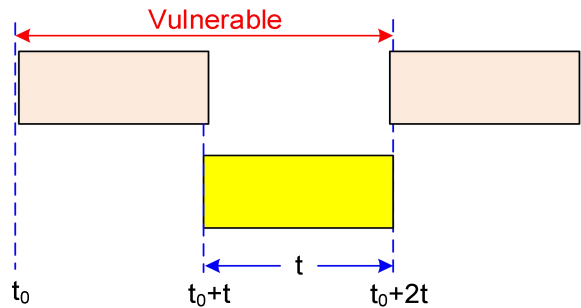
Review(1)

- 关于期中考试：5月6日-9日、闭卷、90分钟、英文试卷、简答题和计算题、无补考
- 核心问题：多个站点如何共享传输介质
- 解决方案：信道分配技术



Review(2):random access protocols

- Pure ALOHA: Continuous time, no carrier sense
- Slotted ALOHA: Slotted time, no carrier sense



Vulnerable Period: $2t$

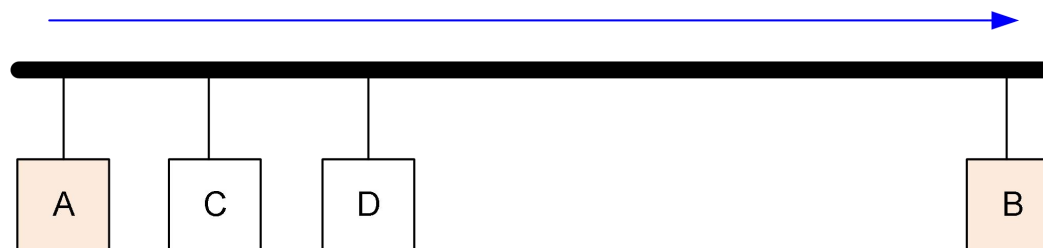
Max utilization(S): 18.4%

Vulnerable Period: t

Max utilization: 36.8%

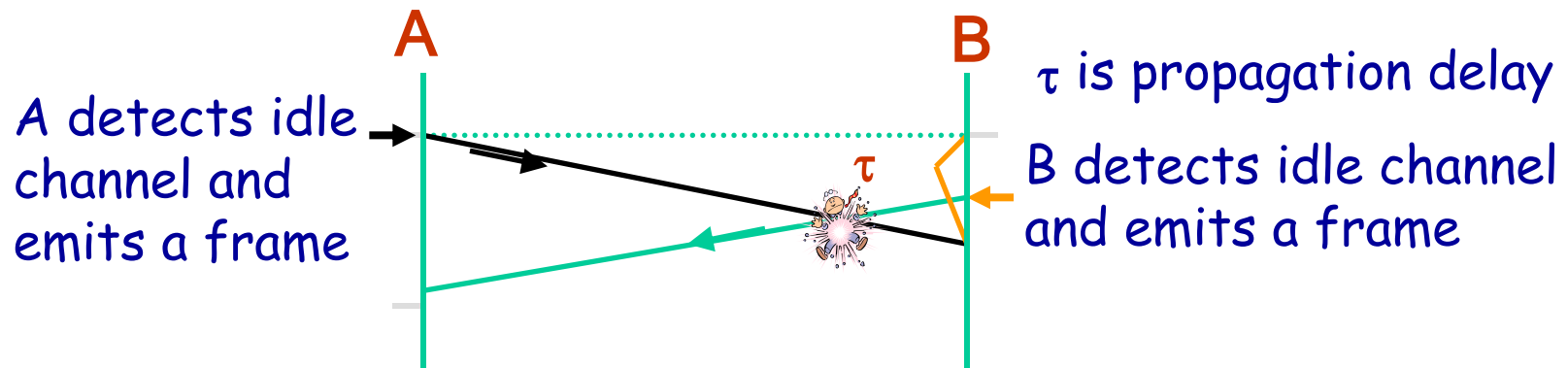
- CSMA: Continuous time, carrier sense

◆ Vulnerable period: Vulnerable Period= ?

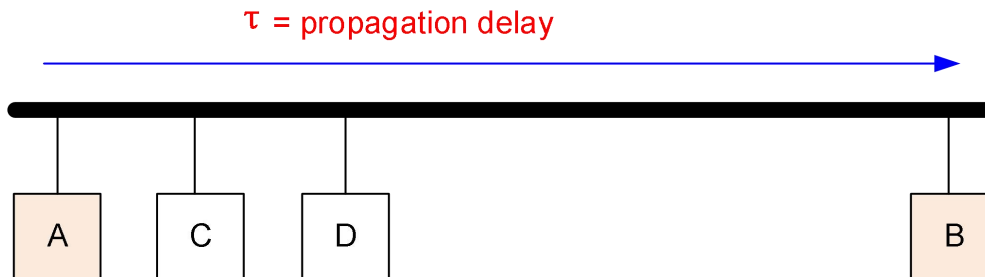


Vulnerable Period of CSMA

假设站A和站B分别位于电缆的两端，A和B之间的传播时延就是电缆的端到端传播时延 τ



Vulnerable Period = Propagation delay (τ)
(Propagation delay $\tau <$ transmission delay t)



Nonpersistant vs. Persistent CSMA

■ Nonpersistent CSMA(非坚持CSMA)

If the channel is busy, station waits for a random of time and try to sense again.

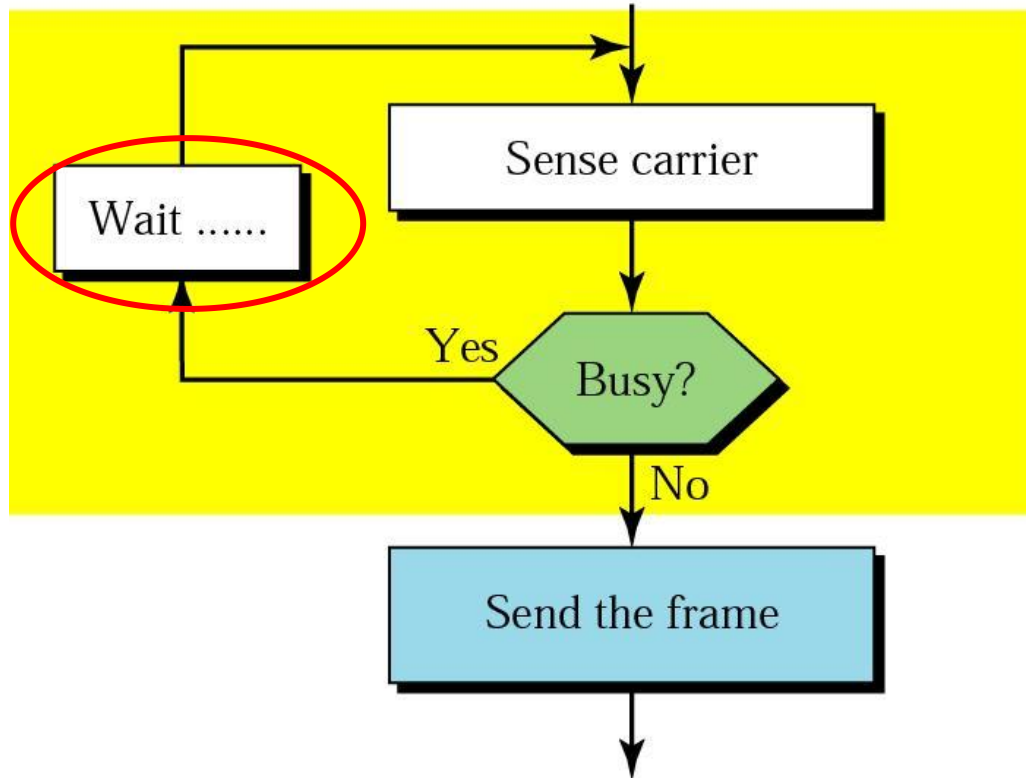
■ Persistent-CSMA(坚持CSMA)

■ If the channel is busy, waits until it becomes idle, then transmits frame immediately

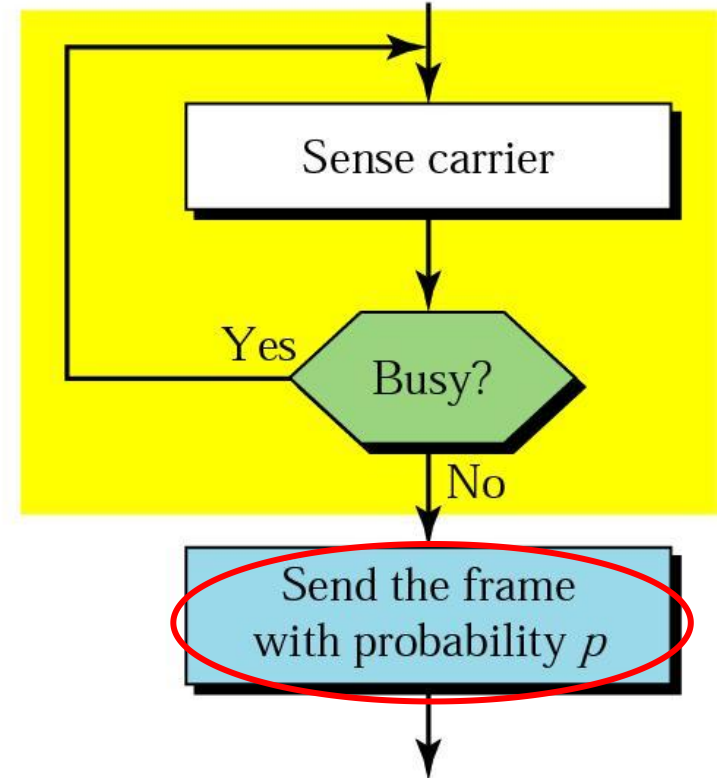
- 1-persistant CSMA: Sending a frame when detecting that channel is idle (with probability 1)
- p-persistent CSMA: Sending a frame with probability p when detecting idle channel, i.e. delay transmission for a random time with probability $1-p$

Nonpersistent vs. Persistent CSMA(cont.)

Nonpersistent strategy

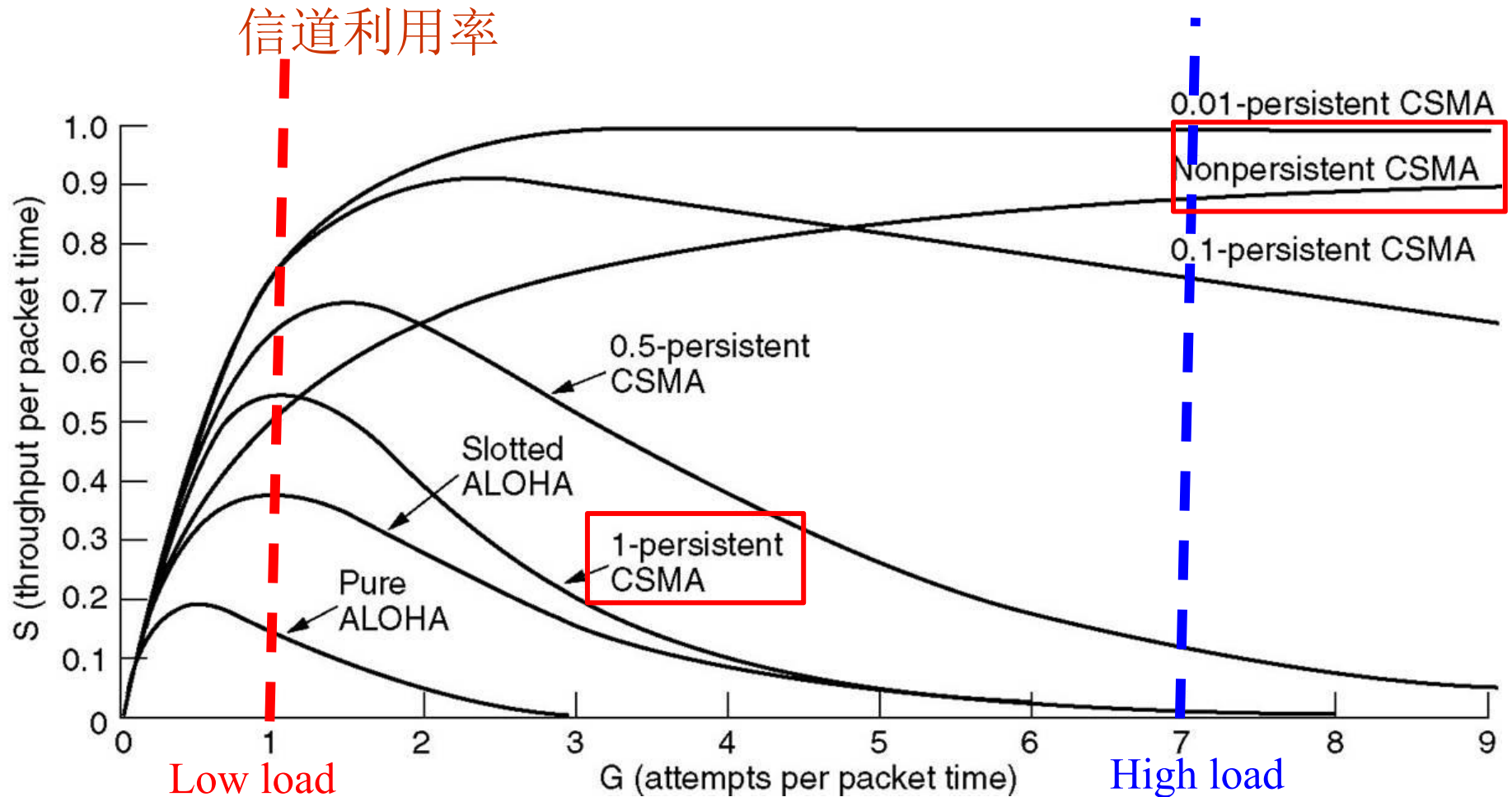


Persistent strategy



$p=1$: 1-persistent CSMA
 $p<1$: p-persistent CSMA

Channel Utilization of ALOHA and CSMA



CSMA Comparison

- 1-persistent CSMA
 - High throughput and low delay when low load
 - Low throughput when high load
- Nonpersistent CSMA
 - High throughput when high load

评价MAC协议的指标:

- ◆ 低负载（轻载）情况下的时延
- ◆ 高负载（重载）情况下的吞吐量（或信道利用率）

低负载适合使用竞争方法

Summary of above random access protocols

- Pure ALOHA, Continuous time, no carrier sense
Vulnerable period: $2t$
- Slotted ALOHA, Slotted time, no carrier sense
Vulnerable period: t
- CSMA, Carrier sense Vulnerable period: Propagation delay τ
在局域网中, τ 远小于 t
 - ◆ Nonpersistent CSMA
 - ◆ 1-persistent CSMA
 - ◆ p-persistent CSMA
- MAC协议指标: throughput, delay
- Similarity of above 3 protocols:
 - ◆ Acknowledged connectionless service (data & Ack)
 - ◆ Only when timer expires, does the sender know that error (including collision) occurs and then start retransmission

(2) CSMA/CD (with Collision Detection) (帶冲突检测的载波监听多点接入)

■ Basic idea

- ◆ **Carrier sensing**(载波监听): **listen** to the channel while a packet is being sent. (避免)

Q1: Who detects? Q2:How to detect? Q3:How long to detect?

- ◆ **Collision detection**(冲突检测): when collision is **detected**, cease transmission and transmit **jam** signal(强化信号) rather than finish transmitting the whole frame. (检测)

Q4:How long to wait?

- ◆ **backoff**(退避): After jam, wait random time then start again (恢复)

Why?

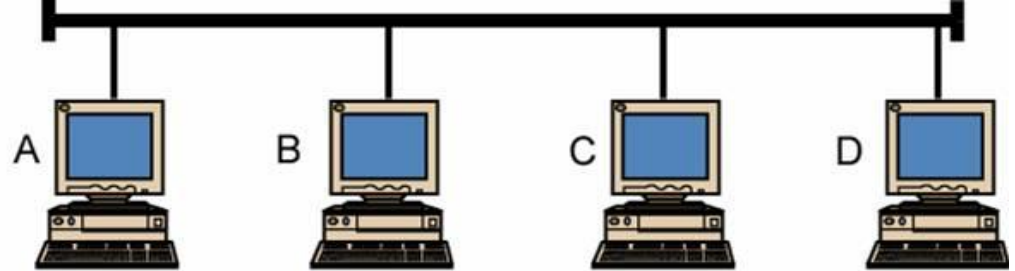
■ Half-duplex system

■ Unacknowledged connectionless service (No Ack)

■ Saving time and bandwidth.

■ Basis of Ethernet LAN.

CSMA/CD Operations



TIME t_0

A's transmission

C's transmission

Signal on bus

TIME t_1

A's transmission

C's transmission

Signal on bus

TIME t_2

A's transmission

C's transmission

Signal on bus

Detecting collision,
C ceases transmission

TIME t_3

A's transmission

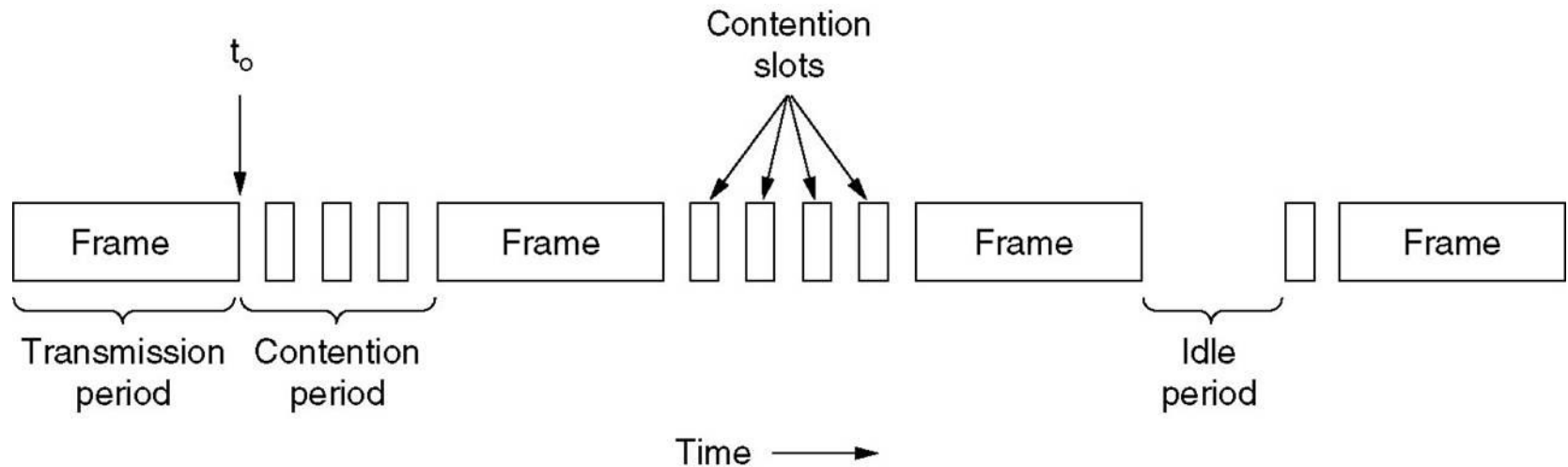
C's transmission

Signal on bus

[Armstrong]

CSMA/CD: States

- CSMA/CD can be in one of three states: contention, transmission, or idle.

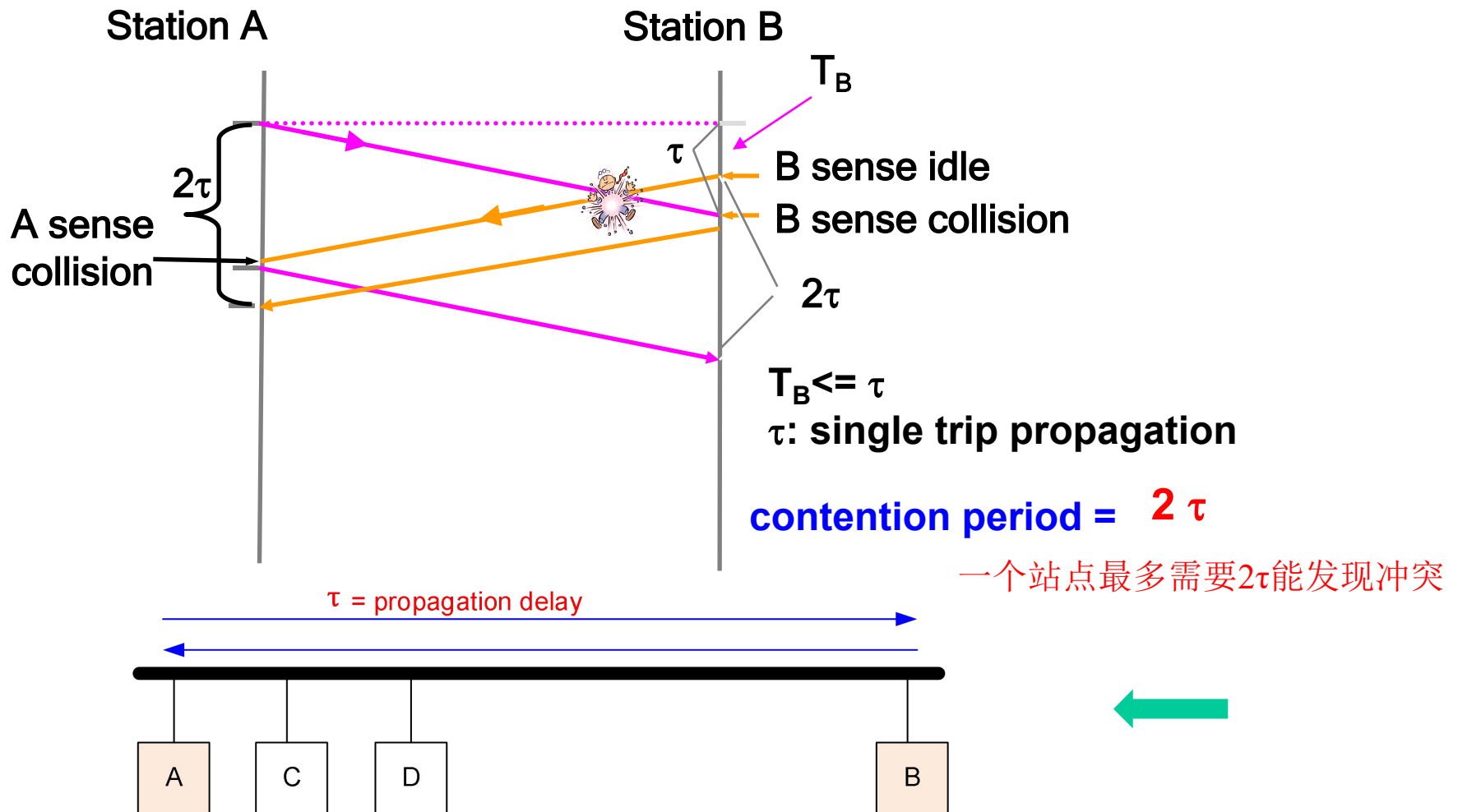


争用期

CSMA/CD: Collision Detection

- Q1: Who detects the collision?
 - ◆ The sender
- Q2: How to detect the collision?
 - ◆ Collision detection is an analog process
 - ◆ Constraint conditions
 - The power of receiving signal
 - The modulation mode of signals
- Q3: How long to detect the collision →
 - ◆ Content period: 2τ
- Q4: How long to wait for retransmission? →
 - ◆ Binary Exponential Backoff Algorithm

CSMA/CD: Contention period



1-persistent CSMA/CD: Retransmission Waiting Time

- Backoff Time(退避时间)
 - ◆ Choose a random number r , and after a delay of r , check if the medium is clear.
 - If the medium is clear, transmit
 - If the medium is in use, wait until free.
- If a collision occurs on a subsequent attempt to send the same frame, repeat
 - ◆ Keep track of the number of attempts;
 - ◆ if a preset maximum attempt number (often 16) is reached, abort
- Exponential backoff(指数退避): the maximum delay time r is doubled between transmission retries

1-persistent CSMA/CD:

Retransmission Waiting Time(cont.)

■ Binary Exponential Backoff Algorithm

- ◆ Random pick a number r from $\{ 0, 1, 2, \dots, 2^k-1 \}$
- ◆ **Backoff time** = $r \times$ basic backoff_time ($51.2\mu\text{sec}$)
- ◆ With $k = \min[\text{number of attempt}, 10]$

Basic backoff_time: $51.2 \mu\text{sec}$

1st collision, random number 0~1

2nd collision, random number 0~3

9th collision, random number 0~511

10th collision, random number 0~1023

11th collision, random number 0~1023

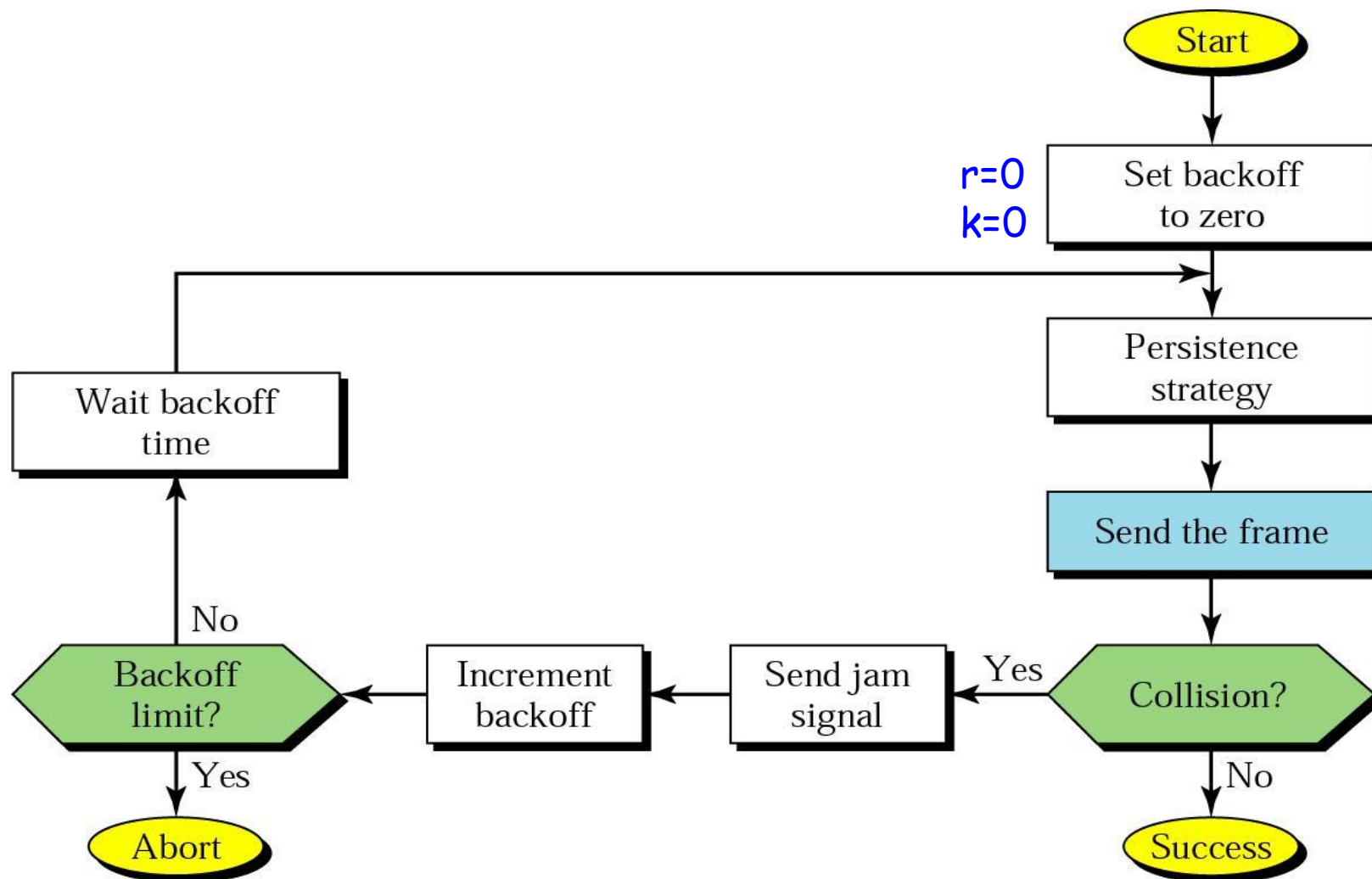
...

16th collision, random number 0~1023

After 16 collisions, reports failure

Further recovery is up to higher layers

Procedure of CSMA/CD



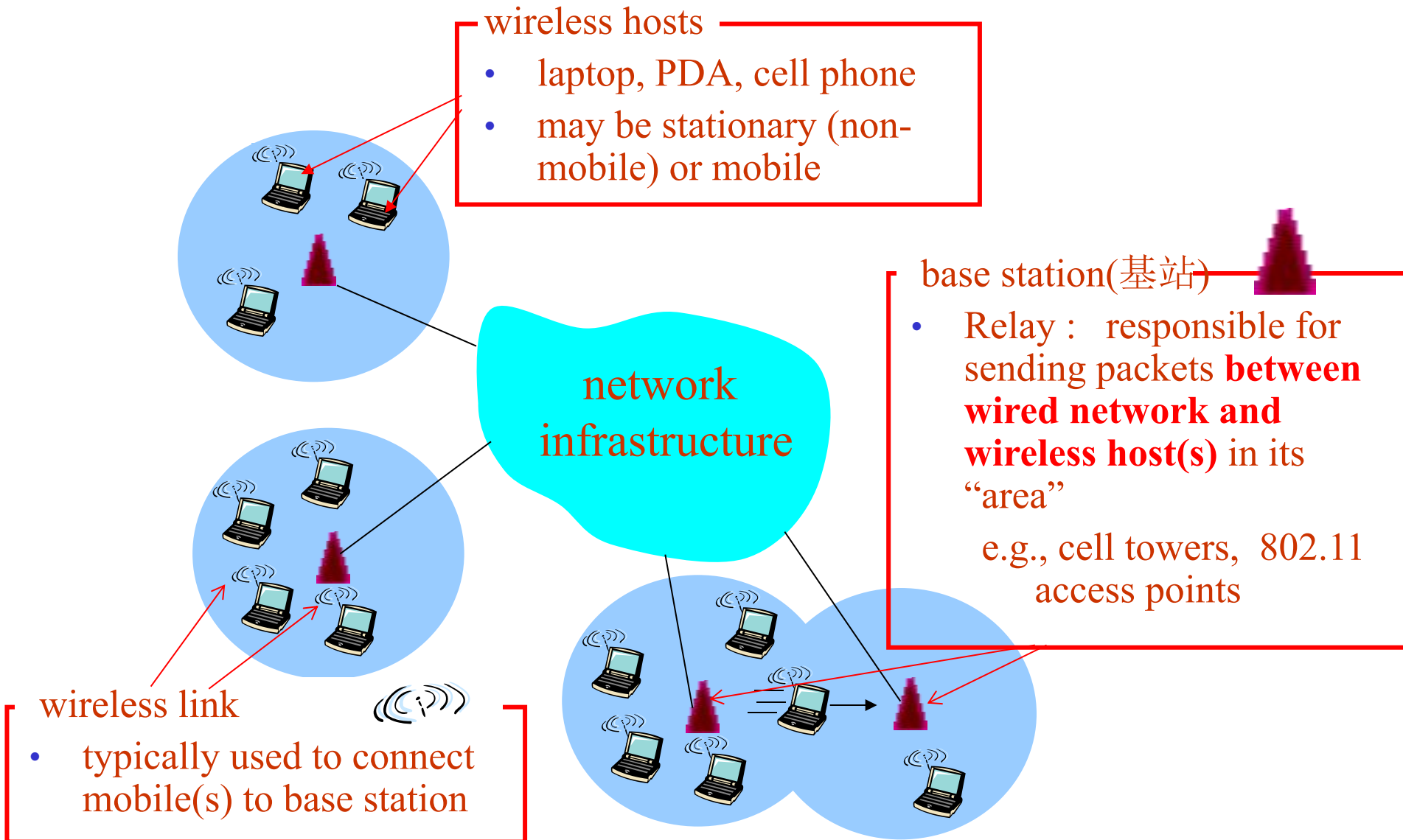
CSMA/CD

- 带冲突检测的载波监听多路访问协议(reduce collision)
- Step1-载波监听: Carrier sensing(发前先监听)
- Step2-冲突检测: Collision detecting(边发边检测冲突)
 - ◆ 如何检测: 收到了非自己发送的数据
 - ◆ 检测时间: 2τ
 - ◆ 检测到冲突的反应: 停止发送, 并发送jam信号
- Step3-随机退避: Binary Exponential Backoff
 - ◆ 重发次数越多, 可能等待的时间越长
 - ◆ 最多16次
- unacknowledged connectionless(data frame only)
- 性能: 帧时($\dagger$)远大于传播时间时(τ), 信道效率高

Multiple Access Protocols

- 4.2.1 ALOHA
- 4.2.2 Carrier Sense Multiple Access Protocols
- 4.2.5 Wireless LAN Protocols

(1) Elements of a wireless network



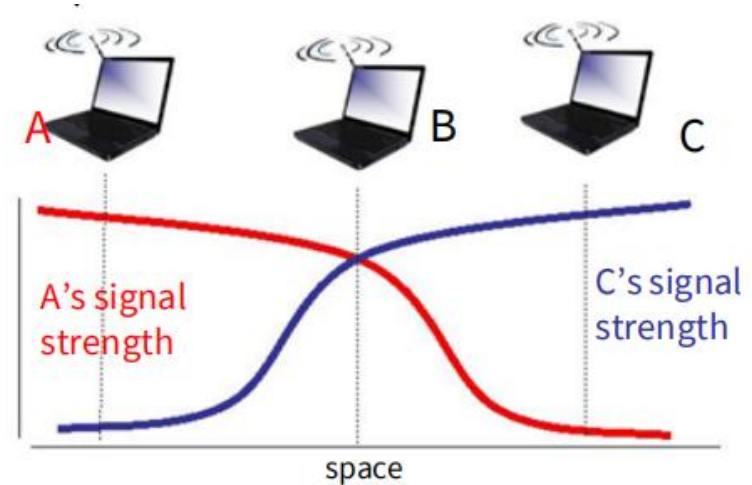
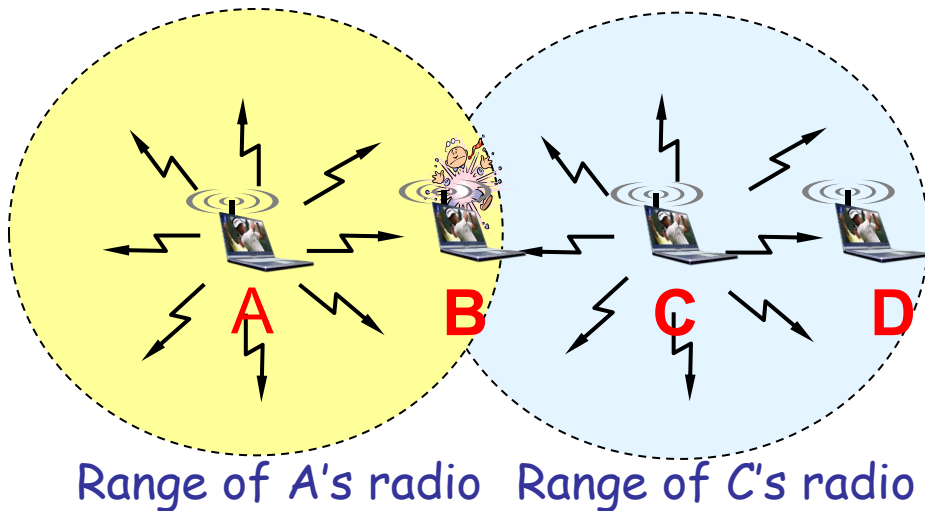
(2) Wireless LAN: Questions

■ Radio Transmitter & Receiver

- ◆ Radio transmitters have some **fixed range**.
 - In some wireless LANs, not all stations are within range of one another, thus **collision may not be detected**
- ◆ **Interference at the receiver**, not at the sender
 - received signals may be 1,000,000 times weaker than transmitted signals
- ◆ Multiple wireless senders and receivers create additional problems
 - Hidden terminal problem ✓
 - Exposed terminal problem

Hidden terminal (隐藏终端) problem

- A is transmitting to B
- C can not hear(sense) the signal, so C is sending to B at same time
- Collision at B (receiver)

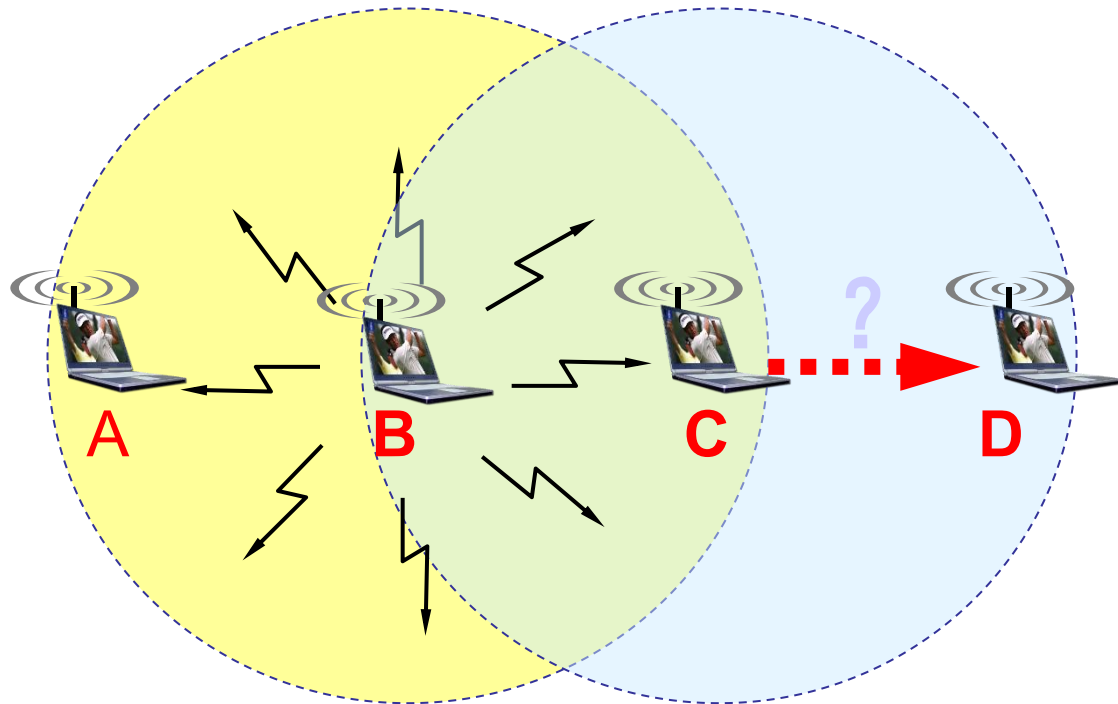


Signal attenuation:

- B, A hear each other
- B, C hear each other
- A, C can not hear each other, interfering at B.

Exposed terminal (暴露终端) problem

- B is sending to A
- C senses the signal, and **falsely** concluded that it can not send to D



Range of B's radio Range of C's radio

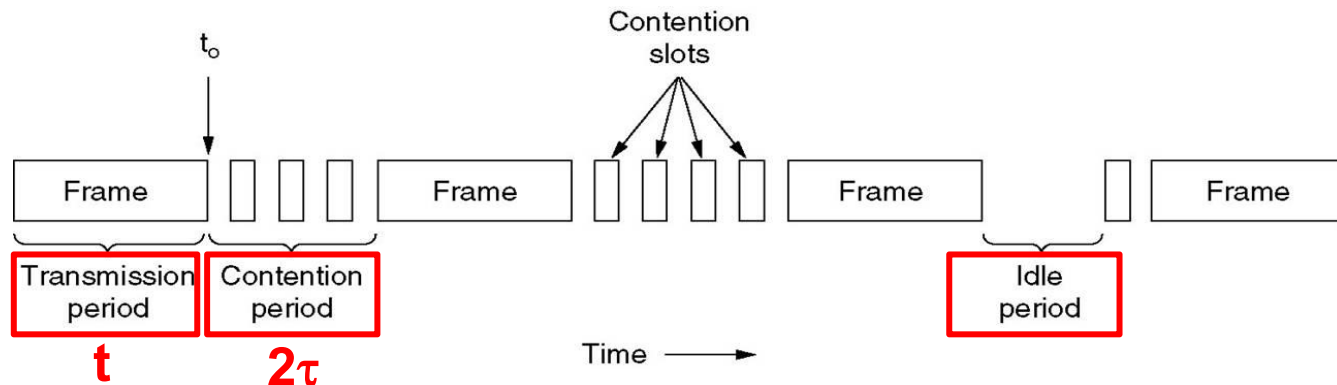
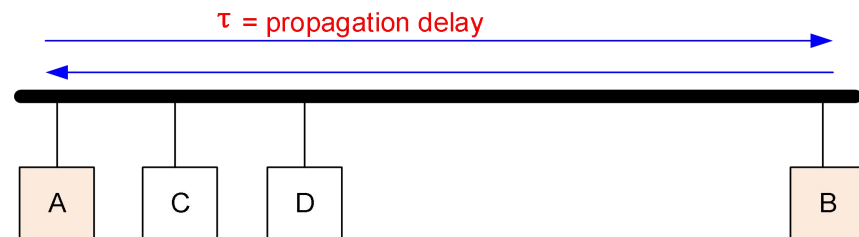
Review

CSMA

- ◆ Carrier sense, Vulnerable Period(脆弱期)= τ
- ◆ Nonpersistent CSMA
- ◆ 1-persistent CSMA, p-persistent CSMA

CSMA/CD

- ◆ three phrases
 - Carrier sensing
 - Collision detection: 检测时间(2τ)、jam signal
 - backoff: Binary Exponential Backoff Algorithm
- ◆ Vulnerable period= τ , contention period(争用期)= 2τ



(2)Wireless LAN:Solutions

■ CSMA/CD is not appropriate

- collision, interference
- Hidden terminal & Exposed terminal...

■ WLAN

- Acknowledged connectionless(ACK+retransmission)
- Collision Avoidance

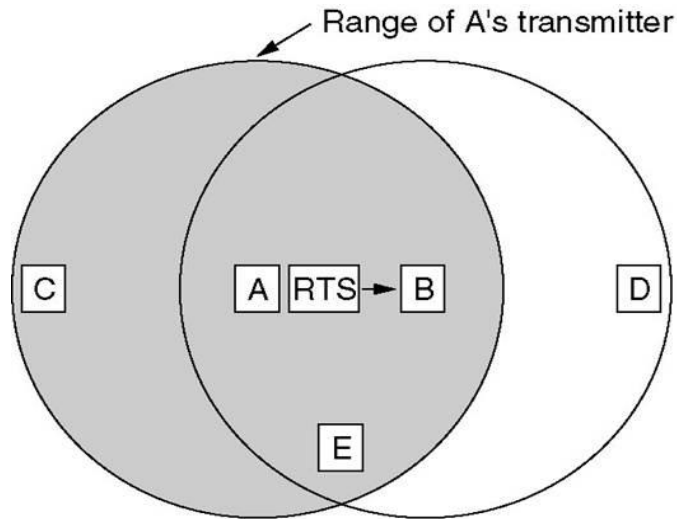
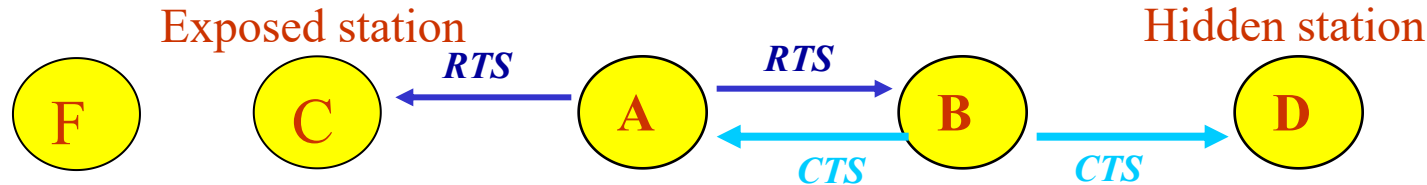
■ Stations use a handshake to determine which one can send

- ◆ **MACA**: Multiple Access with Collision Avoidance
- ◆ **CSMA/CA**: CSMA with Collision Avoidance
(IEEE802.11 Wi-Fi)

(3) MACA (冲突避免多路访问)

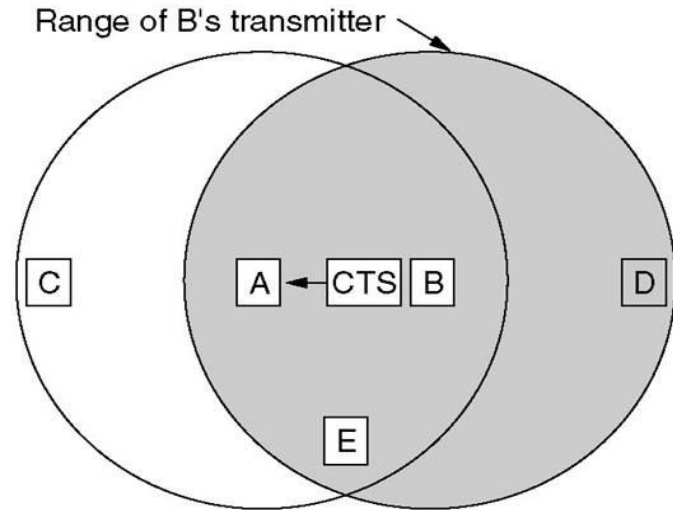
- Multiple Access Collision Avoidance
 - To further enhance reliability, four-frame exchange may be used
 - ◆ Source issues a Request to Send (RTS) frame to destination
 - ◆ Destination responds with Clear to Send (CTS)
 - ◆ After receiving CTS, source transmits data
 - ◆ Destination responds with ACK
- RTS(S to R) -> CTS(R to S) -> DATA(S to R) -> ACK(R to S)**
- RTS and CTS are used to **reserve channel**(预约信道)
 - ◆ RTS alerts all stations within range of source
 - RTS (30 bytes, contains data length of the data frame to send)
 - ◆ CTS alerts all stations within range of destination

(3)MACA:Operation Example



Hearing **RTS**, **C** and **E** keep silent

RTS alerts all stations
within range of source

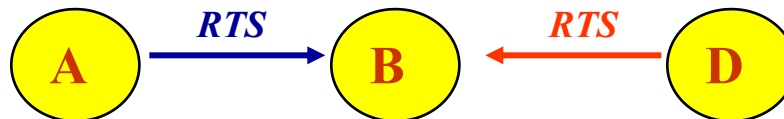


Hearing **CTS**, **D** and **E** keep silent
solving hidden station problem

CTS alerts all stations
within range of destination

(3)MACA: Collisions

- 2 stations send **RTS** frames to one station at the same time
- Unsuccessful transmitter (does not hear a **CTS** within expected time interval) **waits a random time** and tries again
- The algorithm used is **binary exponential backoff**(二进制指数退避)
 - ◆ Similar to **CSMA/CD**

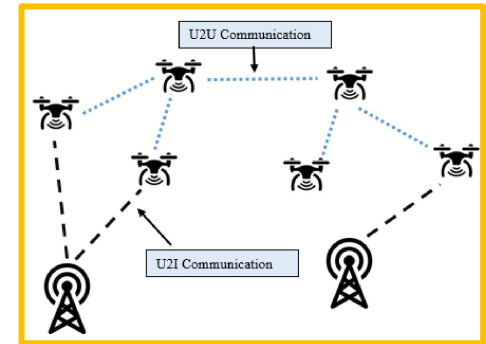


Related researches for MAC

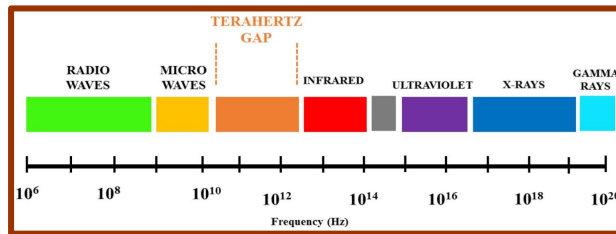
MAC Protocols for

- ◆ Terrestrial Wireless Sensor Networks (TWSNs)
- ◆ Underwater Wireless Sensor Networks (UWSNs)

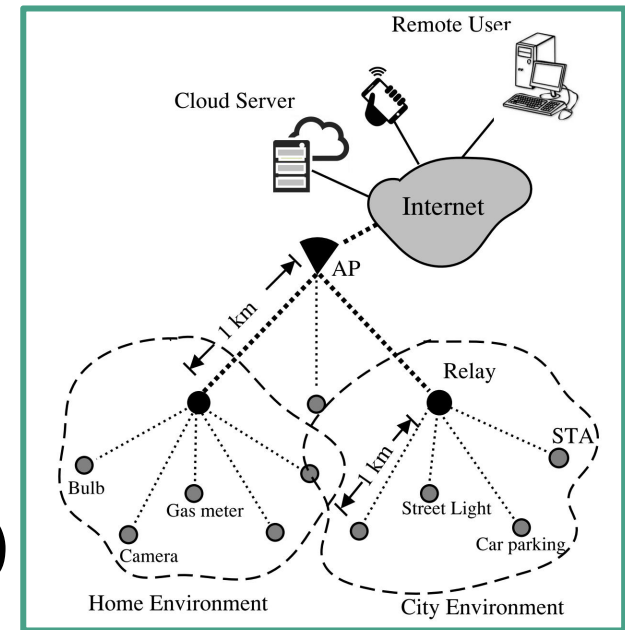
Parameters	TWSNs	UWSNs
Common Communication Modality	Radio waves	Acoustic waves
Propagation Speed	300,000,000 m/s	1,500 m/s
Transmission Range	10 m - 100 m	Up to 10 Km
Frequency	908 - 928 MHz	10 Hz - 100 KHz



- ◆ VENETs, UAV Networks
- ◆ IoT communications
- ◆ Terahertz Communication



- ◆ Visible Light Communication(VLC)



Outline

- 4.1 Channel Allocation Problem
- 4.2 Multiple Access Protocols
- (補)LAN Reference Model: IEEE 802 RM
- 4.3 Ethernet
- 4.4 Wireless LANs
- 4.8 Data Link Layer Switching

(1) Introductions to LANs

■ Features of LAN

- ◆ Limited number of stations and range
- ◆ Higher data rate
- ◆ Lower error rate
- ◆ Supporting broadcasting and multicasting
- ◆ Single administration

■ Topologies

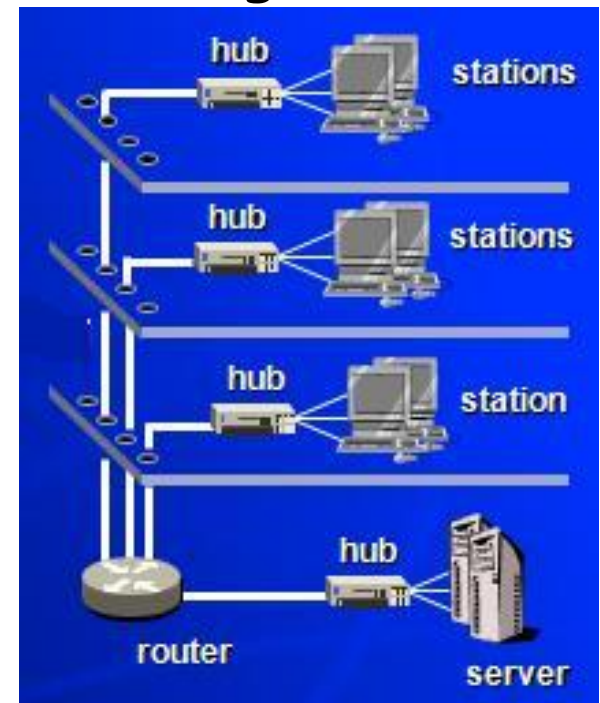
- ◆ Bus, Star, Ring, Tree, ...

■ Transmission Media

- ◆ Twisted pair, Coaxial Cable, Fiber, RF

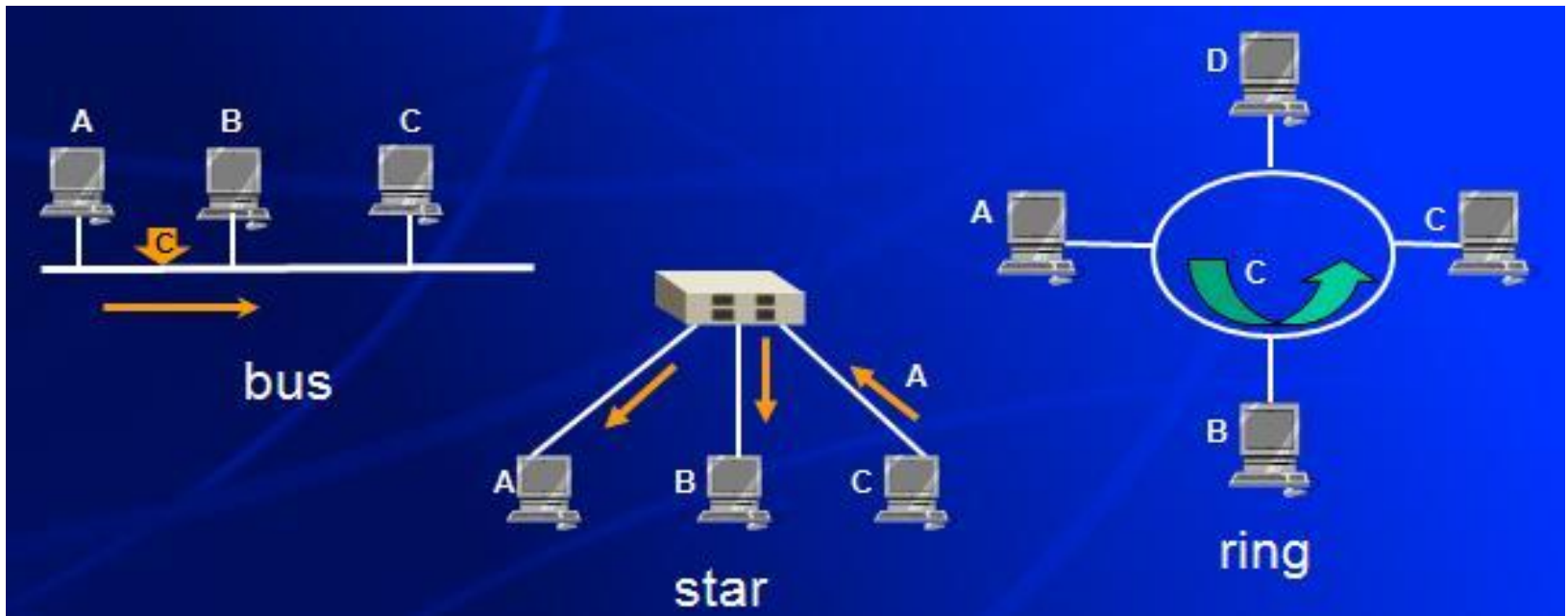
■ Media Access Methods

- ◆ Following MAC protocols to share a common channel

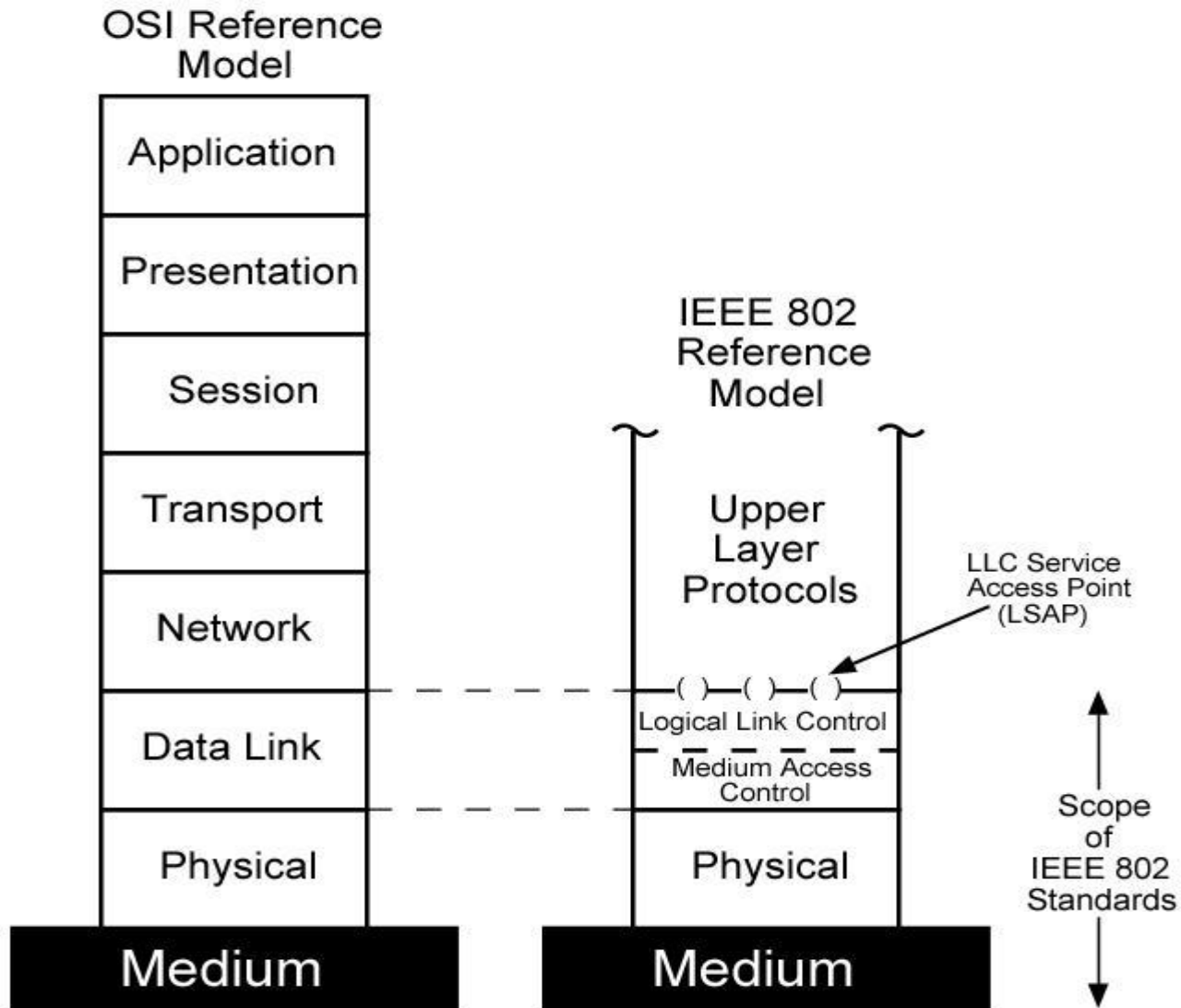


Typical topologies

- The topology of a network defines the structure of its logical layout
 - ◆ **Bus/Tree**: All stations connected to common channel
 - ◆ **Star**: All stations connected to central node (eg. HUB)
 - ◆ **Ring**: A station connects to neighbor with point-to-point link

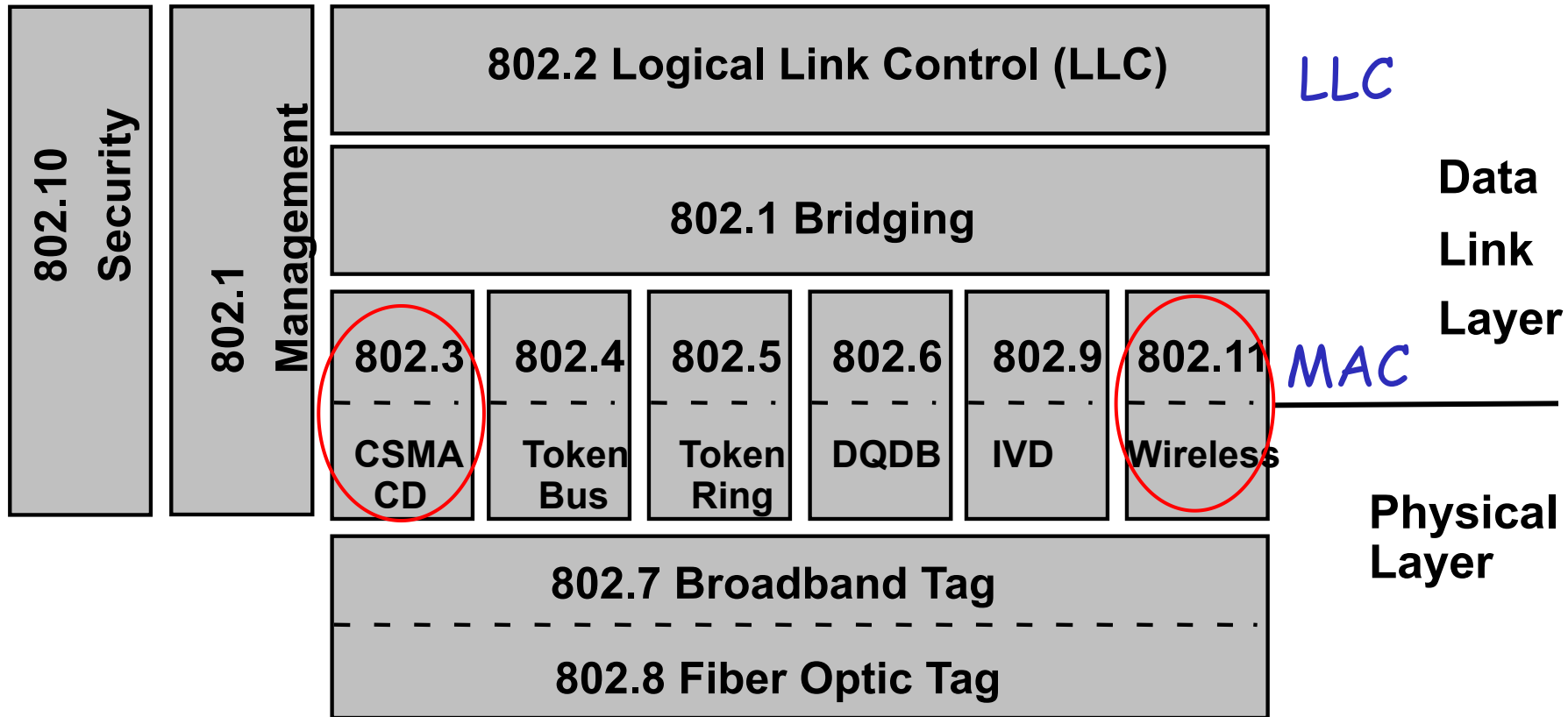


(2) IEEE 802 Reference Model



[Stallings]

(3) IEEE 802 Standards



(4) IEEE 802 Layers

■ Physical Layer

- ◆ Signal encoding/decoding
- ◆ Bit transmission/reception
- ◆ Transmission medium and topology

■ MAC (Media Access Control, 媒体/介质访问控制)子层

- ◆ Framing
- ◆ Error checking
- ◆ Physical Addresses: *eg. MAC addresses*
- ◆ Sharing of medium: **Multiple Access Control**
- ◆ LAN switching, VLAN

■ LLC (Logic Link Control, 逻辑链路控制)子层

- ◆ Hides difference of MAC layers, providing **a uniform interface** to network layer
- ◆ **Flow control** and **error control**
- ◆ Inter-connecting LANs with bridge(网桥)

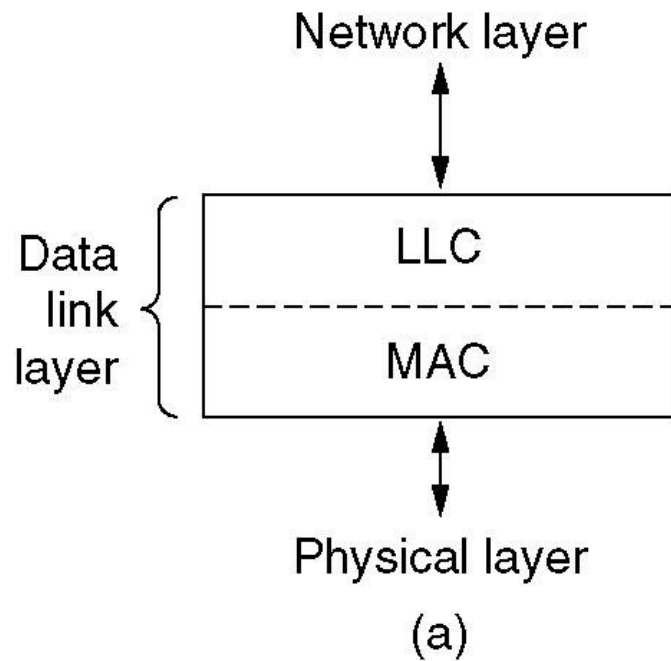
MAC Overview

- Sender: Assembly of data into frame with address and error detection fields
 - Receiver: Disassembly of frame
 - ◆ Address recognition
 - ◆ Error detection
 - Govern access to transmission medium
 - For the same LLC, several MAC options may be available
- (不同的LAN, MAC子层不同, LLC子层相同)

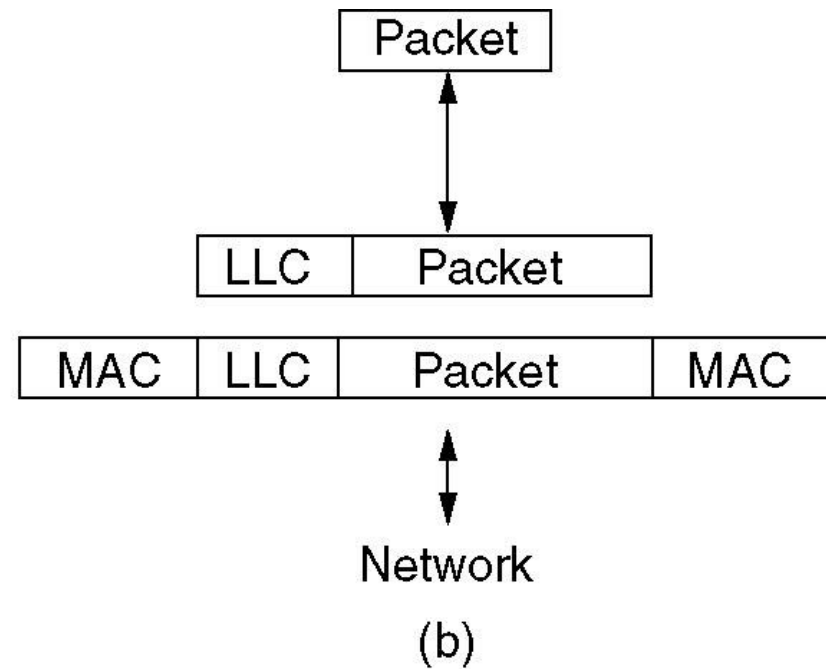
802.2: LLC

- Run on top of 802.3 & other 802 protocols
 - ◆ Provides a single format and interface to the network layer
 - ◆ Hides differences between the various kinds of 802 physical networks
- Provides three service options
 - ◆ Unreliable datagram service
(best-efforts datagram service)
 - ◆ Acknowledged datagram service
(Error-controlled)
 - ◆ Reliable connection-oriented service
(Error-controlled & flow-controlled)

Position of LLC

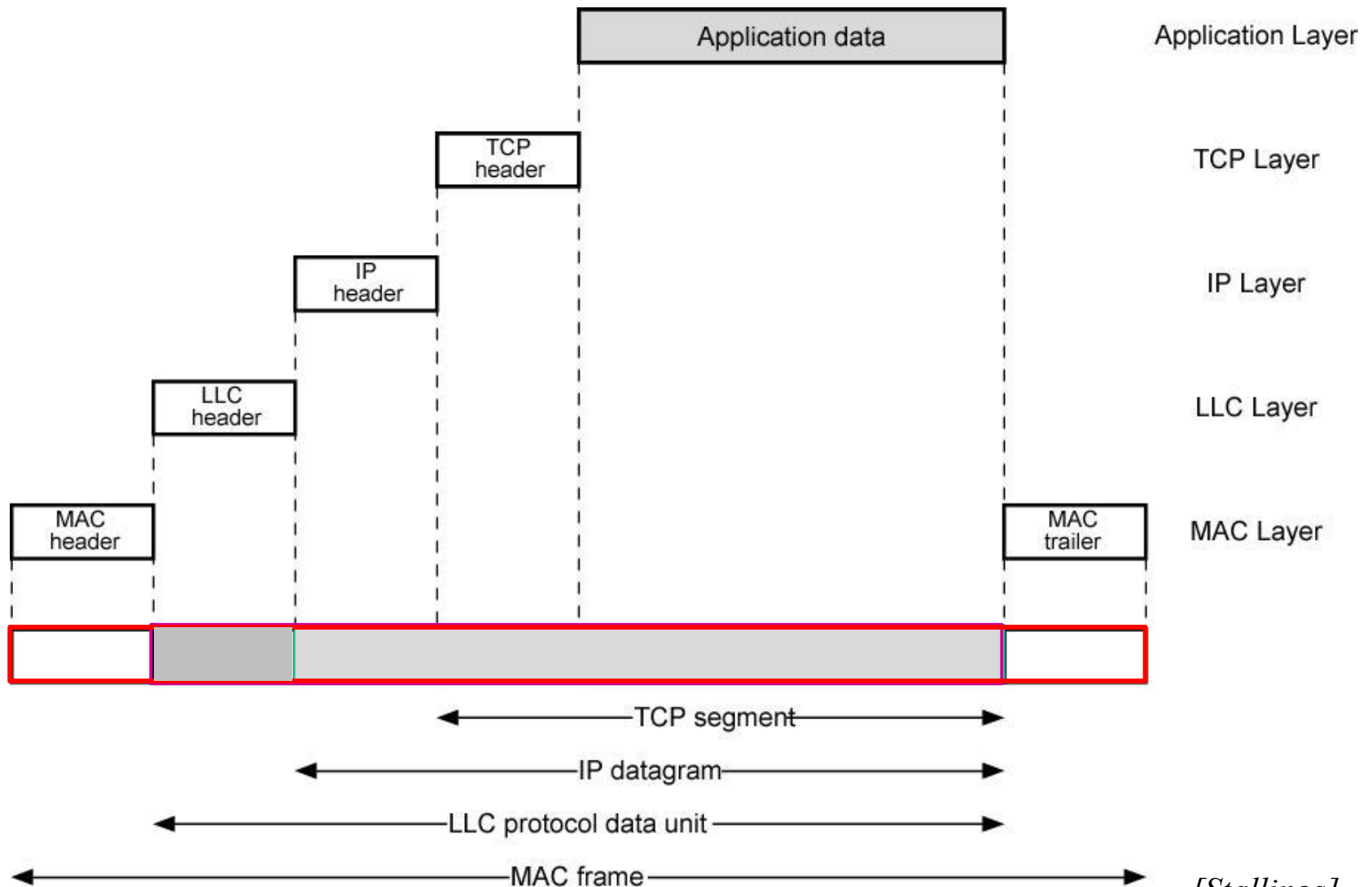


(a) Position of LLC.



(b) Protocol formats.

LAN Protocols in Context (封装)



[Stallings]

Outline

- 4.1 Channel Allocation Problem
- 4.2 Multiple Access Protocols
- (補)IEEE 802 Reference Model
- 4.3 Ethernet
- 4.4 Wireless LANs
- 4.8 Data Link Layer Switching

(1)Early History

1973	Xerox PARC begins development of bus topology LAN , 2.94Mbps
1976	Successful CSMA/CD system to connect 100 workstations on 1 km cable
1980	Digital Equipment Corp, Intel Corp, and Xerox release de-facto Ethernet DIX standard , 10Mbps
1980	IEEE 802 Standardization process begins
1985	IEEE 802.3 Standard Published
1995	Grand Junction Network Company brings up the structure of 100Mbps Ethernet
1998	Standard of Gigabit Ethernet addressed
2002	1 st Standard of 10Gigabit Ethernet published

(2)Classification of Ethernet

DIX Ethernet	10Mbps Ethernet (Classic Ethernet)	CSMA/CD
IEEE802.3		
IEEE802.3u	100Mbps Ethernet (100Base-T Fast Ethernet and Auto-Negotiation)	CSMA/CD or switch
IEEE802.3z	Gigabit Ethernet (1000Base-SX / -LX /-CX)	
IEEE802.3ab	Copper Gigabit Ethernet (1000Base-T)	
IEEE802.3ae	10Gbps Ethernet	
IEEE802.3ba	40/100Gbps Ethernet	Switch, no CSMA/CD

Ethernet

- 4.3.1 Classic Ethernet: Physical Layer
- 4.3.2 Classic Ethernet: MAC Sublayer Protocol
- 4.3.3 Ethernet Performance
- 4.3.4 Switched Ethernet
- 4.3.5 Fast Ethernet
- 4.3.6 Gigabit Ethernet
- 4.3.7 10G Ethernet

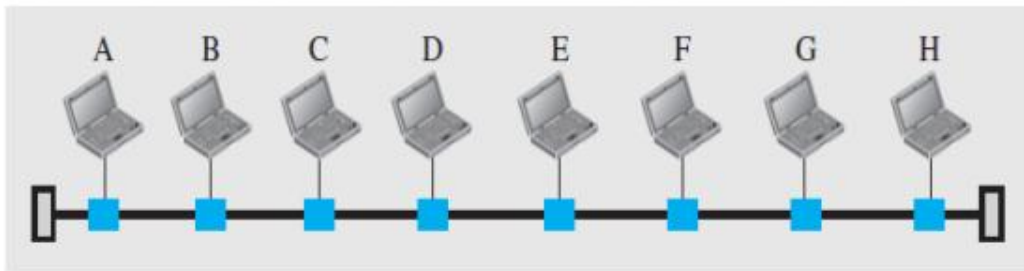
(1) Physical Topology

■ **Bus:** Popular through mid 90s

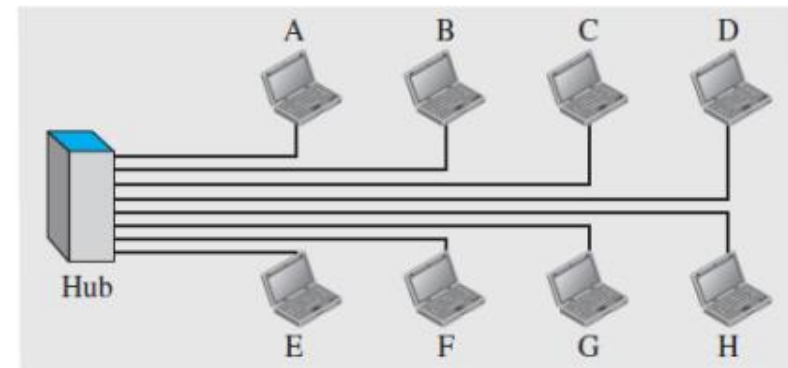
- ◆ All nodes are in same **collision domain**. So all nodes receive sent frames and can collide with each other

■ **Star:** Prevails today

- ◆ Hub: physical topology is star (logical transmission is bus)
- ◆ Switch: active **switch** in center



Bus topology using Coaxial cable



Bus topology using a hub

(2) Hubs(集线器)

■ Physical Layer device

- ◆ regeneration signal(digital signal)
- ◆ Providing a **star** connection **physically**, implements a **bus** topology **logically**(one collision domain)

■ Supporting 10Mbps or 100Mbps

■ Configuration limitation

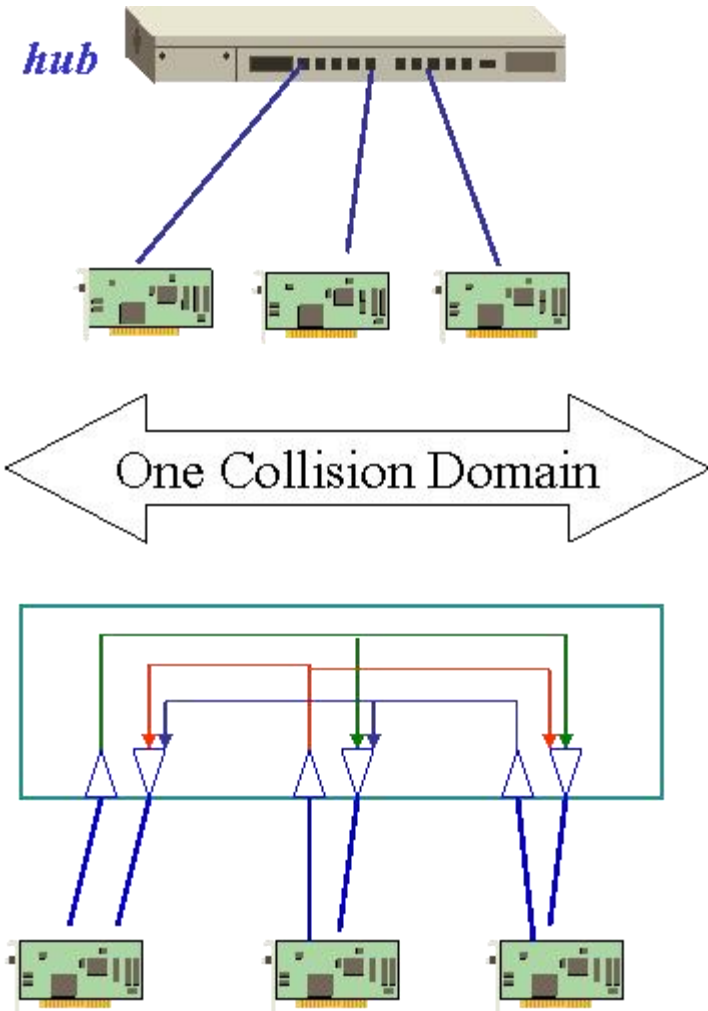
- ◆ up to 3 hubs in a segment

■ Naming principles of Ethernet

100 BASE -T X

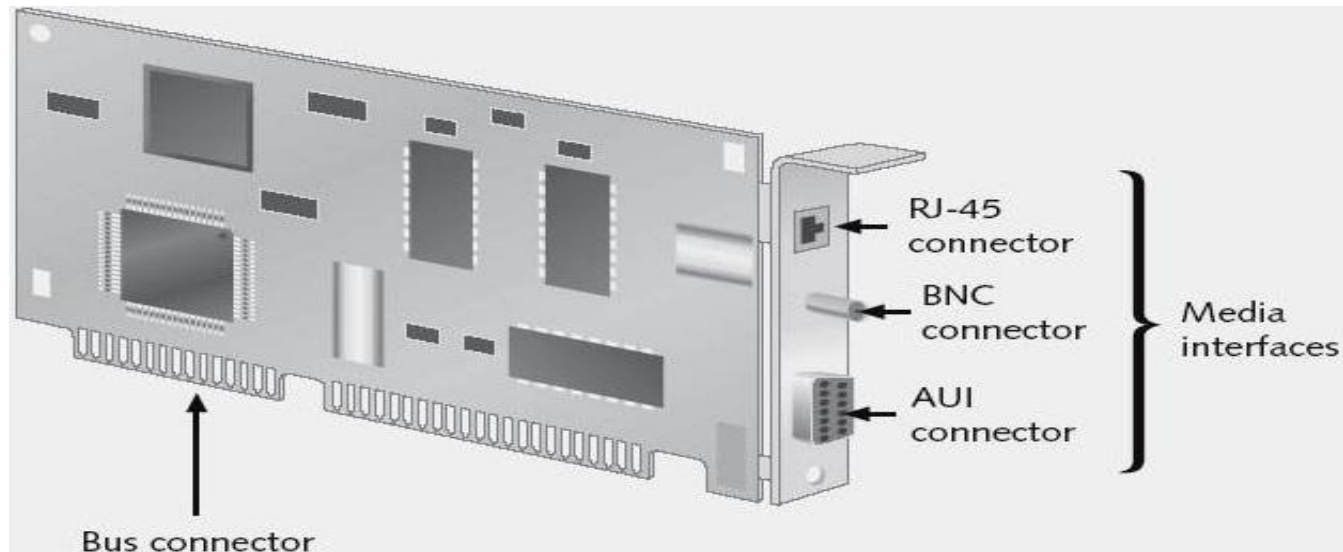
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speed modulation media encoding



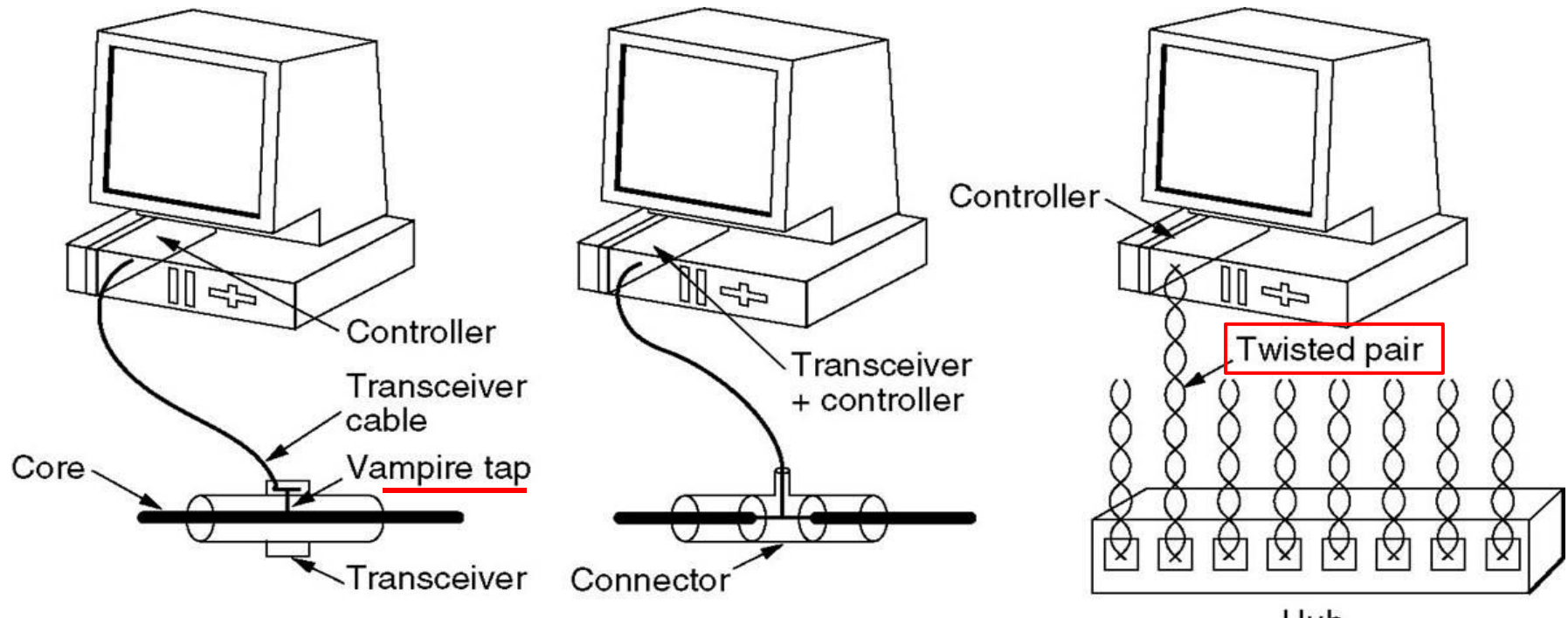
(3) Network Interface Cards (网卡)

- Tasks of NIC
 - ◆ Establishes and manages the computer's network connection
 - ◆ Encoding: translates digital computer data into signals
- NIC provides LLC, MAC and physical layer functions
- Each card has a unique MAC address



(4) Classic Ethernet(传统以太网): 10Mbps

Name	Cable	Max. seg.	Nodes/seg.	Advantages
10Base5	Thick coax	500 m	100	Original cable; now obsolete
10Base2	Thin coax	185 m	30	No hub needed
10Base-T	Twisted pair	100 m	1024	Cheapest system
10Base-F	Fiber optics	2000 m	1024	Best between buildings



10Base5

Coax

10Base2

10Base-T

(4) Classic Ethernet: 10Base5-Thick Ethernet

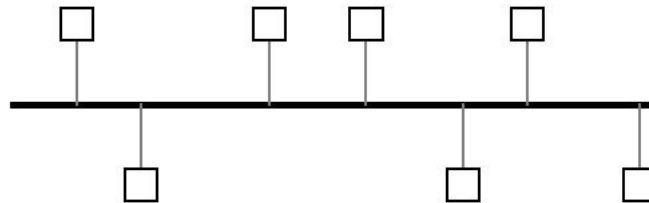
■ Cable

- ◆ 10mm, **Coaxial cable**, 500m
- ◆ Connections to it are generally made using **vampire taps**, in which a pin is very carefully forced halfway into the coaxial cable's core

■ Transceiver cable

- ◆ Connects the **transceiver** to an interface board in computer
- ◆ Up to 50 meters long

■ Bus Topology: Truly shared media



(4) Classic Ethernet: 10Base5-Thick Ethernet(cont.1)

■ Interface board

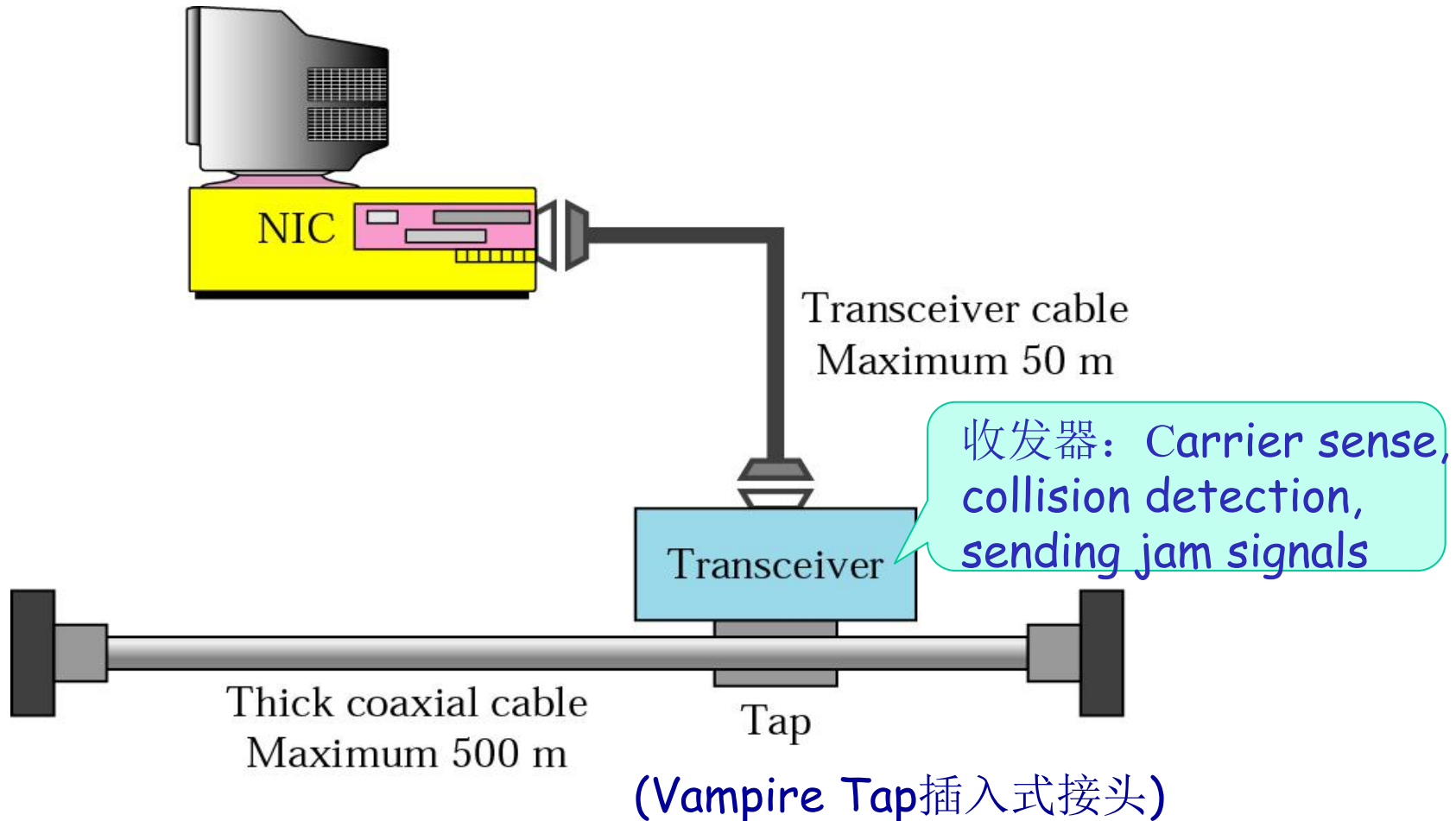
- ◆ Transmits and receives frames, assembling data into proper frame format, processing checksums
- ◆ Manage a pool of buffers for incoming frames, a queue of buffers to be transmitted, direct memory transfers with the host computers

■ Transceiver(CSMA/CD)

- ◆ Handle carrier detection and collision detection
- ◆ For collision, puts a special invalid signal(jam) on cable to ensure that all other transceivers also realize that a collision has occurred

(4) Classic Ethernet: 10Base5-Thick Ethernet(cont.2)

■ Connection of a Station using 10Base5



Computer Networks, MAC Sublayer

(4) Classic Ethernet: 10Base2-Thin Ethernet

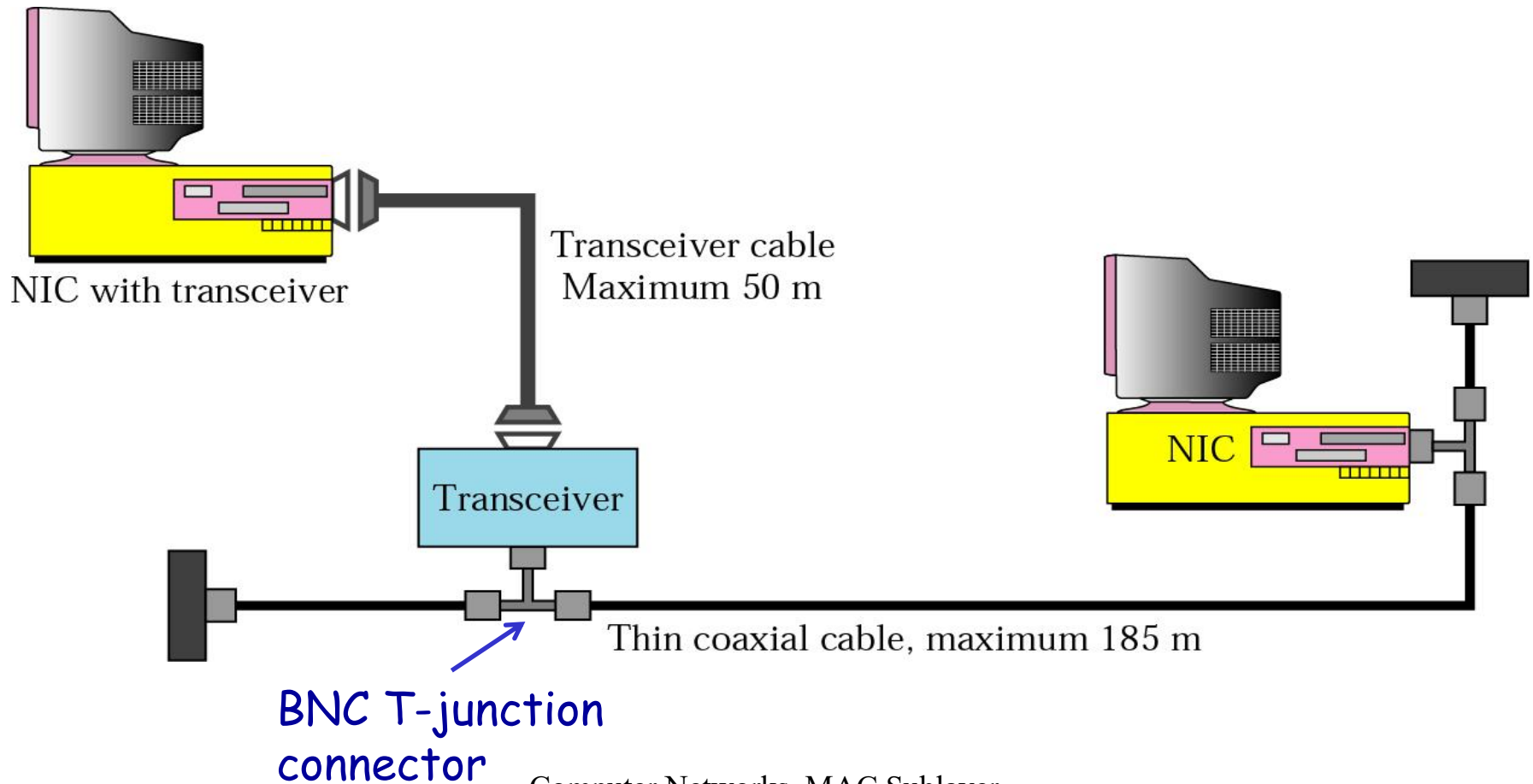
- Cable: 5mm, Coax, 185m
- BNC T-junction connector
- Transceiver electronics are put on the NIC



BNC T-junction connector

(4) Classic Ethernet: 10Base2-Thin Ethernet(cont.)

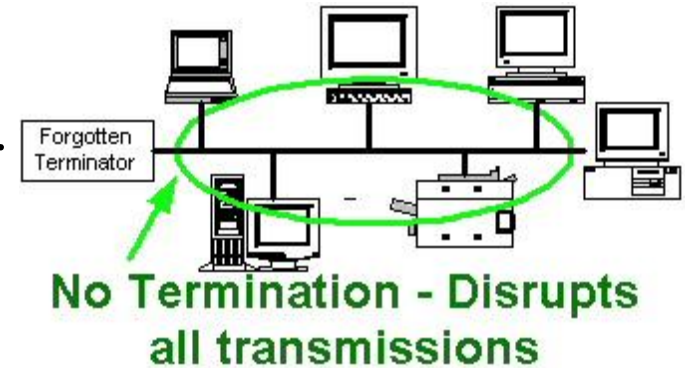
■ Connections of Stations using 10Base 2



Bus Topology Evolved to Star Topology

■ Problems with Bus Topology

- ◆ Break in Coax cable can sever service to multiple nodes
- ◆ Fault in Coax cable can disrupt service to all nodes
- ◆ Adding/Removing nodes disrupts network



■ Star Topology

- ◆ Driven by cabling issues
 - Single cable breaks/faults effect only one node
 - Emergence of cheap unshielded twisted pair (UTP) cable
- ◆ Introduces "**Hub**" that isolates faulty nodes/cables

(4) Classic Ethernet: 10BaseT

- 10Base-T: No shared cable at all

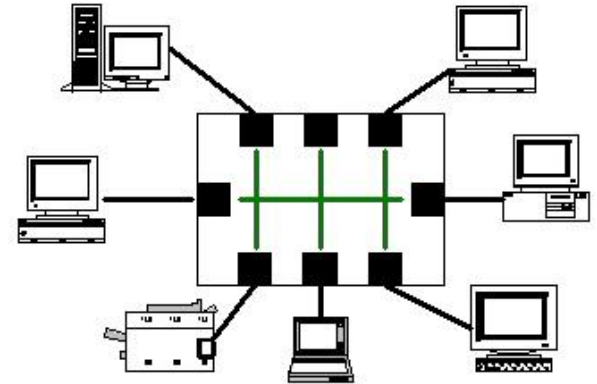
- ◆ Twisted pair cables

- HUB(集线器)

- ◆ A box full of electronics

- ◆ Station is connected to HUB by a **dedicated** cable (Max 100m)

- **Star Topology** with **Logical Bus**(任意站发送, 所有站接收)



- RJ-45, UTP5 (Unshielded Twisted Pairs)

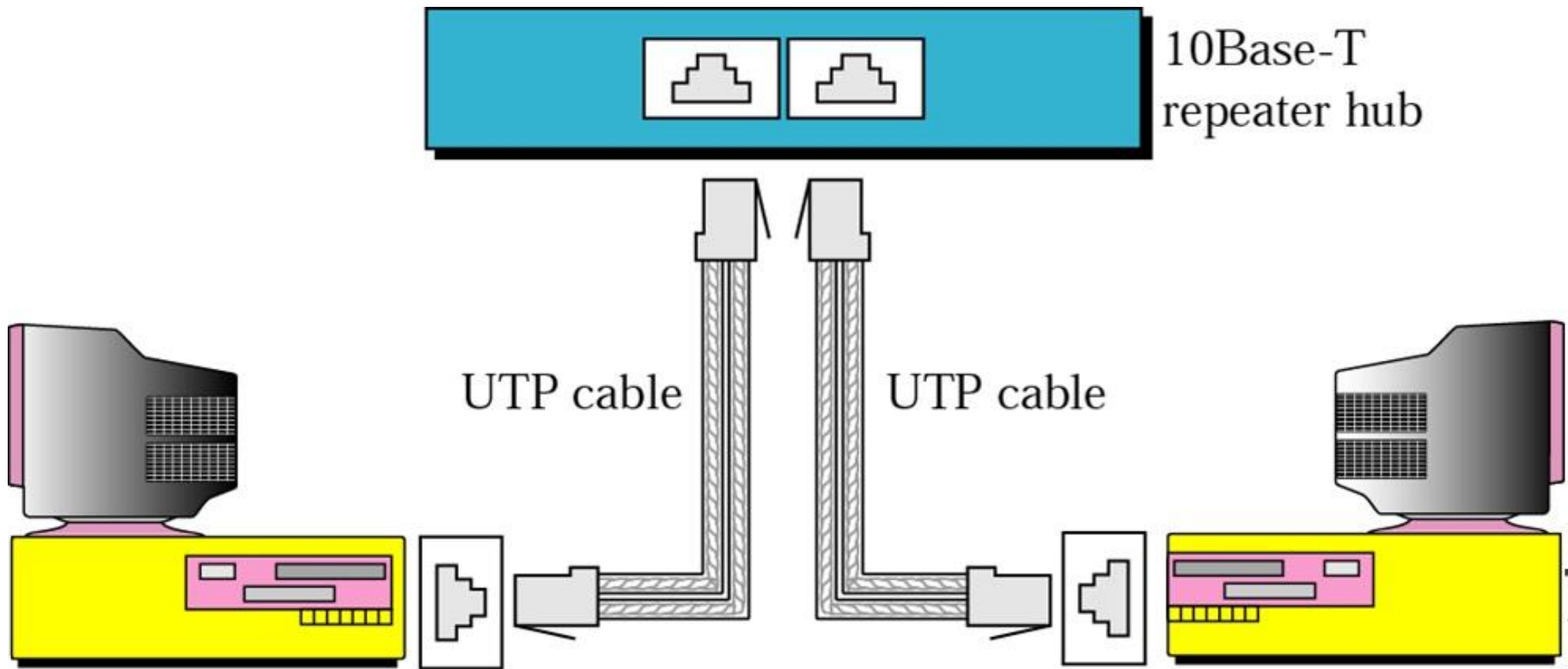
- ◆ 两对线, 收发各一对

- ◆ 差分传输: 减少电磁干扰

- ◆ 同一线对的绞合: 减少近端串扰

(4) Classic Ethernet: 10BaseT(cont.1)

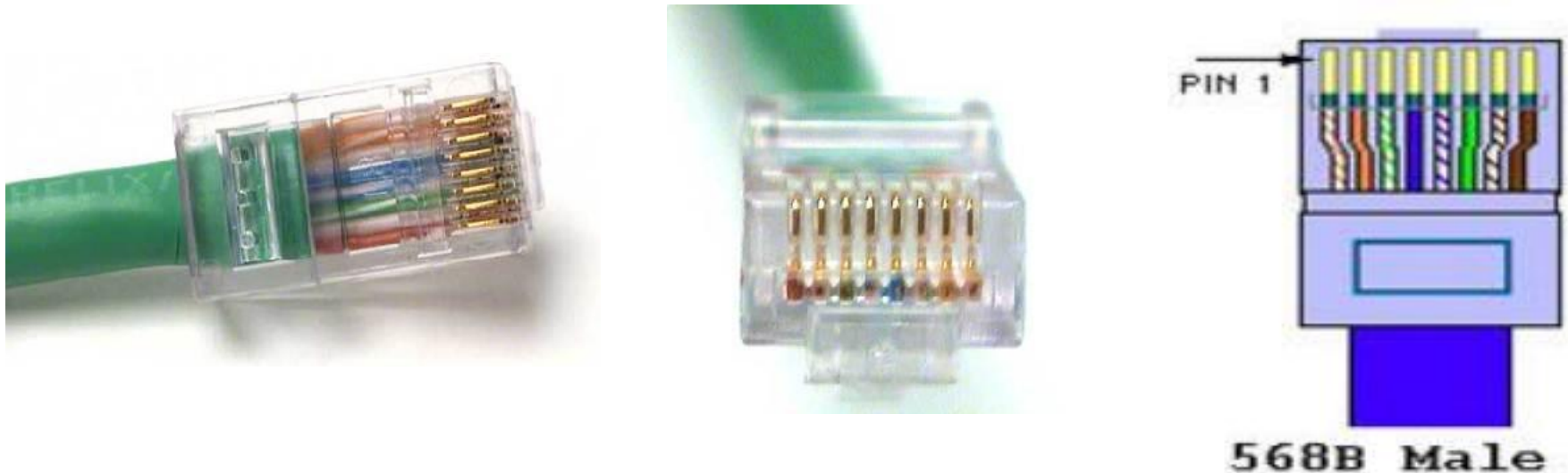
■ Connection of Stations



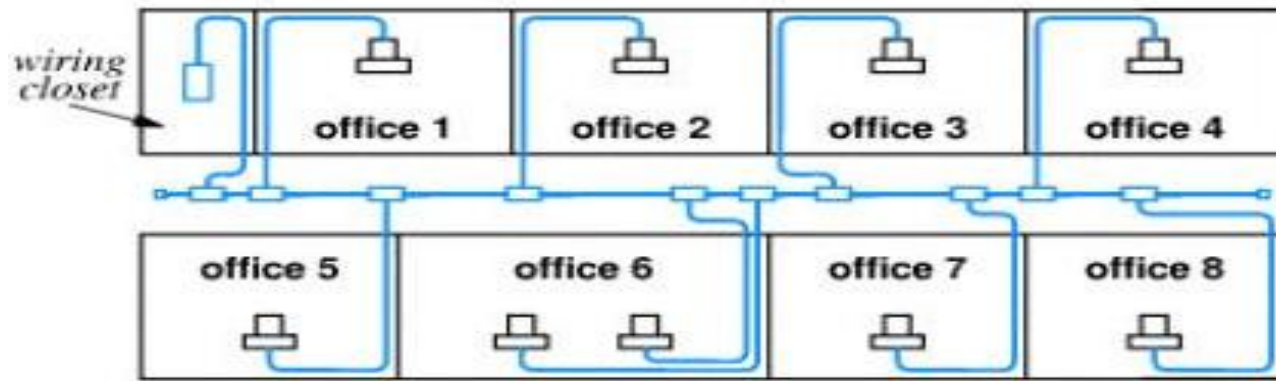
(4) Classic Ethernet: 10BaseT(cont.2)

■ Connection Media: UTP Implementation

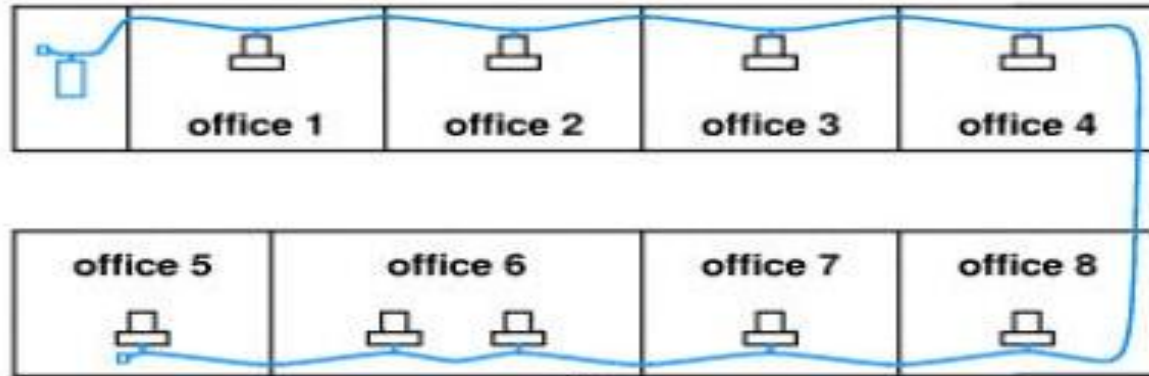
- ◆ **RJ-45 connectors** are used for UTP cable
- ◆ RJ stand for registered jack(标准插座), and the number 45 refers to a specific wiring sequence



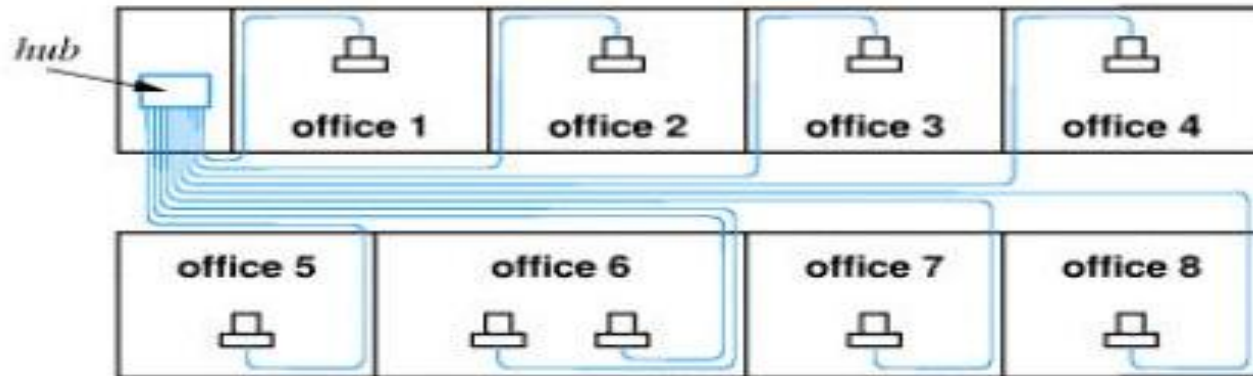
Cabling of 10Mbps Ethernet



(a) 10Base5



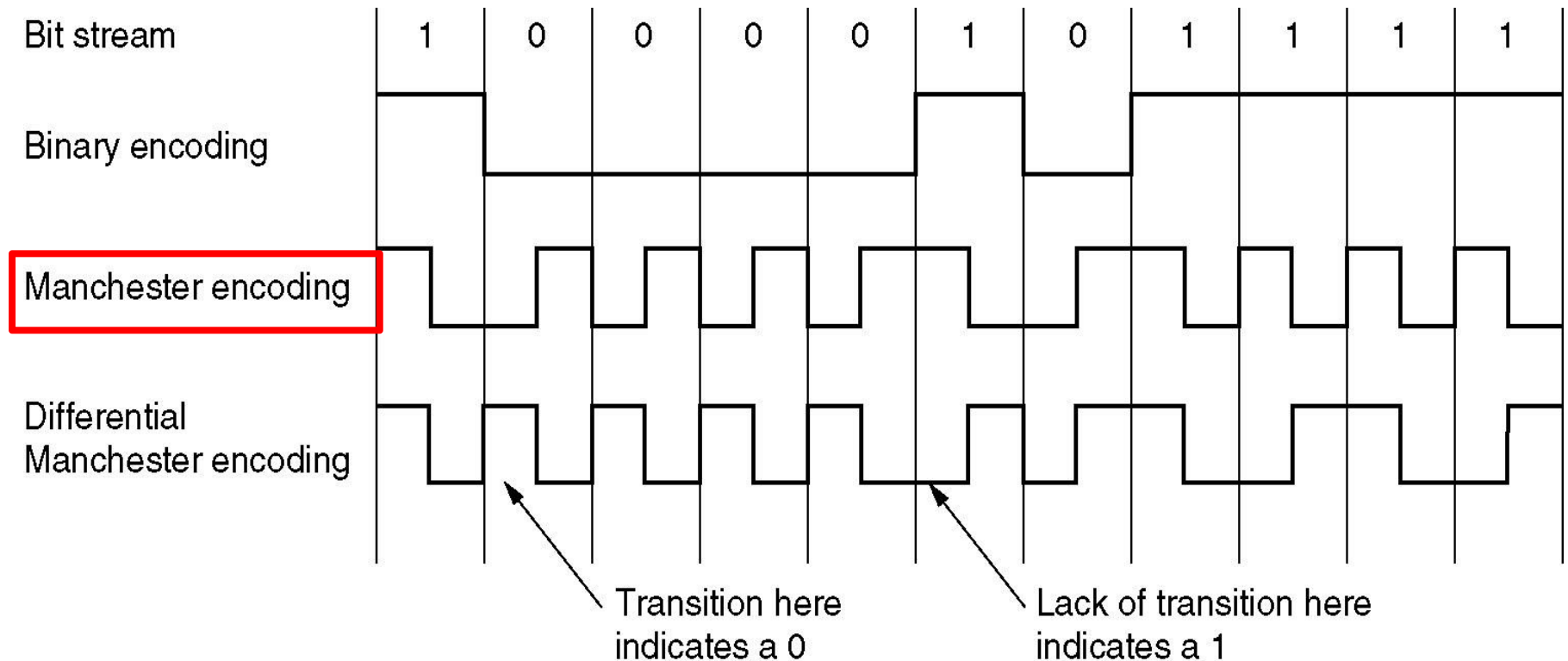
(b) 10Base2



(c) 10BaseT

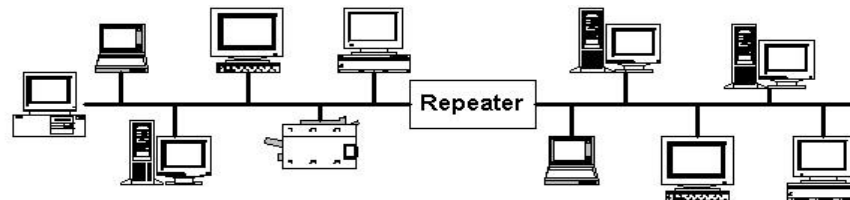
(4) Classic Ethernet: Manchester Encoding

■ Transition in the middle of each bit



(5) Classic Ethernet: Repeater

- Ethernet has a **maximum cable length** per segment (unamplified length)
- **Repeater**(Interconnection device)
 - ◆ Connecting dissimilar media/PHY types (eg. 10base-T and 10base-2)
 - ◆ **Physical layer** device
 - ◆ **Half Duplex ONLY**: **receivers**, **amplifies**(i.e. **regenerates**) and **retransmits** signals in both directions
 - ◆ Cable segments connected by repeaters is no different from a single cable(except for **a small delay introduced by the repeaters**)



Digital World:
Regeneration

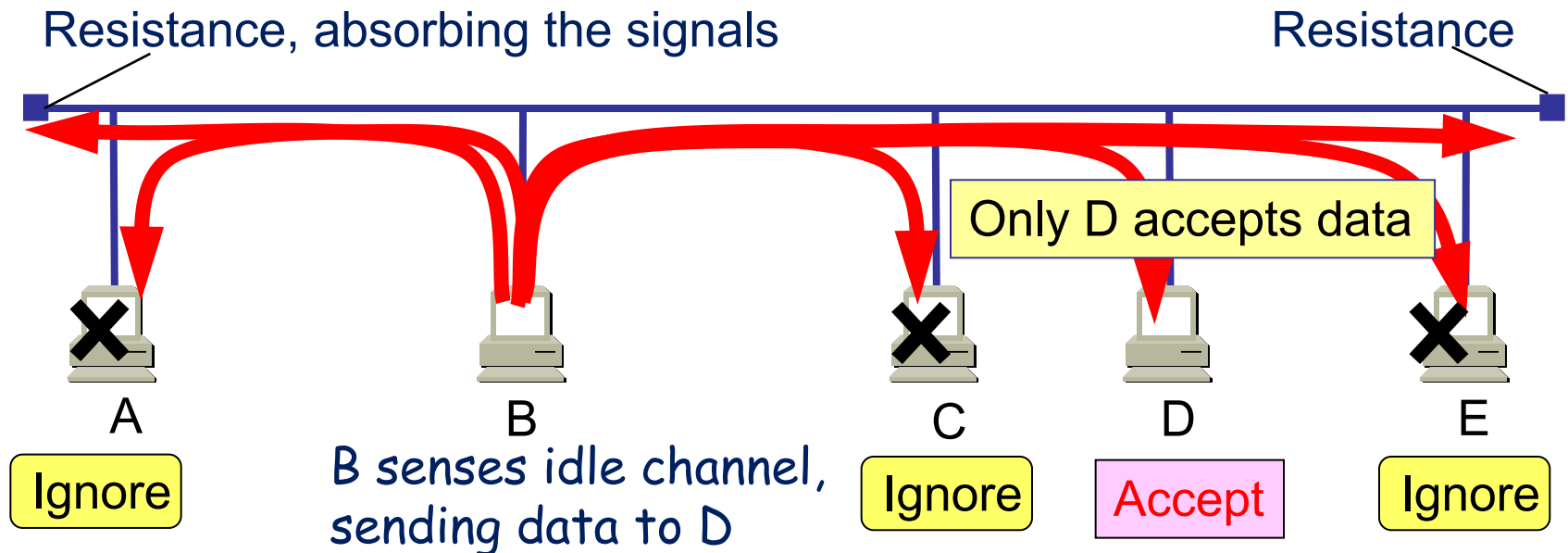


Ethernet

- 4.3.1 Classic Ethernet: Physical Layer
- 4.3.2 Classic Ethernet: MAC Sublayer Protocol
- 4.3.3 Ethernet Performance
- 4.3.4 Switched Ethernet
- 4.3.5 Fast Ethernet
- 4.3.6 Gigabit Ethernet
- 4.3.7 10G Ethernet

(1) Overview of Ethernet / IEEE 802.3

- Broadcasting link: every station can receive data sent by any other station
 - ◆ Addressing is necessary to identify destination
 - ◆ Only one station copies frame, others ignore the frame



(2) Operation steps of Ethernet / IEEE 802.3: Inter Frame Gap (帧间间隔)

- Also called InterFrame Gap
- 9.6μs for 10M bps Ethernet
 - ◆ More precisely, 96 bit time($96/10M = 9.6\mu s$)
 - ◆ Used for a station to change from transmit mode to receive mode (half-duplex)
 - ◆ In practice, IFG may be much shorter
 - Eg. 4.7μs



(2) Operation steps of Ethernet / IEEE 802.3: 1-persistent CSMA/CD (1-坚持的CSMA/CD)

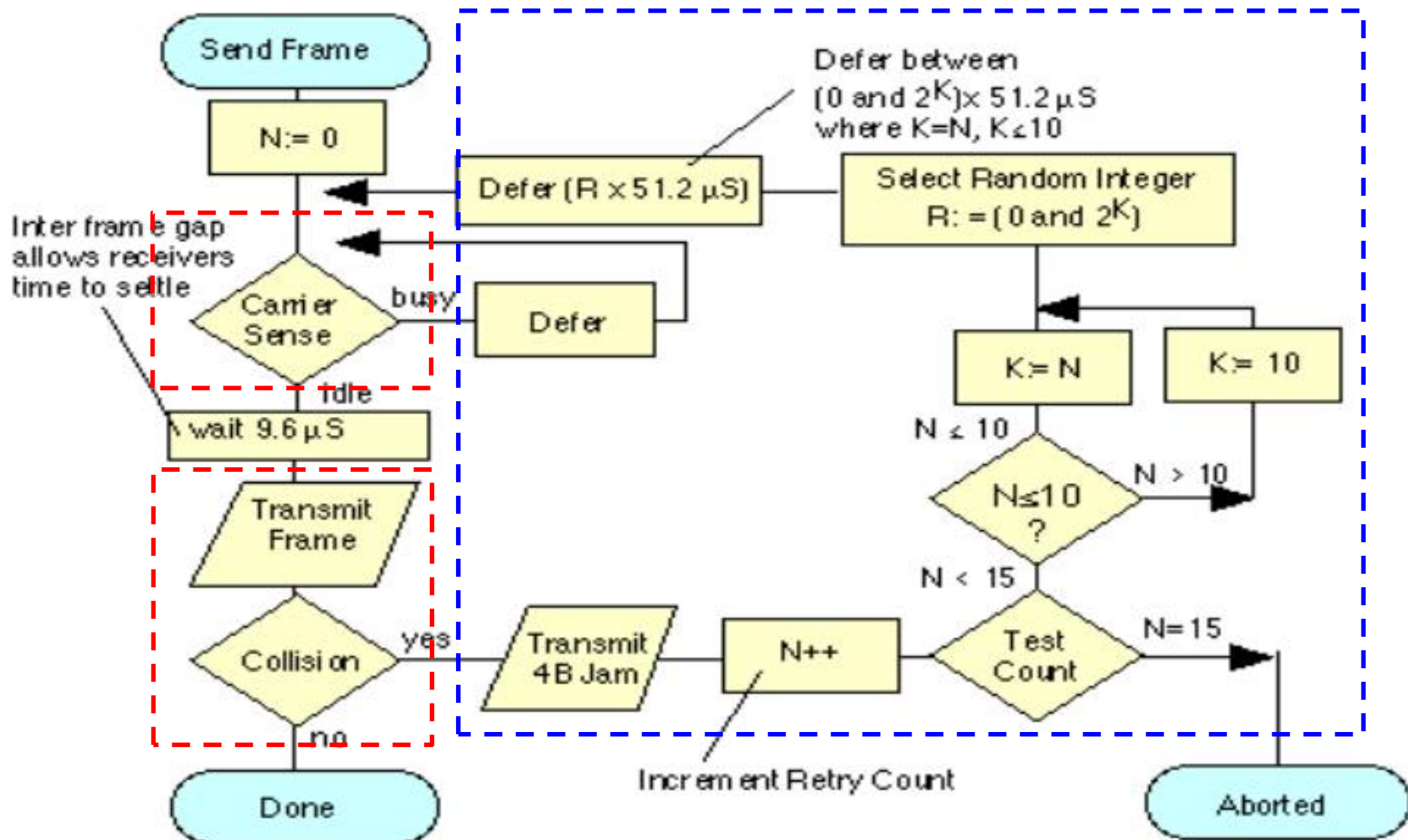
- Step 1. Sensing the channel first, "Listen before talking". 监听信道是否空闲
- Step 2. If channel is quiet for IFG period, then transmit. "Talk if quiet". 如果空闲时间超过IFG, 则发送
- Step 3. If channel is busy, monitor the channel until it is quiet for IFG period before transmitting. "Wait for quiet before talking." 如果信道忙, 持续监听
- Step 4. Monitor the channel continuously during transmission to detect collisions. "Listen while talking". 边发送边检测冲突

(2) Operation steps of Ethernet / IEEE 802.3: 1-persistent CSMA/CD (cont.1)

- Step 5. If collision occurs, the first node recognizes it will send 4 bytes **jam** to ensure that all other stations detect the collision. Transmitters should stop immediately, receivers should reject data. "A buzzer sounds off indicating we have more than one talker at a time." 如果发现冲突，发送jam信号
- Step 6. All nodes wish to transmit should recalculate their backoff period and must wait a random period called "**backoff**(退避)" and attempt again. After 16 collisions, drop the frame to be transmitted. 收到jam信号，所有站点重新计算退避时间并退避一段时间后重新监听信道

(2) Operation steps of Ethernet / IEEE 802.3: 1-persistent CSMA/CD (cont.2)

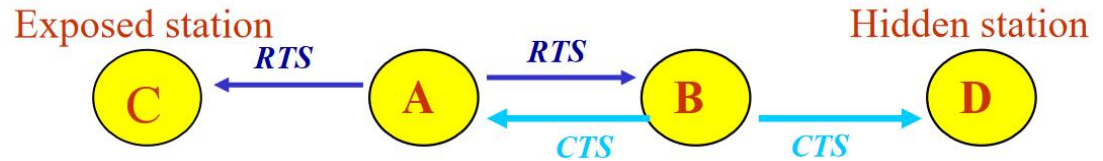
■ Flow chart of Ethernet / IEEE 802.3



Review(1)

◆ WLAN

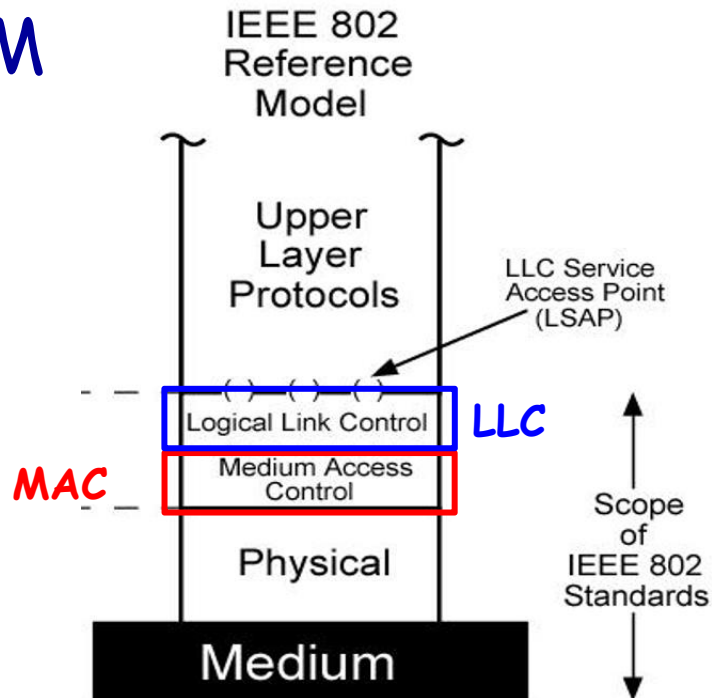
- Acknowledged connectionless(ACK+retransmission)
- Collision Avoidance: handshake



- MACA: RTC-CTS-Data-ACK

◆ IEEE802 RM

- ◆ PHY
- ◆ MAC
- ◆ LLC



Review(2)

Classic Ethernet: 10Mbps

◆ Topology: **Bus evolved to Star**

◆ **Bus**

◆ 10BASE5(VampireTap), 10BASE2(BNC T-junction)

◆ Interconnection device: **Repeater**

PHY

◆ **Star**--10BASE-T(RJ45), **Hub**

◆ Components: **NIC**

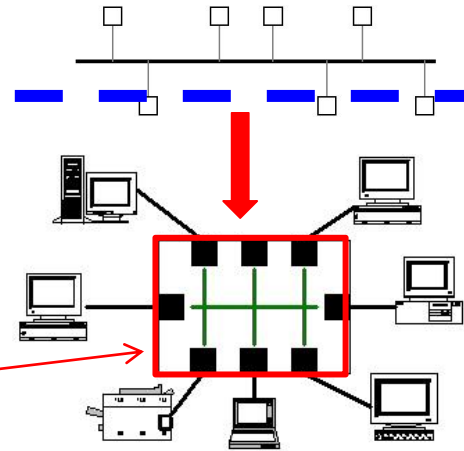
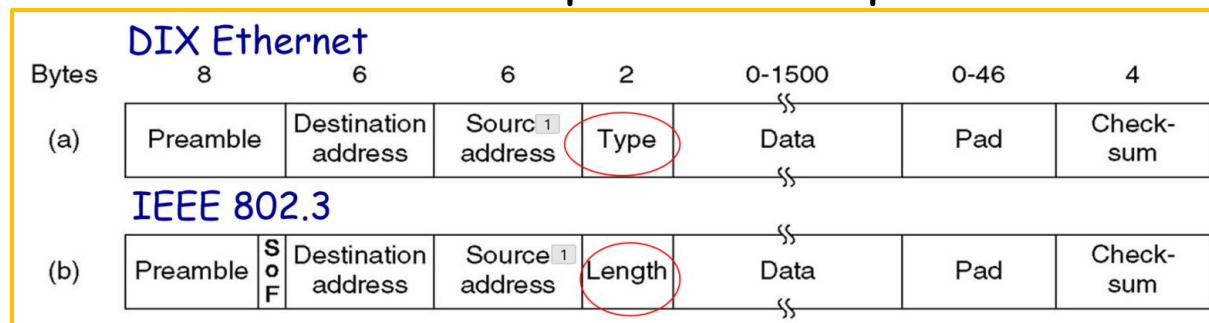
◆ Line coding: **Manchester**

◆ Framing: **Physical layer coding violations**

◆ Media Access Control: **CSMA/CD**

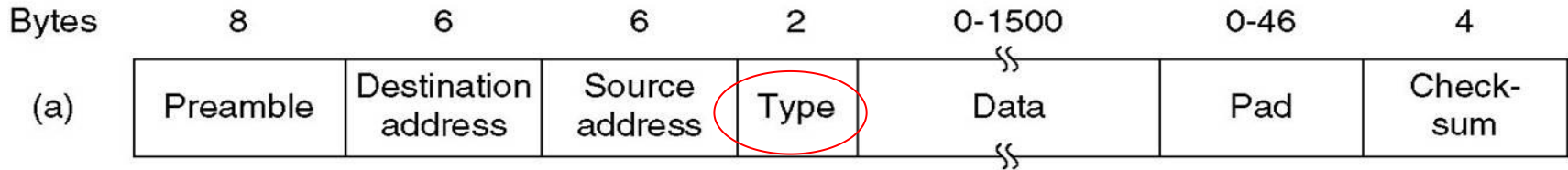
MAC ➤ 1-persistent CSMA/CD, Operation steps of Ethernet

◆ PDU

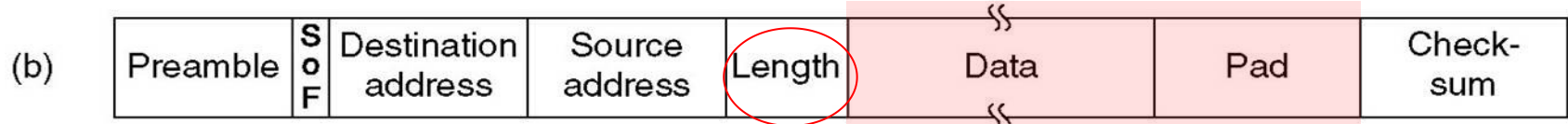


(3) Frame Structure

DIX Ethernet



IEEE 802.3



- Preamble(前导码): 8bytes, 101010...101011, used for clock synchronization, **SO**F(Start Of Frame)-mark byte boundary for MAC
- Address: **MAC address**
- Type: indicates the higher layer protocol, e.g., IP, Novell IPX,... (0x0806:ARP, 0x0800:IP, 0x8137:IPX) (>0X600)
- Length: number of bytes of **Data+Pad** (<0X600, 1536B)
- Data: maximum 1500 bytes, **minimum 46 bytes** (Why?)
- Pad: Only used when necessary to **extend frame to 64 bytes**
- Checksum: CRC, checked at receiver, if error is detected, the frame is simply dropped

(4) MAC Address (Physical Address)

- Must be **unique on every LAN**
- MAC address: 6 bytes, denoted in hex number
 - ◆ eg. 00-60-97-98-1E-6B
- First 24-bits identify the vendor(**OUI: Organizationally Unique Identifier**)
- second 24-bits is a serial number assigned by the vendor
 - ◆ eg. 00-60-97 indentifies 3COM CORPORATION (<http://standards.ieee.org/>)



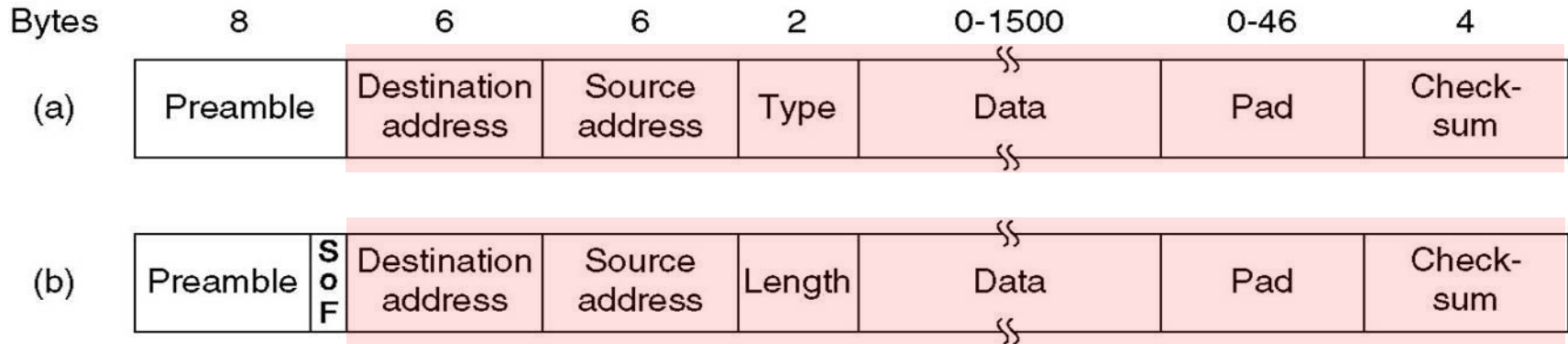
006097981E6B

EA=006097981E6B
3COM CORPORATION
MADE IN U.S.A.



- Source Address: Unicast address
- Destination Address
 - ◆ Unicast Address
 - ◆ **Broadcast Address**(=[FF:FF:FF:FF:FF:FF](#)) "to all nodes"

(5) Frame length



Maximum frame length

- ◆ 1518 bytes, Data length 1500 bytes
- ◆ Based on the fact that a transceiver needs enough RAM to hold an entire frame and RAM was expensive in 1978

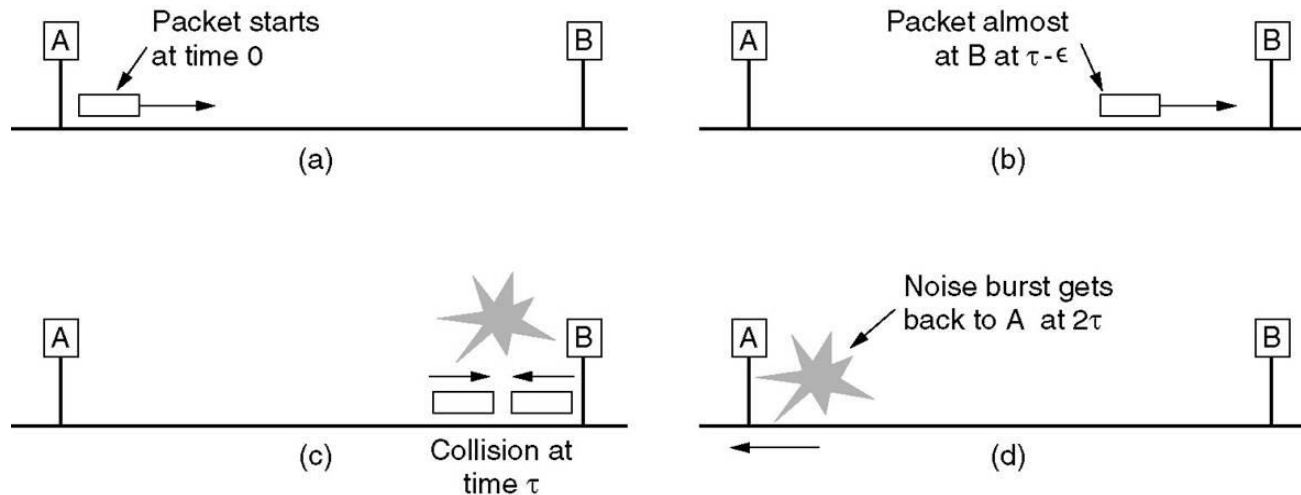
Minimum frame length (要求会计算)✓

- ◆ 64 bytes, Valid frames must be at least 64B
- ◆ If data portion is less than 46 bytes, Pad field is used to fill out the frame to the minimum size

(5) Frame length: Minimum Frame Size(64 Bytes)

■ Reason: to give a host enough time to detect collisions

◆ If host has less than 46 bytes to send, the adaptor pads (adds) bytes to make it 46 bytes



■ The transmission time shall be longer than 2τ .

◆ For a 10Mbps LAN with max length of 2,500 meters and 4 repeaters, the round-trip time(= 2τ) is about 51.2 μ s, i.e. 512 bit time, 64Byte time.

Summary: Ethernet/IEEE 802.3 Parameters

Slot time(2τ) 冲突	51.2 μs
Attempt Limit	16 (tries)
Backoff Limit	10 (exponent)
Jam Size	32 bits
Max Frame Size	1518 bytes
Min Frame Size	64 bytes
Address Size	48 bits (6 bytes)

Ethernet

- 4.3.1 Classic Ethernet: Physical Layer
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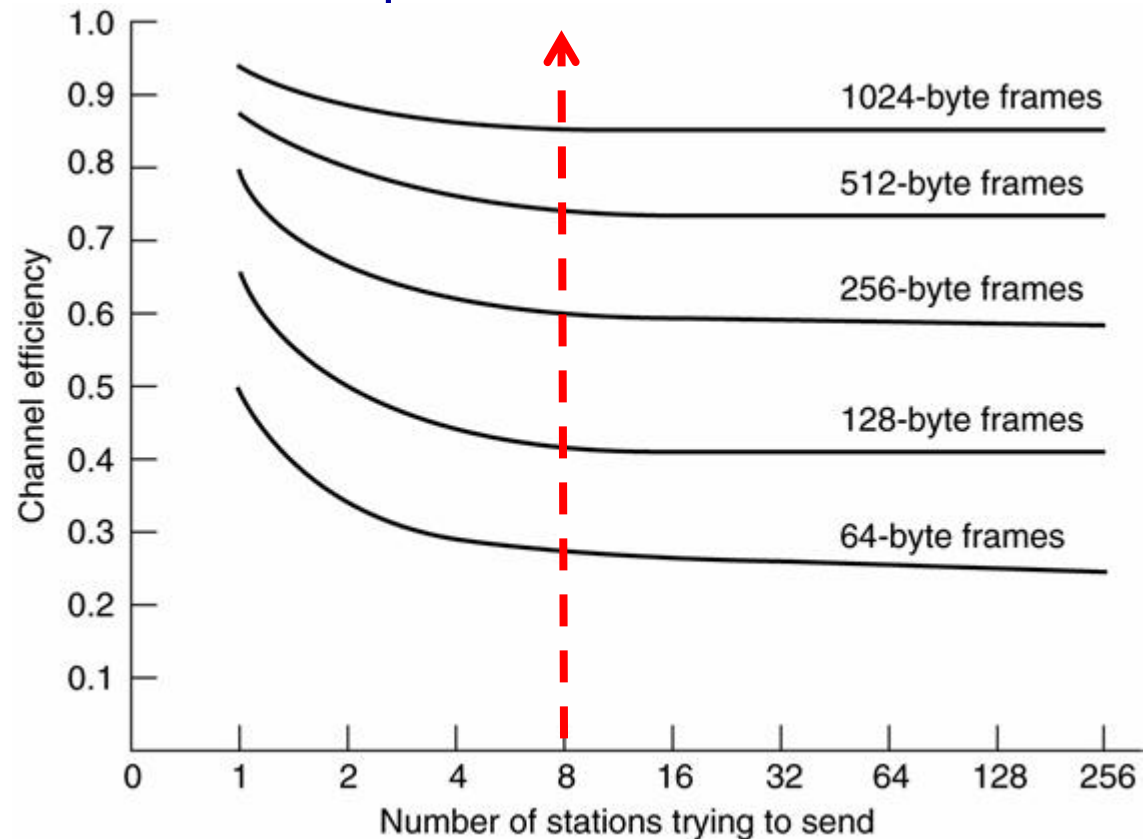
4.3.3 Performance of IEEE802.3

Efficiency of Ethernet at 10 Mbps with 512-bit slot times

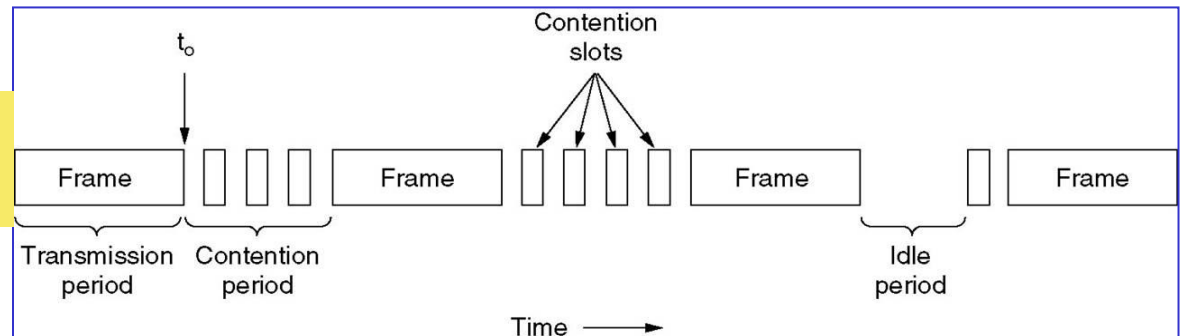
Channel efficiency =

$$\frac{1}{1+2BL_e/cF}$$

- frame length F
- network bandwidth B
- cable length L

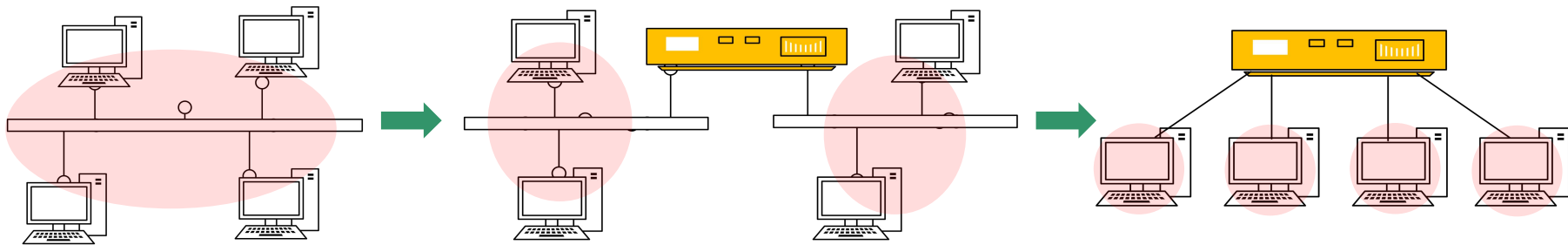


Channel efficiency = $t/(t+w)$
 w 为平均竞争间隔



How to increase speed of data transmission in Ethernet?

- (1) Increasing the data rate
 - ◆ Fast Ethernet and Gigabit Ethernet
- (2) Decreasing collision
 - ◆ Decreasing number of computers in a LAN
 - Using bridges(网桥) to connect LANs
 - ◆ Using switched Ethernet(交换式) instead of shared Ethernet(共享式, classic Ethernet)

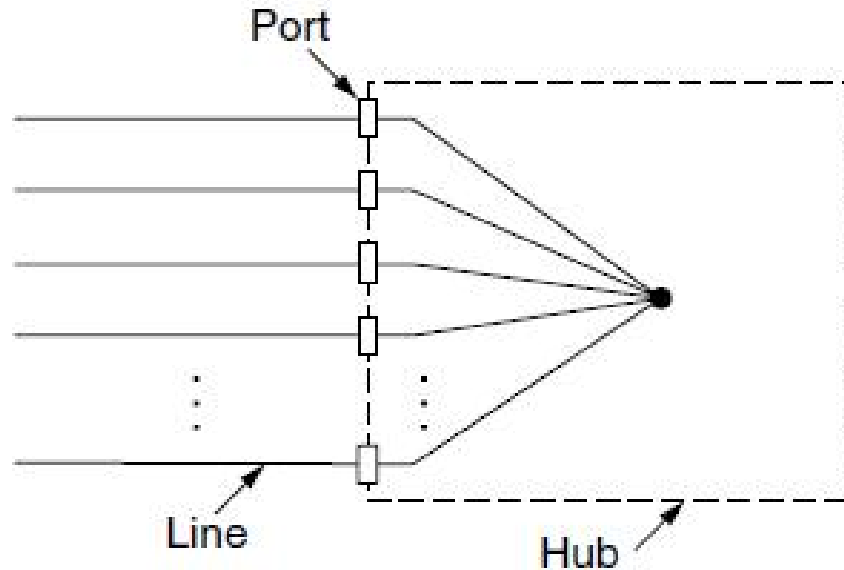


Computer Networks, MAC Sublayer

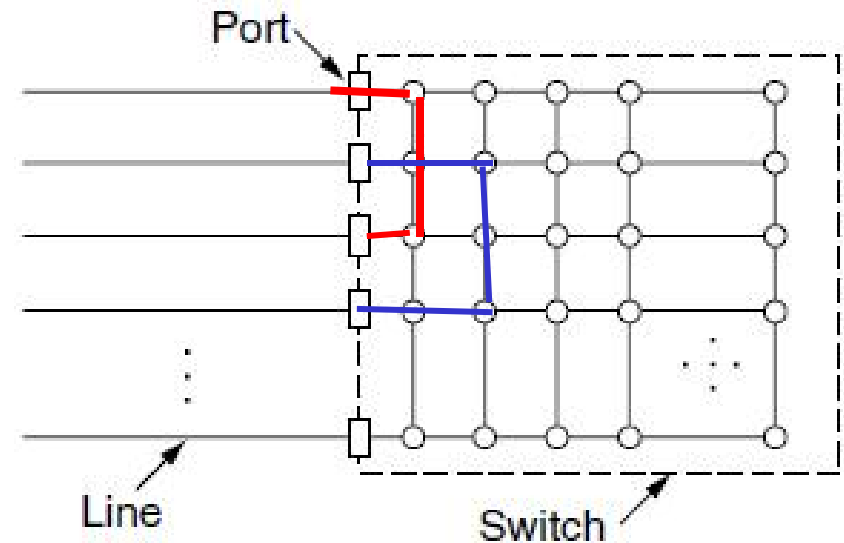
Ethernet

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- 4.3.7 10G Ethernet

(1) HUB(集线器) vs. Switch(交换机)



Hubs are logically equivalent
to the single long cable
(逻辑上是总线拓扑)



A switch contains a **high-speed backplane** that
connects all of the ports

(2) Switched Ethernet (交换式以太网)

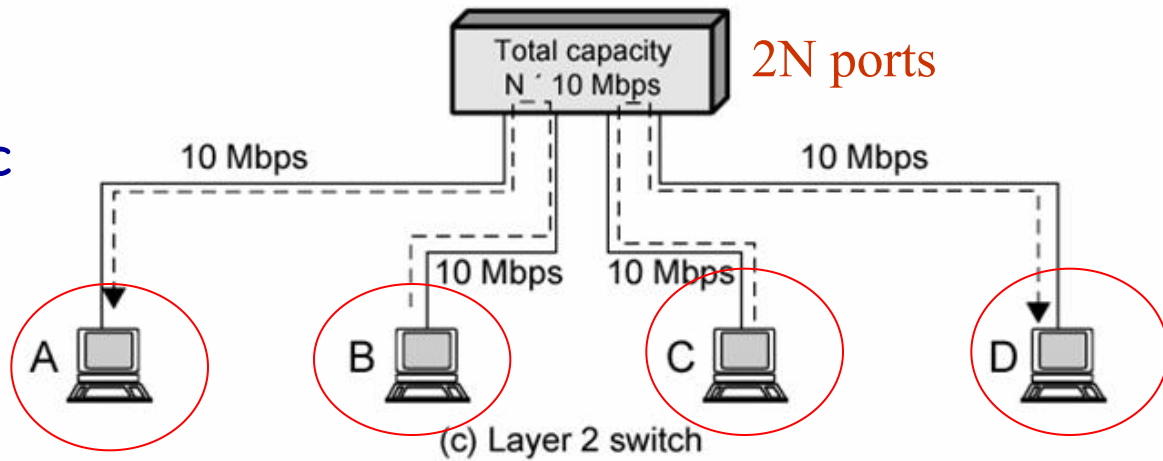
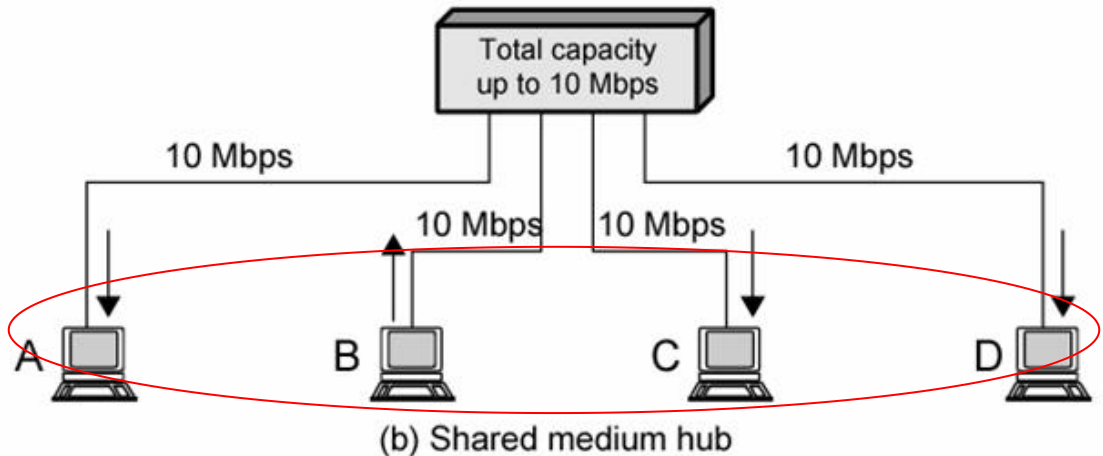
- Central **switch** instead of hub

- Incoming frame from particular station switched to appropriate output line

- Unused lines can switch other traffic, more than one station transmitting at a time

- Multiplying capacity of LAN

- More secure** than classic Ethernet



Collision Domain

[Stallings]

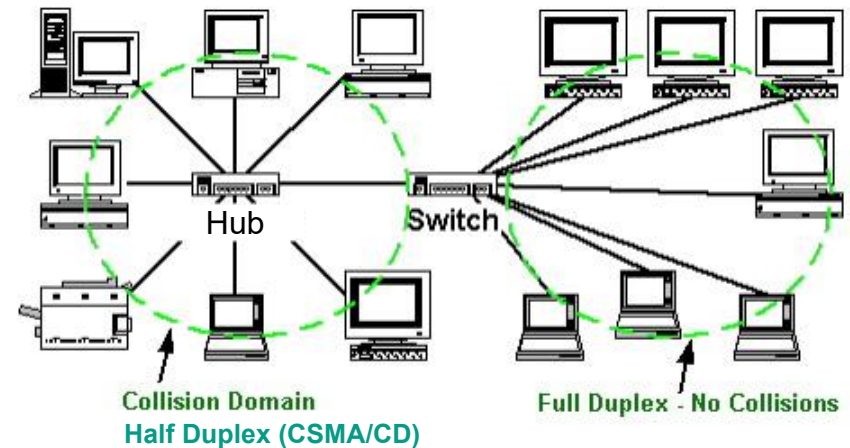
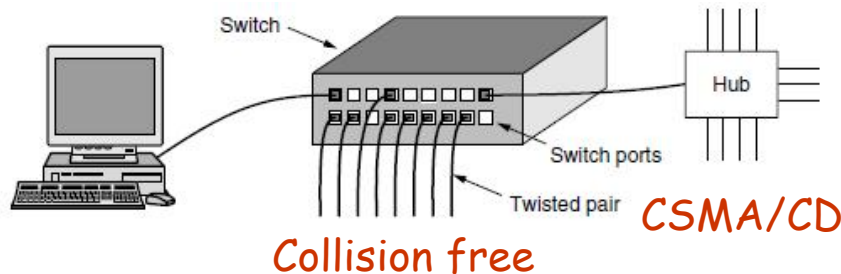
(3) Switches(交换机)

■ Data Link Layer device

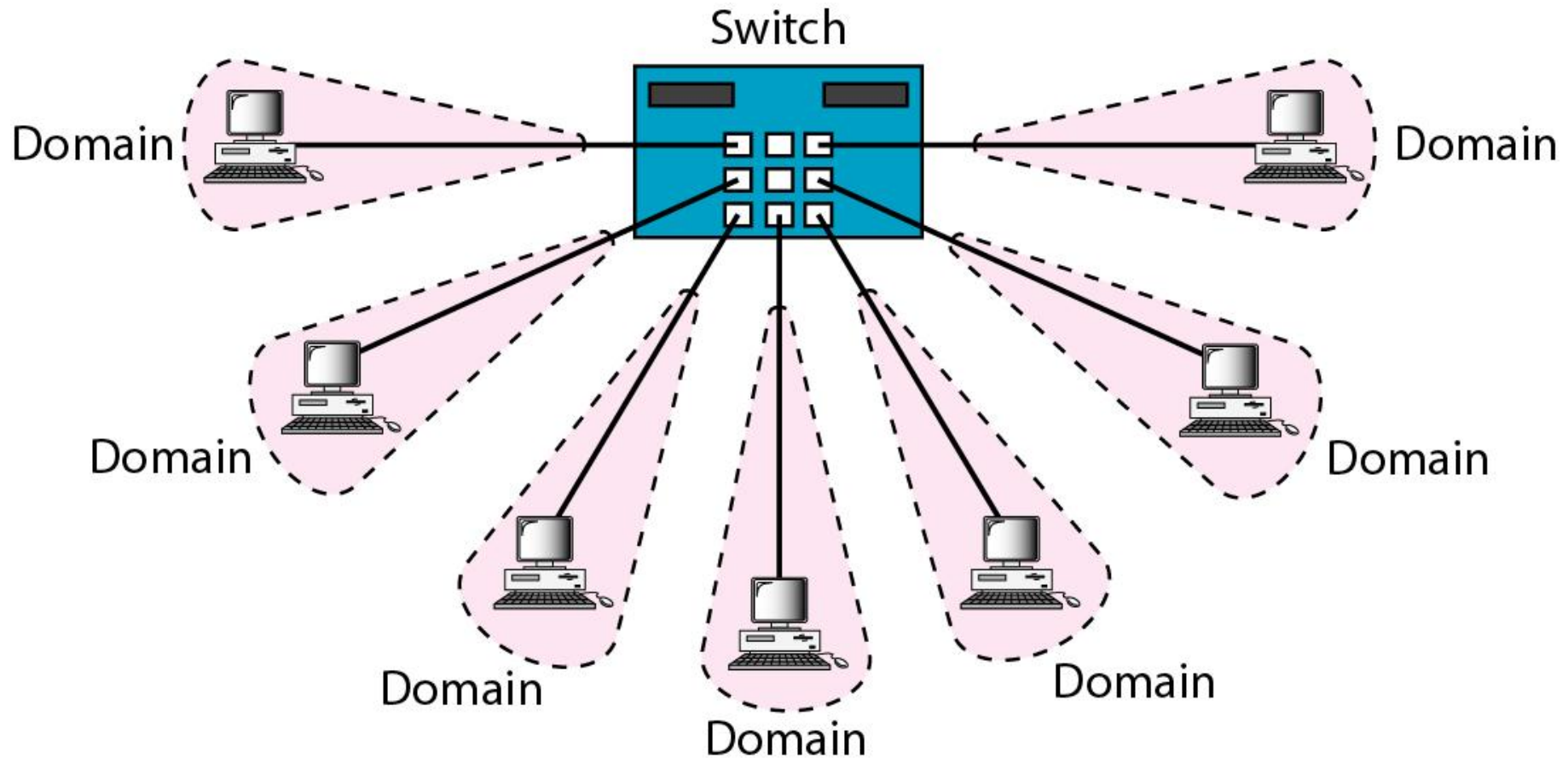
- ◆ Switching frames on destination MAC address
- ◆ Providing dedicated channels between each pair of users

■ Collision domain

- ◆ With only one port per collision domain, collisions are impossible and performance is improved



Collision Domain of Switched Ethernet



Switches(补)

■ Cut-through

- ◆ 内部采用**基于硬件的交叉矩阵**，**边接收**数据帧边根据帧头的目的地址**直接转发**
- ◆ 缺点：无法检查数据帧是否出错、输入/输出需要相同速率

■ Stored-Forward

- ◆ 缓存整个数据帧，检查数据包是否正确，过滤掉**碰撞的残帧**和**CRC错误帧**，再根据目的**MAC**地址转发
- ◆ 优点：支持差错检测、支持不同输入/输出端口间交换

	转发前读取的字节(B)	延迟	丢弃错误数据帧
Cut-through	14(8+6)	短	无
Stored-Forward	全部	长	残帧、错误帧

Ethernet

- 4.3.1 Classic Ethernet: Physical Layer
- 4.3.2 Classic Ethernet: MAC Sublayer Protocol
- 4.3.3 Ethernet Performance
- 4.3.4 Switched Ethernet
- 4.3.5 Fast Ethernet
- 4.3.6 Gigabit Ethernet
- 4.3.7 10G Ethernet

(1) Overview of fast Ethernet: 100M

- IEEE802.3u
- Keep all the old frame formats, interfaces, and procedural rules of IEEE802.3
- Reduce the bit time from 100 ns to 10 ns
- Minimum frame length remains 64B
- All fast Ethernet systems use hubs and switches; cables with vampire taps or BNC connectors are not permitted

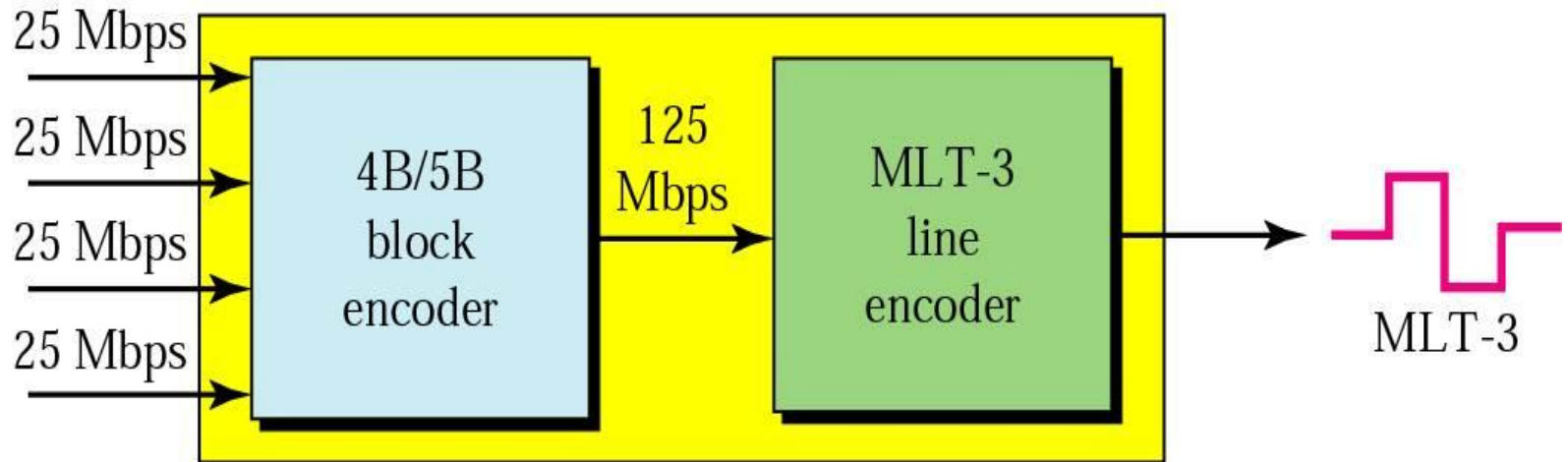
Name	Cable	Max. segment	Advantages
100Base-T4	Twisted pair	100 m	Uses category 3 UTP
100Base-TX	Twisted pair	100 m	Full duplex at 100 Mbps
100Base-FX	Fiber optics	2000 m	Full duplex at 100 Mbps; long runs

(2) Fast Ethernet: 100Base-T4

- Uses a signaling speed of 25 MHz
- Requires 4 twisted pairs (UTP3)
 - ◆ One is always to the Hub
 - ◆ One is always from the Hub
 - ◆ The other two are switchable to the current transmission direction-- Half duplex
- Coding
 - ◆ Manchester encoding is not used
 - ◆ Ternary signals(三元信号) are sent during a clock period
 - ◆ With 25MHz signaling speed, at least 33.3-Mbps per channel(8B/6T)

(3) Fast Ethernet: 100Base-TX

- Uses a signaling speed of 125 MHz
- Requires 2 twisted pairs (UTP5)
 - ◆ One is to the hub
 - ◆ One is from the hub (Full duplex)
- Coding (自阅)
 - ◆ 4B/5B: $125 * 4/5 = 100\text{Mbps}$



MLT: Multi Level Transmit

Fast Ethernet: Summary

Specification		Frequency (MHz)	Encoding	number of wires	cable for sending	Full duplex	Max seg. length
10BASE-T	UTP3 UTP5	20	Manchester	2 pairs	1 pair	Yes	100m
100BASE-T4	UTP3	25	8B6T-NRZ	4 pairs	3 pairs	No	100m
100BASE-TX	UTP5	125	4B5B-MLT3	2 pairs	1 pair	Yes	100m
100BASE-F	Fiber	125	4B5B-NRZ	2 pairs	1 pair	Yes	2000m

(4) Fast Ethernet: Interconnection Devices

■ Hub (Half-duplex communication)

- ◆ Incoming lines are logically connected, forming a **single collision domain**
- ◆ The system works just like **classic Ethernet (CSMA/CD)**
- ◆ Hubs are **not permitted with 100Base-FX**

■ Switch (Full-duplex communication)

- ◆ Incoming frame is buffered and passed to the destination
- ◆ 100Base-FX cables are too long for collision algorithm, must be connected to switches

(4) Fast Ethernet: Interconnection Devices(Cont.)

■ Autonegotiation(自动协商机制)

- ◆ 10-Mbps & 100-Mbps Autoconfiguration
- ◆ 803.3u provides a way for two stations to negotiate
 - the speed (10 or 100 Mbps)
 - the duplexity (half or full)
- ◆ Most fast Ethernet products use this feature to **autoconfigure** themselves
- ◆ Autonegotiation is **mandatory for 1000BASE-T and faster**

*[Wikipedia]: Autonegotiation is the procedure by which two connected devices **choose common transmission parameters**, e.g. **speed** and **duplex mode**. Autonegotiation was initially an optional feature, first introduced with 100BASE-TX (1995 IEEE 802.3u Fast Ethernet standard), and is backward compatible with 10BASE-T*

Ethernet

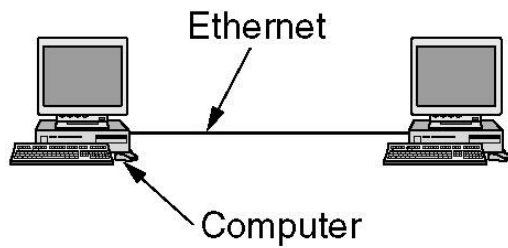
- 4.3.1 Classic Ethernet: Physical Layer
- 4.3.2 Classic Ethernet: MAC Sublayer Protocol
- 4.3.3 Ethernet Performance
- 4.3.4 Switched Ethernet
- 4.3.5 Fast Ethernet
- 4.3.6 Gigabit Ethernet
- 4.3.7 10G Ethernet

(1) Overview

- IEEE 802.3z
- Goals: Make Ethernet go 10 times faster
- Remain backward compatible with all existing Ethernet standards
 - ◆ Unacknowledged datagram service with both unicast and multicast
 - ◆ Use 48-bit addressing scheme already in use
 - ◆ Maintain the same frame format, including the minimum and maximum frame sizes

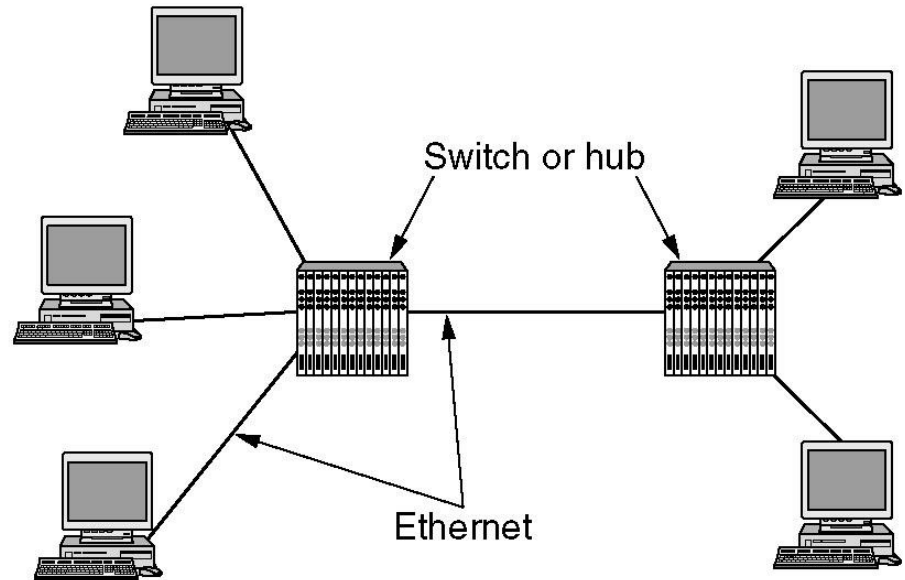
(1) Overview(Cont.)

- Allows for point-to-point links and shared broadcast channels



(a)

A two-station Ethernet



(b)

A multistation Ethernet

Full-duplex: computer connecting to switch
Half-duplex: computer connecting to hub

(2) Switch

■ Switch (Full-duplex Mode)

- ◆ Since no contention is possible, the **CSMA/CD protocol is not used**
- ◆ Maximum length of the cable is determined by signal strength issues （最大电缆长度取决于信号衰减程度）
- ◆ Switches are free to mix and match speeds.
- ◆ **Autoconfiguration** is supported

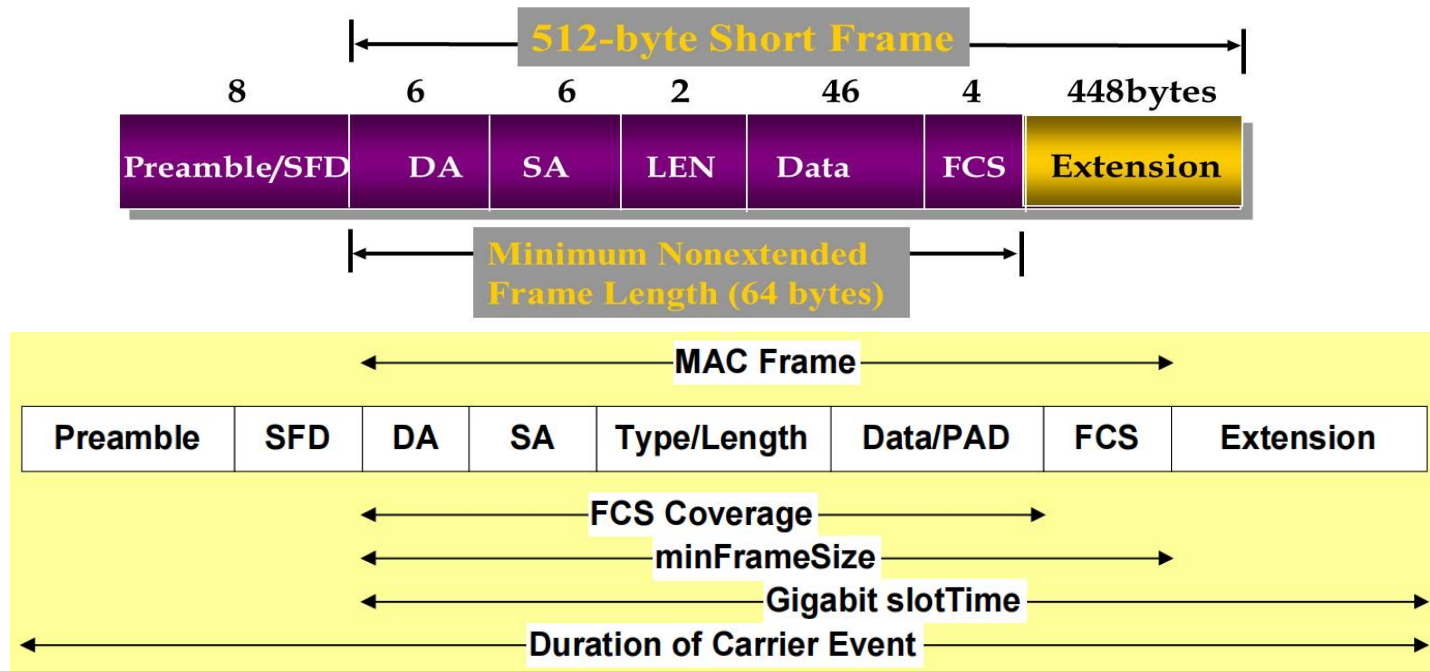
(3) Hub

- Hub (Half-duplex Mode)
- Simulating classic Ethernet
 - ◆ Collisions are possible, so **CSMA/CD** protocol is required
- Problem: A radius of 25 meters
 - ◆ Without modification to the Ethernet MAC, the allowed diameter at 1Gbps (10x faster Ethernet) would be ~20m (10x smaller)
 - ◆ To fix this, the **minimum transmission size must be increased** such that an acceptable distance (~200m) is covered

(3) Hub: Carrier extension

Carrier extension(载波扩展)

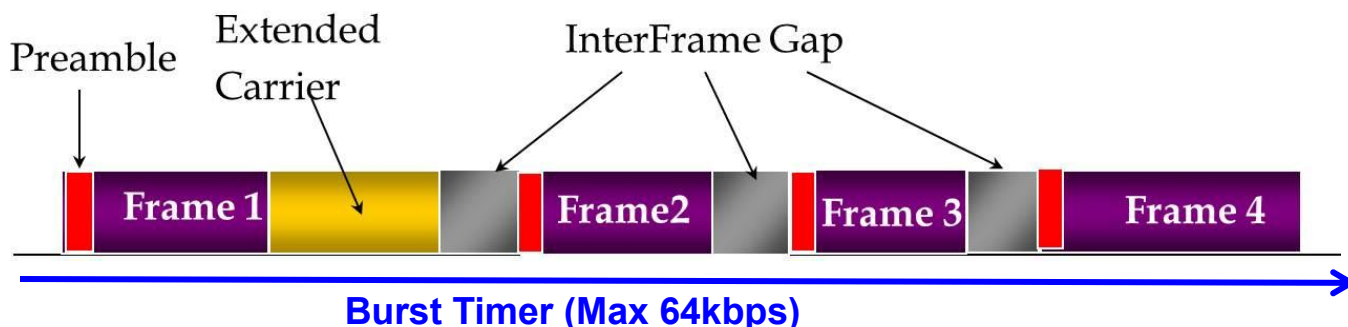
- ◆ Add padding **after** the normal frame to extend the frame to **512 bytes (=4096bits)** (Content period)
- ◆ Padding is added by the sending hardware and removed by the receiving hardware



Remain the radius of the network to 200 meters

(3) Hub: Frame bursting

- Extension fields create waste in small packets
 - This waste cannot be eliminated but can be reduced by frame bursting
- Frame bursting(帧突发)
 - Allows a sender to transmit a concatenated sequence of multiple frames in a single transmission
 - The first frame use carrier extension method, subsequent short frames need not be extended, and only be sent continuously
 - Maximum size transmitted burst ~64Kbits(8192Bytes)



以太网类型	双工模式	最短帧长	关键机制
千兆以太网	半双工	64 字节	载波延伸、帧突发
千兆以太网	全双工	无	无需冲突检测

Ethernet JUMBO Frames(補)

■ BAD IDEA

■ Why didn't Ethernet change the frame size?

- ◆ Half-duplex 1 Gig Ethernet use extension bits to pad a frame out to 512Bytes to enable a larger collision domain
- ◆ change the min to 512Bytes, and max to 4K or 9K?

■ Because

- ◆ Such equipment would **NOT work** with any **previously deployed equipment**
- ◆ NOT a smart move for the **WORLDS** most popular networking technology
- ◆ IEEE 802.3 will **NEVER** adopt a larger frame size, as they are **COMMITTED** to supporting the installed **base**.

(4) Gigabit Ethernet cabling

Name	Cable	Max. segment	Advantages
1000Base-SX	Fiber optics	550 m	Multimode fiber (50, 62.5 microns)
1000Base-LX	Fiber optics	5000 m	Single (10 μ) or multimode (50, 62.5 μ)
1000Base-CX	2 Pairs of STP	25 m	Shielded twisted pair
1000Base-T	4 Pairs of UTP	100 m	Standard category 5 UTP

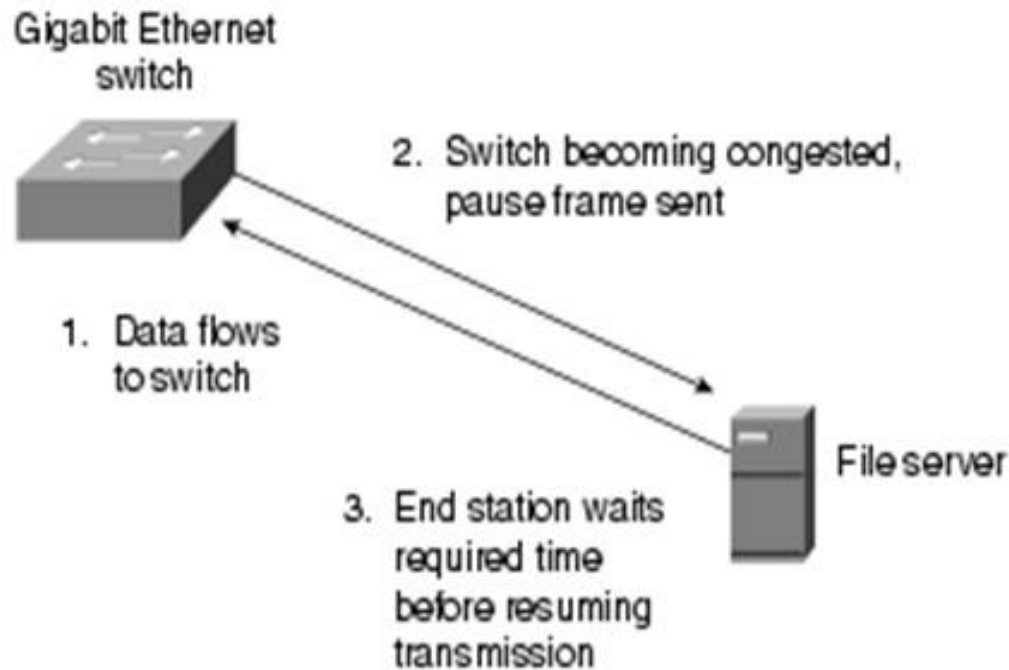


(5) Flow Control

- Introduced for Fast and Gigabit Ethernet
- Problem: 1 Gbps is quite fast
 - ◆ If a receiver is busy with some other task for 1 ms, 1953 frames may have accumulated, and buffer overruns are very likely
- Flow control
 - ◆ Supported on Full-duplex links only
 - ◆ Sends MAC Control Frames called Pause Frames which are used to enable or disable frame transmission
- PAUSE frames
 - ◆ One end sending a special control frame to the other end telling it to pause for some period of time
 - ◆ Normal Ethernet frames, containing a type of 0x8808
 - ◆ Pauses are given in units of the minimum frame time
 - ◆ For Gigabit Ethernet: 512ns~33.6ms

(5) Flow Control(cont.)

■ Example of Flow Control in Gigabit



Ethernet

- 4.3.1 Classic Ethernet: Physical Layer
- 4.3.2 Classic Ethernet: MAC Sublayer Protocol
- 4.3.3 Ethernet Performance
- 4.3.4 Switched Ethernet
- 4.3.5 Fast Ethernet
- 4.3.6 Gigabit Ethernet
- 4.3.7 10G Ethernet

(1) Application Scenarios

- High-speed, local backbone interconnection between large-capacity switches
- Inside data centers and exchanges to connect high-end routers, switches, and servers
- In long-distance, high bandwidth trunks between offices
- The long distance connections use optical fiber, while the short connections may use copper or fiber

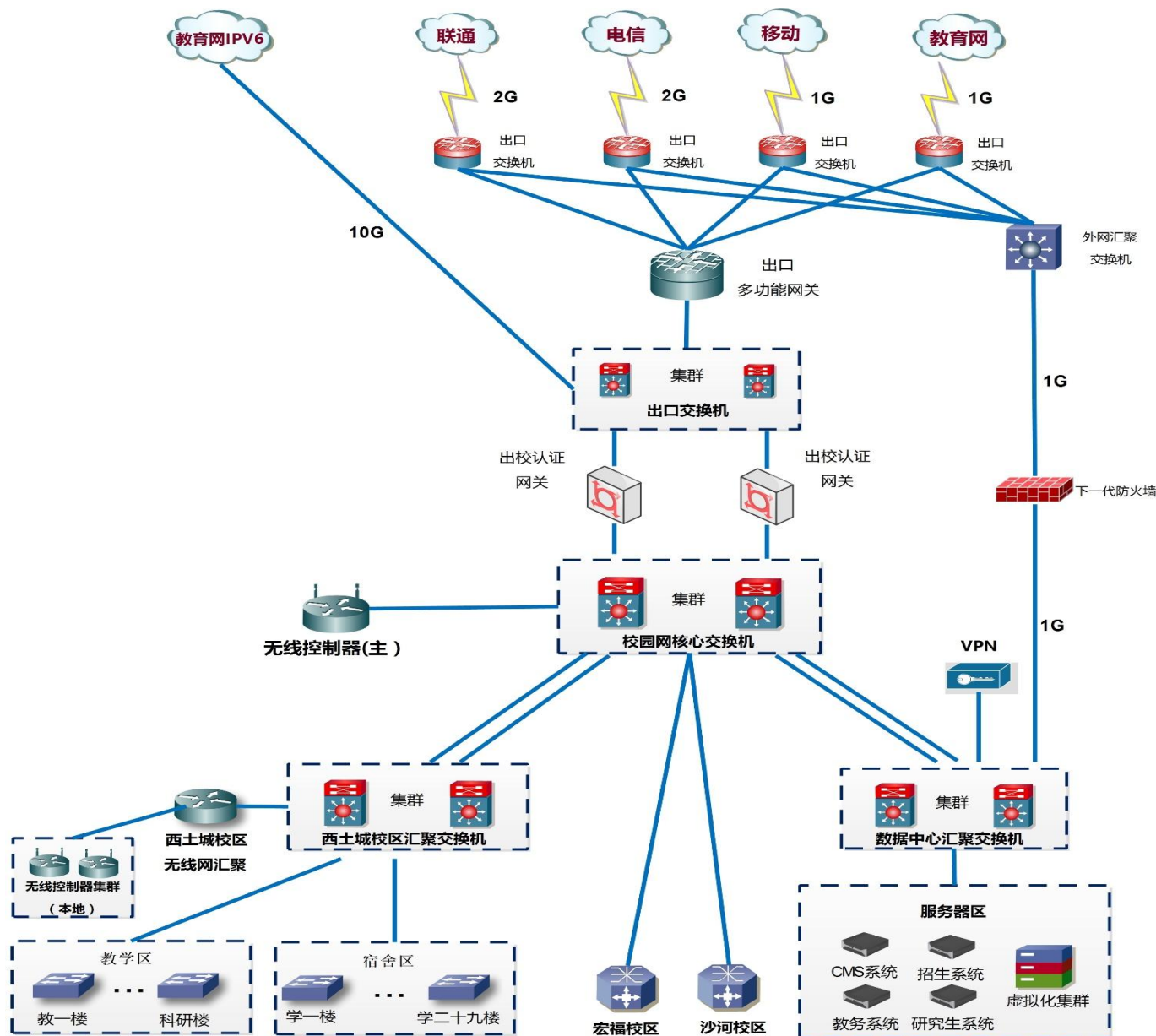
(2) Design Objectives

- Preserve the 802.3 Ethernet frame format at MAC interface
- Support **only full duplex operation**. (Not using CDMA/CD)
- Concentrate on the **details of physical layers** that can **run at high speed**
- Support **autonegotiate** and fall back to the highest speed supported by both ends on the line.

(3) 10-Gigabit Ethernet cabling (自阅)

Name	Cable	Max. segment	Advantages
10GBase-SR	Fiber optics	Up to 300 m	Multimode fiber (0.85 μ)
10GBase-LR	Fiber optics	10 km	Single-mode fiber (1.3 μ)
10GBase-ER	Fiber optics	40 km	Single-mode fiber (1.5 μ)
10GBase-CX4	4 Pairs of twinax	15 m	Twinaxial copper
10GBase-T	4 Pairs of UTP	100 m	Category 6a UTP

(4) Deployment of 10G Ethernet



Summary of Ethernet

■ Ethernet

◆ Media access control: 1-persistent CSMA/CD

◆ MAC address: 48bits

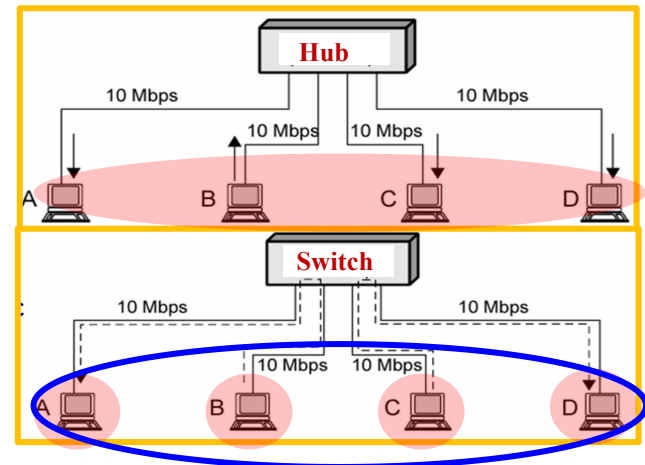
◆ Frame size: 64-1518B

◆ Devices: Hub and Switch

➤ layers and functions

➤ collision domain 

➤ broadcast domain 



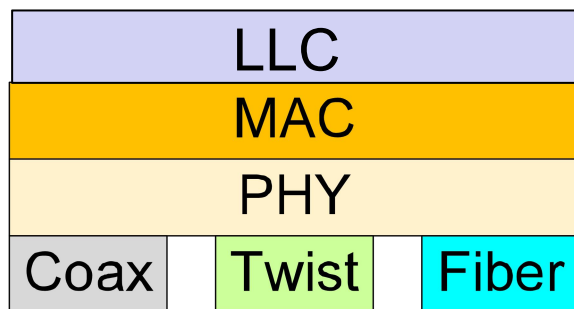
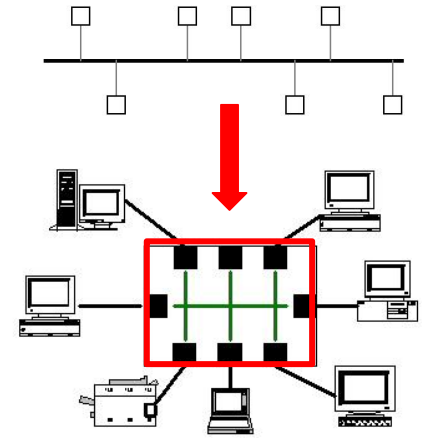
■ Classic Ethernet: 10Base5, 10Base2, 10BaseT

■ Fast Ethernet and Gigabit Ethernet: similarity and difference with 10Mbps Ethernet

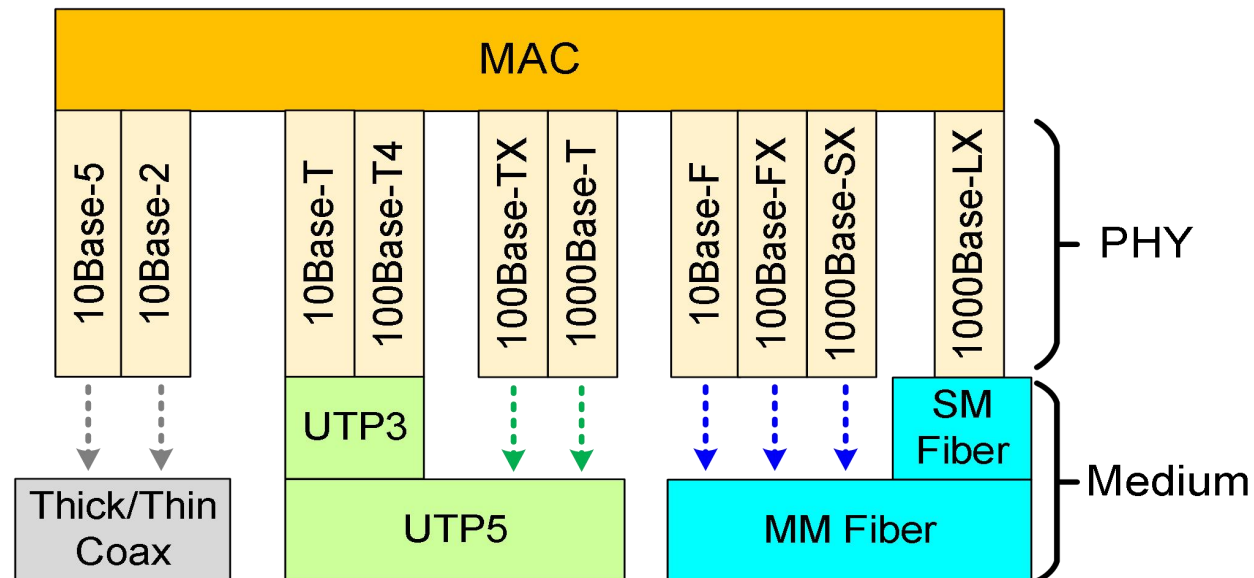
■ 10G Ethernet: usage, objectives

Summary - Ethernet

- Topology: **Bus** evolved to **Star**
- Medium: **Coax**, **Twist**, **Fiber**
- PHY Layer
 - ◆ Components: **NIC**
 - ◆ Coding: **Manchester**, **8B/6T-NRZ**, **4B/5B-MLT3**...
- Protocol Stack



IEEE802模型



	Classic Eth.	Switching	Fast Eth.	Gigabit Eth.	10G Eth.
速率	10Mbps	$\geq 100M$	100Mbps	1Gbps	10Gbps
标准	IEEE802.3				
Topology	总线型、星型	星型	星型,点到点	星型,点到点	星型, 点到点
模式/ 设备	half duplex/ Hub	full duplex/ Switch	半双工:Hub 全双工:Switch	半双工:Hub 全双工:Switch	full duplex/ Switch
Coding	Manchester		4B/5B MLT-3 8B/6T NRZ	8B/10B...	
介质访问控制技术	CSMA/CD	No collision	CSMA/CD(Hub) 或 No collision	CSMA/CD(Hub) 或 No collision	No collision
Frame size	Min-64B, Max-1518B			64B,1518B 载波扩展512B	64B,1518B
其他	1、最大网络长度2500m 2、帧间隔IFG:9.6微秒 3、MAC地址	1、无冲突 2、更安全:过滤、隔离流量	1、最大网络长度200m 2、Auto-negotiation	1、自动协商 2、Carrier Extension,frame bursting, 200m 3、Pause frame	Application Scenarios

The future of Ethernet

- Ethernet's bright (and fast) future
 - ◆ The IEEE Ethernet Working Group approved specifications for 200 Gb/sec and 400 Gb/sec Ethernet
 - ◆ Research is being done for a 1.6 terabyte per second standard

- Supersize Ethernet for AI infrastructures
 - ◆ Ultra Ethernet Consortium, 2023.07

Reasons for Longevity of Ethernet

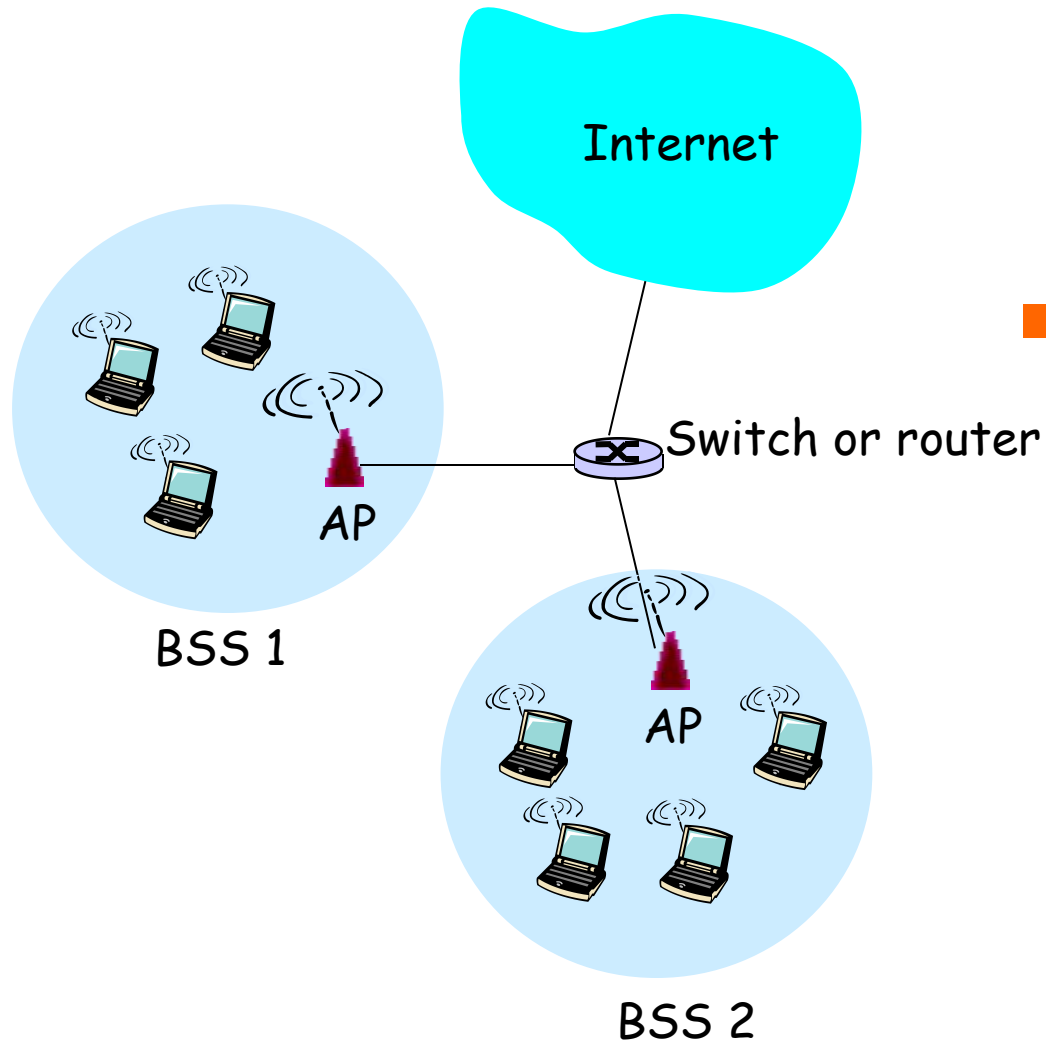
KISS(Keep It Simple, Stupid)

- **Reliable:** star topology with hub or switch
- **Cheap**
- **Easy to maintain**
 - ◆ No software to install
 - ◆ No configuration tables to manage
 - ◆ Adding new hosts is as simple as just plugging them in
- Interworking easily with TCP/IP
- Having most markets of LAN or MAN
- 推荐论文(选读)
 - Ethernet-A survey on its fields of application

Outline

- 4.1 Channel Allocation Problem
- 4.2 Multiple Access Protocols
- (補)IEEE 802 Reference Model
- 4.3 Ethernet
- 4.4 Wireless LANs
- 4.8 Data Link Layer Switching

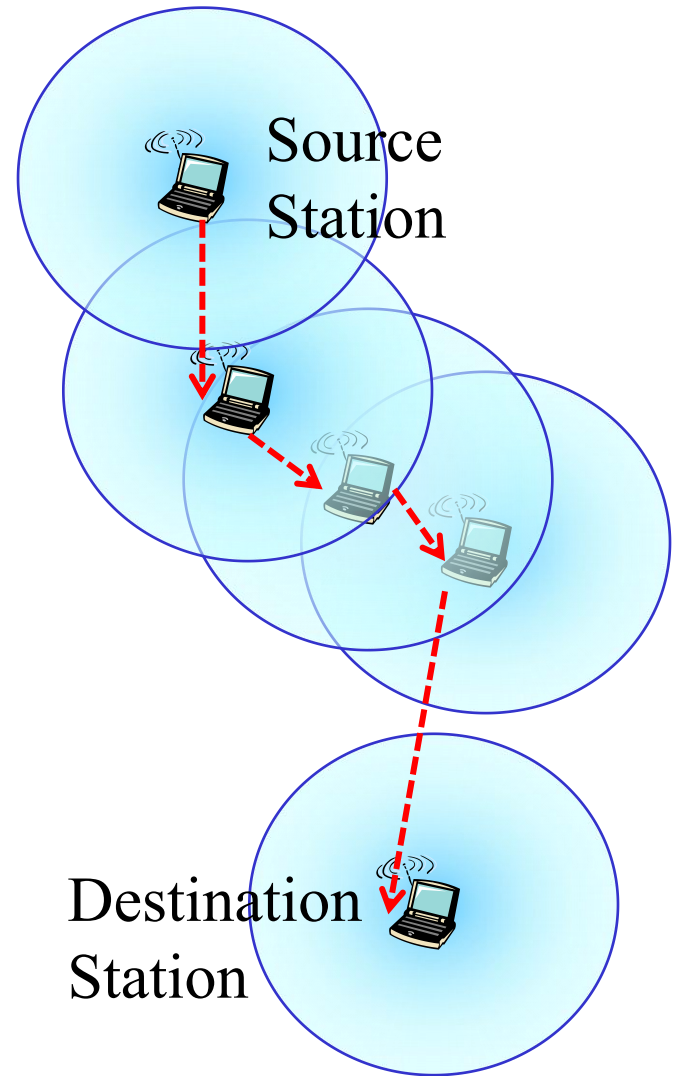
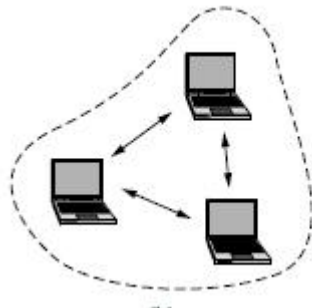
4.4.1 802.11 LAN architecture



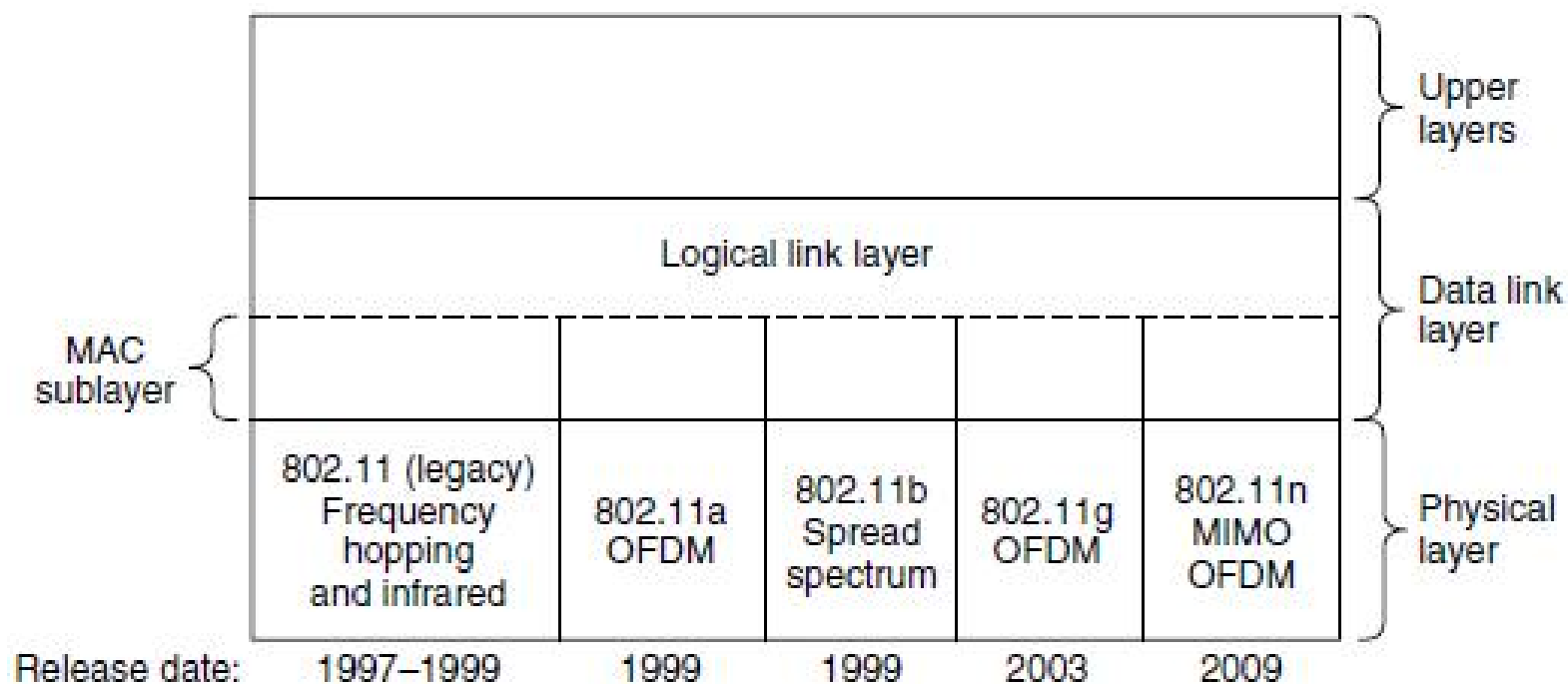
- wireless host communicates with base station
- ◆ base station = access point (AP)
- Basic Service Set (BSS)
 - ◆ wireless hosts
 - ◆ access point (AP): base station

Wireless LAN: Ad hoc Mode

- No base stations
- Nodes can only transmit to other nodes within link coverage
- Nodes organize themselves into a network:
route among themselves



IEEE802.11 Protocol Layer



- **FHSS:** Frequency Hopping Spread Spectrum 跳频扩频
- **DSSS:** Direct Sequence Spread Spectrum 直接序列扩频
- **OFDM:** Orthogonal Frequency Division Multiplexing 正交频分复用
- **MIMO:** Multiple Input Multiple Output 多输入多输出

物理层编码技术不属于课程范畴！

Widely Used 802.11 WLANs

802.11b

- ◆ 2.4 GHz unlicensed radio
- ◆ up to 11 Mbps
- ◆ direct sequence spread spectrum (DSSS) in physical layer
- ◆ widely deployed, using base stations

802.11a

- ◆ 5 GHz range
- ◆ up to 54 Mbps
- ◆ OFDM

802.11g

- ◆ 2.4 GHz range
- ◆ up to 54 Mbps
- ◆ OFDM

802.11n

- ◆ 2.4GHz/5GHz range
- ◆ up to 600 Mbps
- ◆ MIMO(multiple antennae)

-
- ◆ All use CSMA/CA for multiple access
 - ◆ All have base-station and ad-hoc network versions

4.4.3 802.11 MAC Sublayer: Multiple access

■ How to avoid collision?

- ◆ Collision Detection does not work! **Why?**

■ Two Modes of operation

- ◆ **Point coordination function (PCF, 点协调功能)**
 - Controlled by **base station**
 - Polling-based priority
- ◆ **Distributed coordination function (DCF, 分布协调功能)**
 - No central control
 - **CSMA/CA**

(1) 802.11 MAC: Modes of operation

■ PCF (Point Coordination Function点协调功能): Polling(轮询) 集中控制

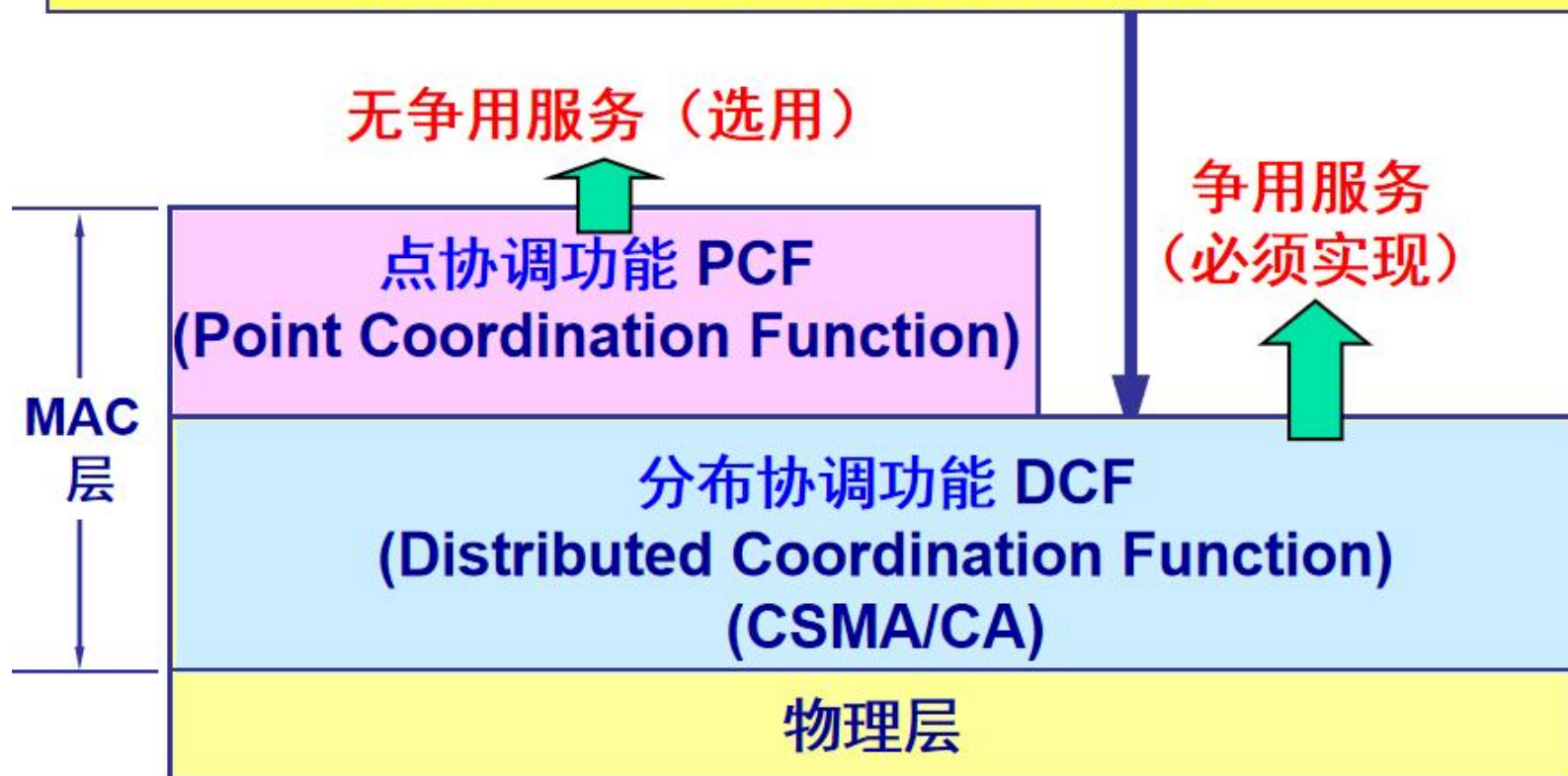
- ◆ Uses **base station** to control all activity in its cell
- ◆ Base station **broadcasts** a **beacon frame**(信标帧) **periodically** (eg.100ms)
- ◆ The beacon frame contains system parameters, such as clock synchronization, etc. It also invites new stations to sign up for polling service
- ◆ Once a station has signed up, it is effectively **guaranteed** a certain fraction of the **bandwidth**

■ DCF (Distributed Coordination Function,分布协调功能)

- ◆ Stations use a **handshake** to determine which one can send
- ◆ **MACA**: Multiple Access with Collision Avoidance
- ◆ **CSMA/CA**:CSMA with Collision Avoidance

802.11 MAC: Modes of operation

DCF 子层在每一个结点使用 **CSMA** 机制的**分布式接入**算法，让各个站通过**争用**信道来获取发送权。
因此 **DCF** 向上提供争用服务。



(2) 802.11 MAC: CSMA/CA

- Collisions: nodes transmitting at same time.
- 802.11: CSMA—sense before transmitting
 - ◆ Don't collide with ongoing transmission by other node
- 802.11: no collision detection!
 - ◆ Difficult to receive (sense collisions) when transmitting due to weak received signals(fading)
 - Radios are always half duplex
 - Received signal is 10^6 times weaker than the transmitted, collision detection mechanism does not work
 - ◆ Can't sense all collisions in any case: hidden terminal
 - ◆ goal: avoid collisions-CSMA/C(ollision)A(voidance)

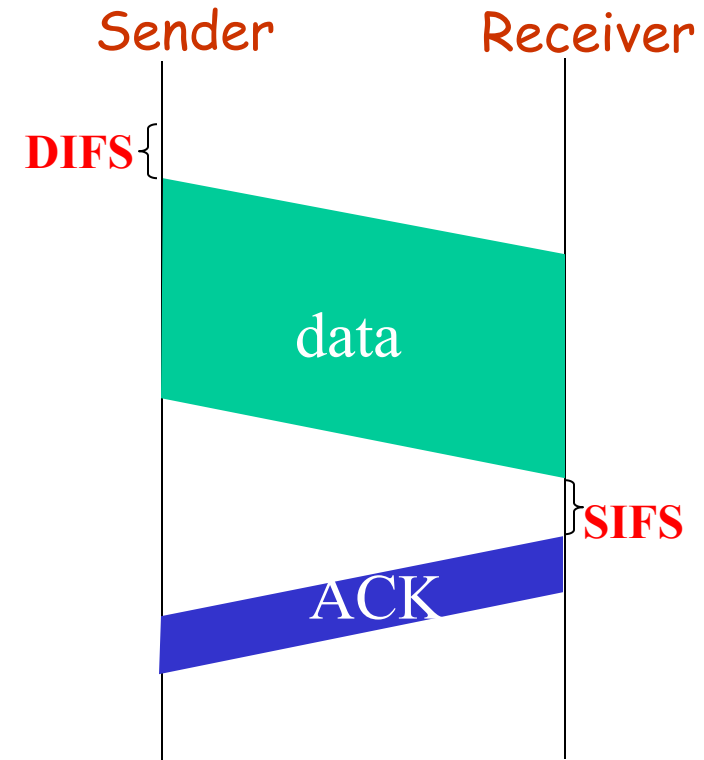
(2) 802.11 MAC : A Brief View of CSMA/CA

802.11 sender

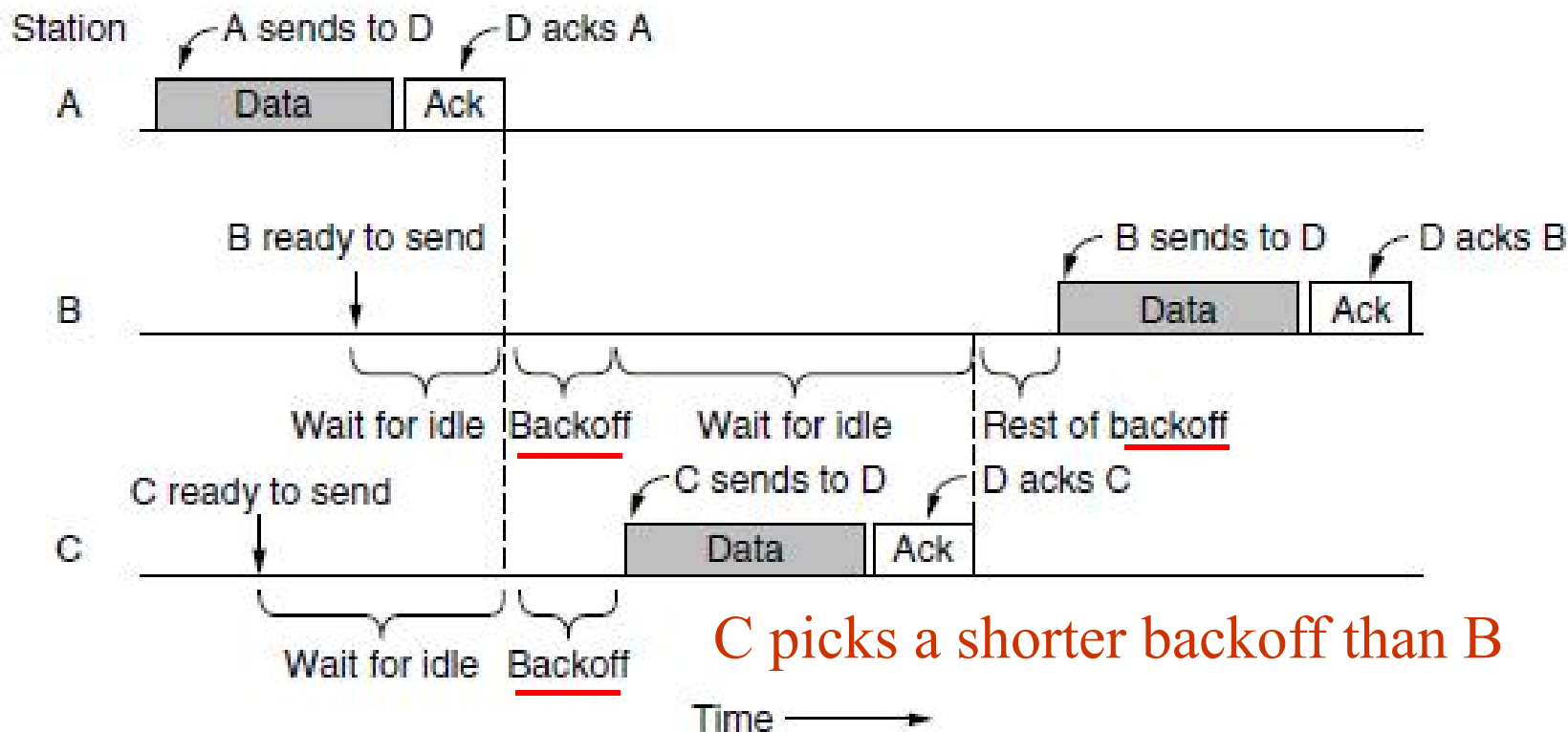
1. Waits until the channel is idle by **sensing** and wait **D(CF)IFS**
2. **Random backoff** (0~15 time slots): counts down idle slots, pausing when frames are sent. **Sends** its frame when the counter reaches 0
3. If no ACK(indicate a **collision** or other error), **increase random backoff** interval(double), repeat 2

802.11 receiver

- if frame received OK,
return ACK(**Explicit ACK**) after
S(hort)IFS



(2) 802.11 MAC : CSMA/CA Example



■ Compared with Ethernet

- ◆ Starting backoffs(退避) early helps to avoid collisions.
- ◆ Acknowledgements are used to infer collisions.

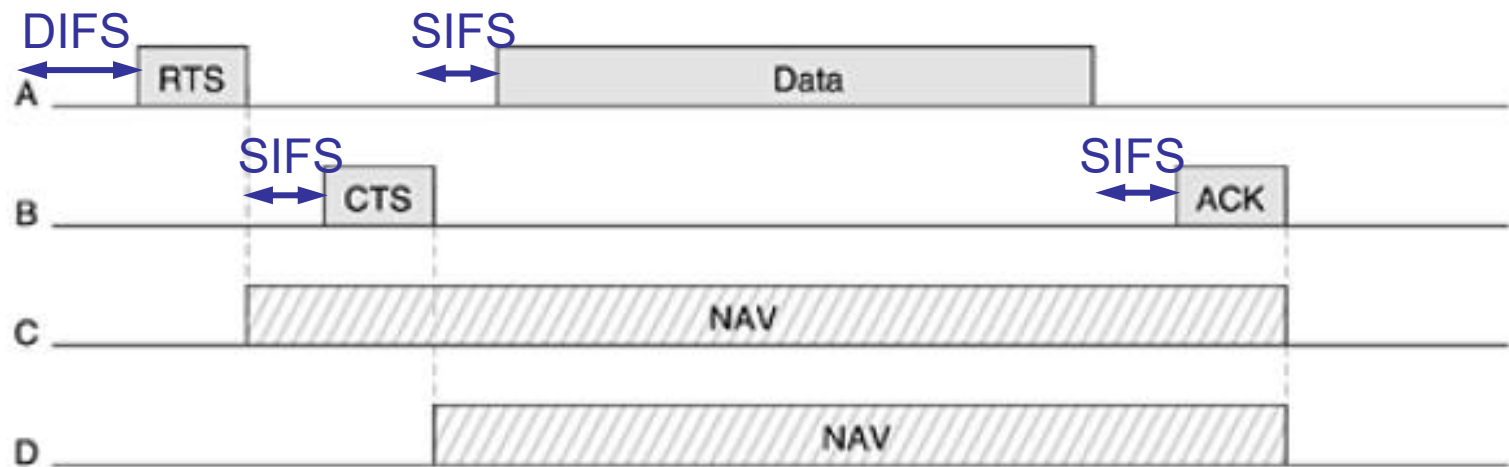
(3) 802.11 MAC: Channel sensing

Physical sensing

- Checks the medium to see if there is a valid signal

Virtual sensing

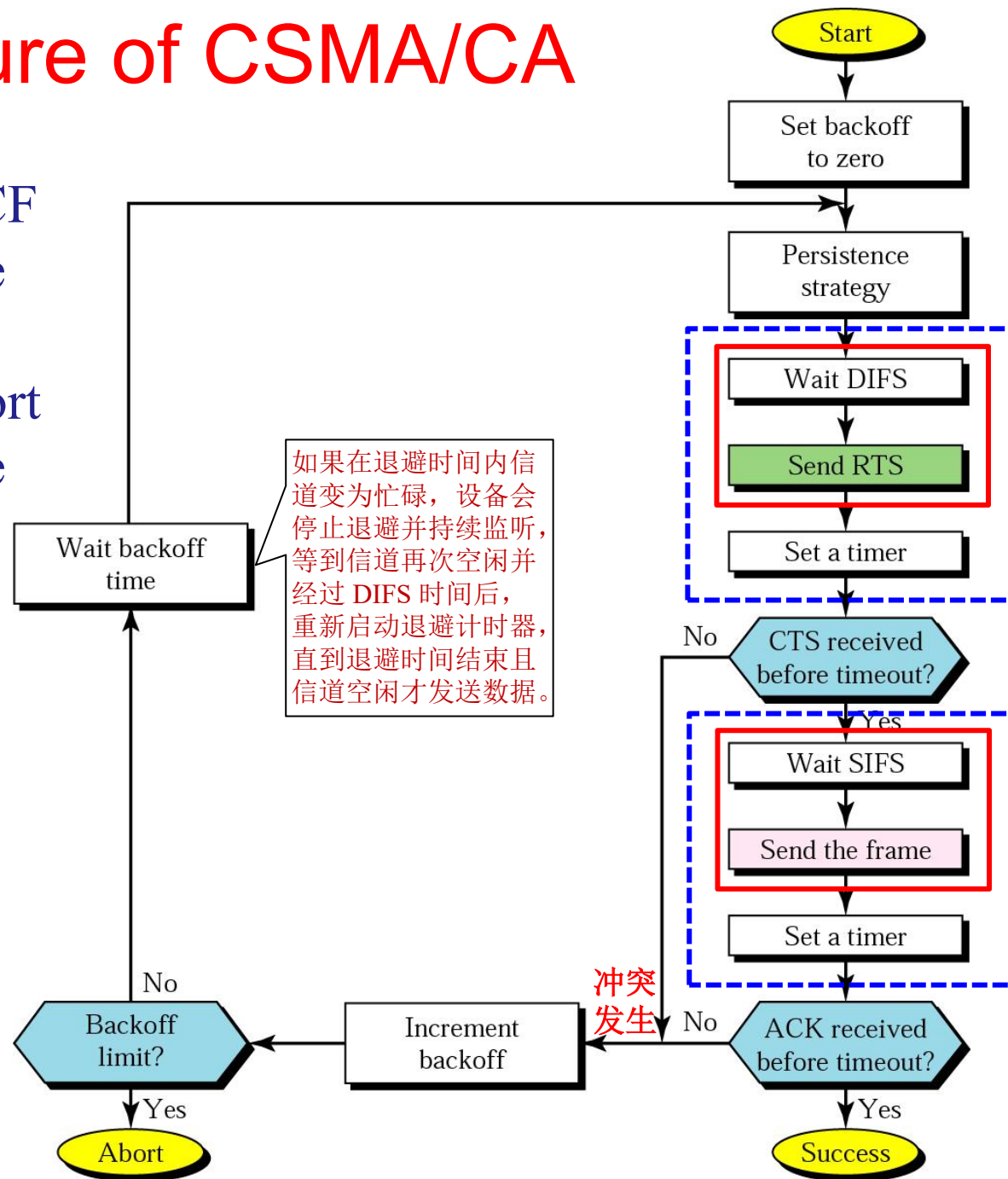
- Each station** keeps a logical record of when the channel is in use by tracking **NAV (Network Allocation Vector)**
- Each frame** carries a NAV
 - NAV says how long the sequence of which this frame is part will take to complete
- Optional RTS/CTS mechanism**



通过NAV确定退避时间，以避免冲突

Procedure of CSMA/CA

- DIFS: DCF Inter-Frame Space
- SIFS: Short Inter-Frame Space



[Forouzan]

(4) 802.11 reliability--Fragment burst(段突发)

- Acknowledgement and retransmissions is of little help in channels with high bit error rate
- Lower the transmission rate
 - ◆ use more robust modulations for a given SNR
- Send shorter frames: **fragment burst**
 - ◆ Frame success probability(S)
 - p is the error rate of 1 bit

$$S=(1-p)^n \quad p=10^{-4}$$

$$n=1518B=12144\text{bit}, \quad S=0.296(<30\%)$$

$$n=506B=4048\text{bit}, \quad S>2/3$$

短帧出错概率低！

$$S = (1 - p)^n$$

$$n = 1518B = 12144 \text{ bits}$$

$$p = 10^{-4} \quad S = 0.296$$

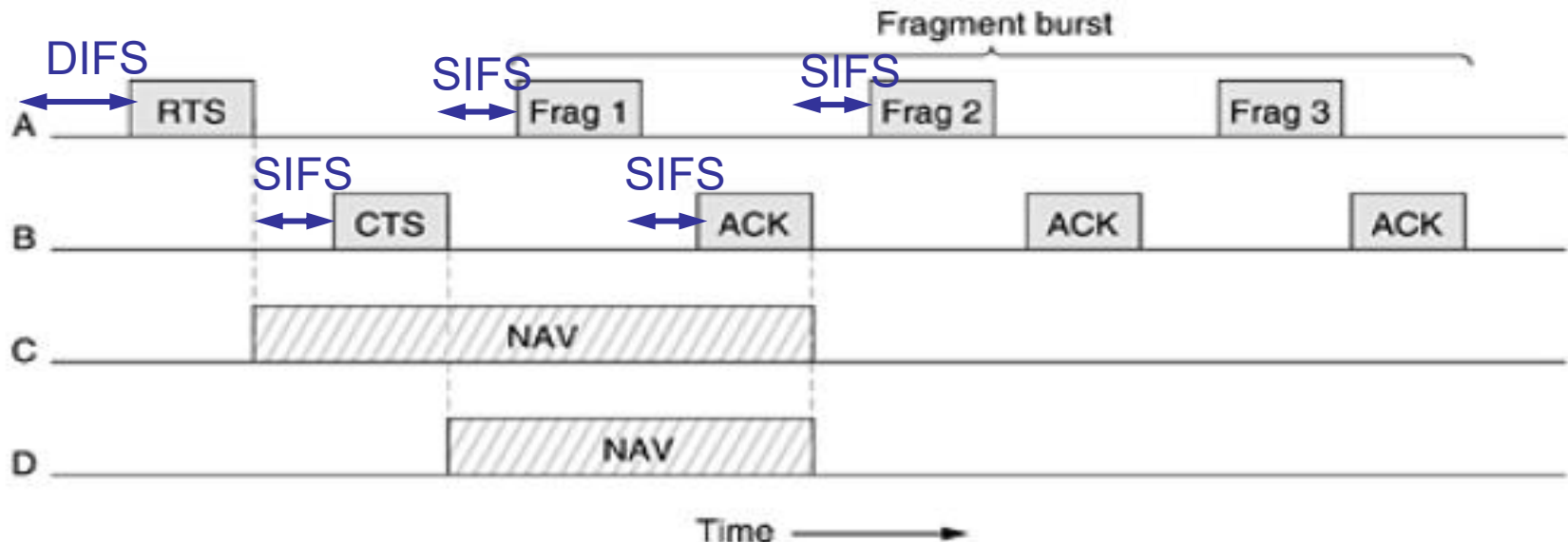
$$p = 10^{-5} \quad S = 0.885$$

$$p = 10^{-6} \quad S = 0.988$$

802.11 MAC: Reliability--Fragment burst(段突发)

■ Fragment burst

- ◆ The solution is to split frame into smaller pieces—**fragments**
- ◆ Each fragment is sent with individual sequence number and checksum by **stop-and-wait** protocol



(5) 802.11 Saving power

■ Beacon frames

- ◆ Clients can enter **power-save mode(省电模式)**;
- ◆ AP will buffer traffic;
- ◆ the client **wakes up for every beacon**, and check the **traffic map** that is sent as part of the beacon
- ◆ the client **sends a poll message** to the AP
- ◆ AP sends the buffered traffic

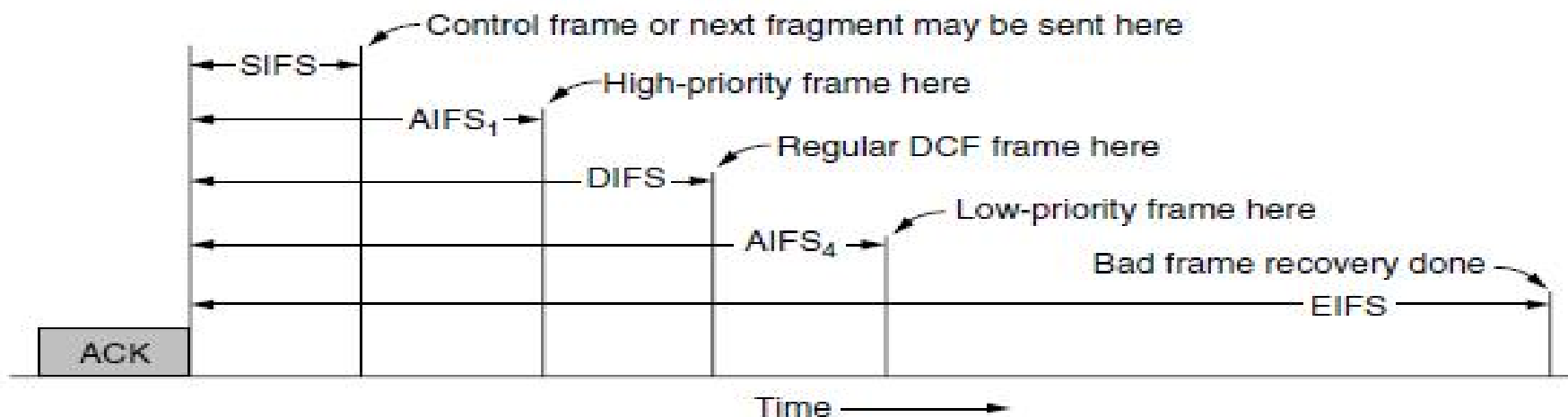
■ APSD (Automatic Power Save Delivery)

- ◆ Clients can enter **power-save mode**;
- ◆ AP will buffer traffic;
- ◆ the client **sends frames** to the AP
- ◆ AP sends the buffered traffic
- ◆ e.g VoIP wireless phone sends and receives frames every 20ms

(6) IEEE802.11e: QoS—Interframe Spacing

Intervals between frames are defined to provide priority services

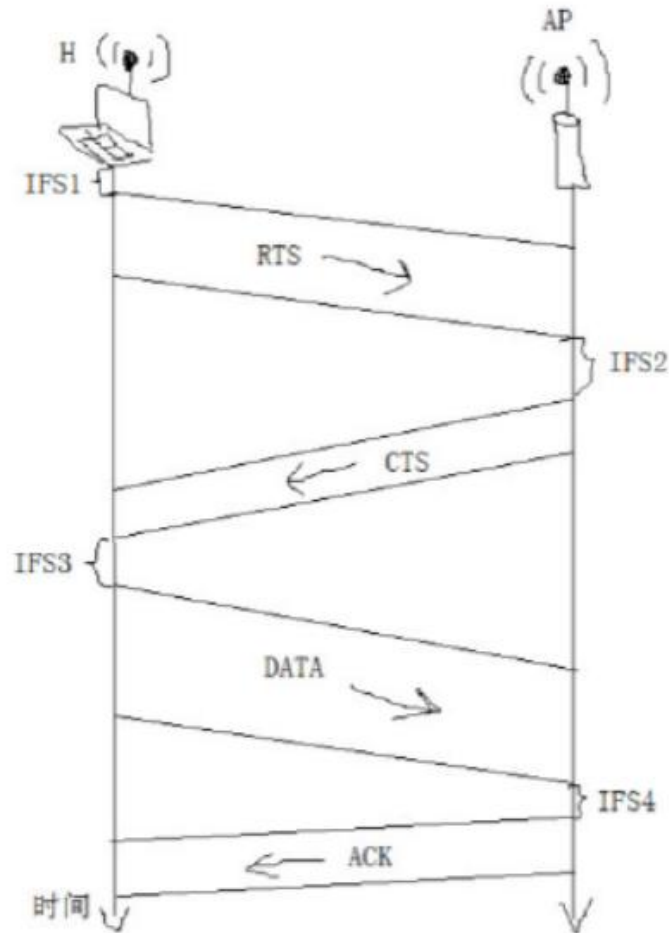
- ◆ **DIFS**: The interval between regular data frames
- ◆ **SIFS**: The shortest interval. It is used to allow the parties in a single dialog the chance to go first (eg. letting the receiver send **ACK**, or the sender transmit a burst of fragments)
- ◆ **AIFS₁** (Arbitration InterFrame Space): The short interval, can be used by the AP to move high-priority traffic
- ◆ **AIFS₄**: The long interval, can be used for low-priority traffic
- ◆ **EIFS**: is used only by a station that just received a bad or unknown frame, to report the problem



通过为不同类型的帧设置不同长度的帧间间隔IFS，实现优先级服务

example

37. 某 IEEE 802.11 无线局域网中主机 H 与 AP 之间发送或接收 CSMA/CA 帧的过程如下图所示，在 H 或 AP 发送帧前所等待的帧间间隔时间 (IFS) 中最长的是：



- A、IFS1
- B、IFS2
- C、IFS3
- D、IFS4

答案：A

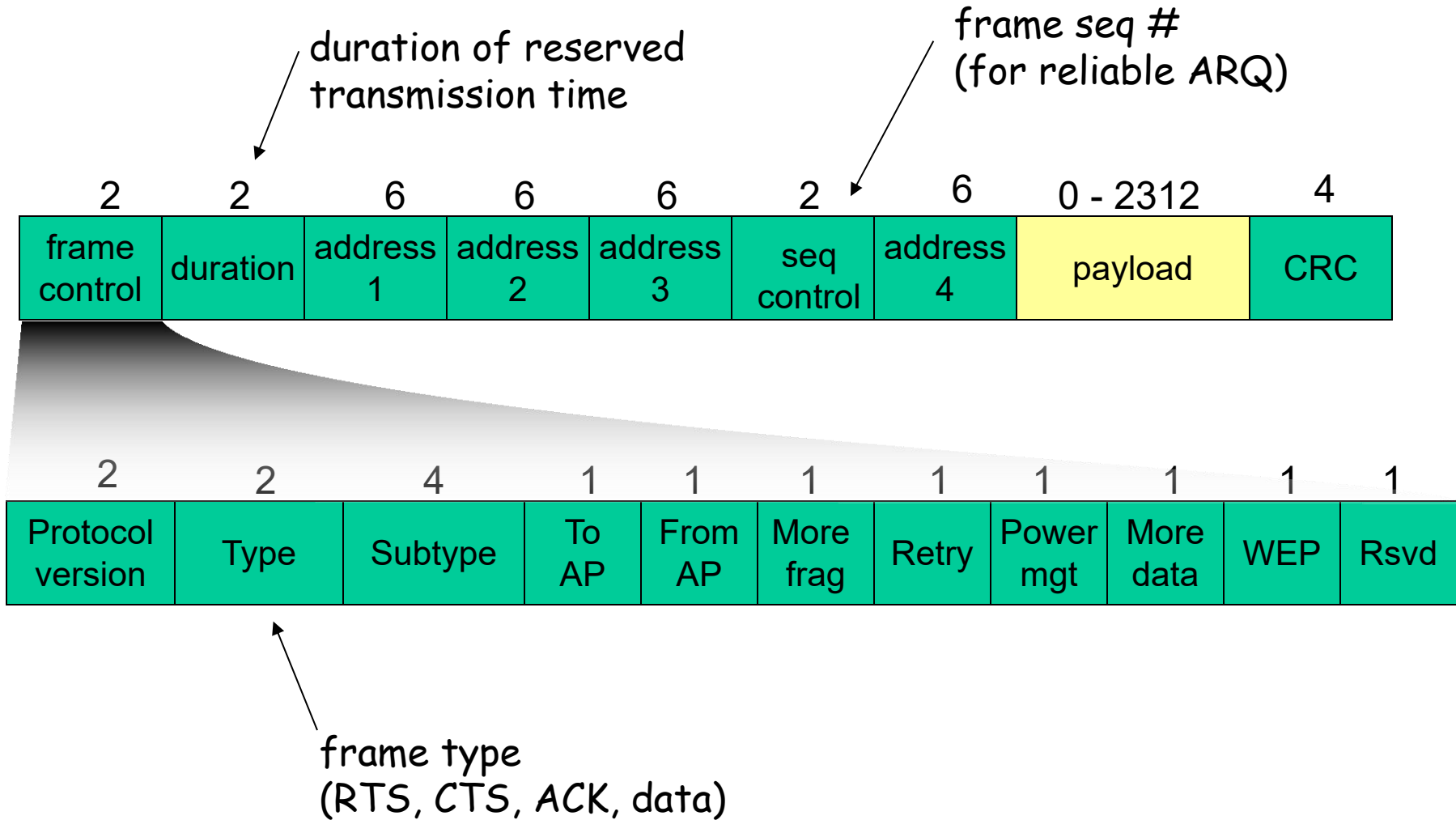
(6) IEEE802.11e: QoS Extension--TXOP

- Transmission Opportunity(发送机会)
- Original problem: Rate anomaly(速率异常)
 - ◆ A station sends one frame at a time
 - ◆ If a fast station and a slow station send at same time, the fast one has to slow down.
 - ◆ Two stations: $c_1=6\text{Mbps}$, $c_2=54\text{Mbps}$, average rate= 5.4Mbps , $c=1/((1/c_1)+(1/c_2))$
- With transmission opportunities, each station gets an equal amount of airtime, not an equal number of frames.
 - ◆ $C_1=6M/2$, $c_2=54M/2$, $c=c_i/n$

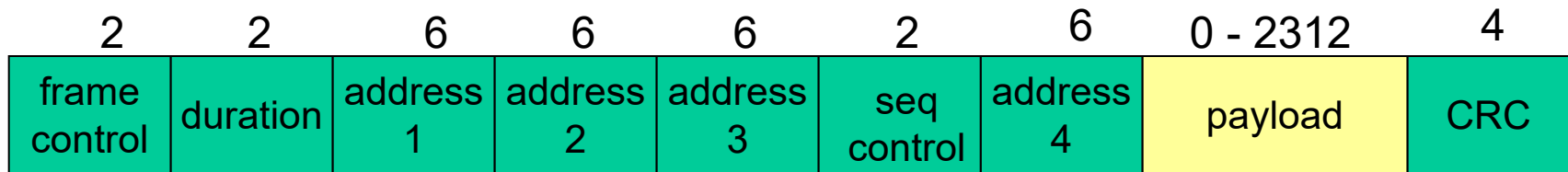
4.4.4 MAC Layer Frames

- Data Frames
- Control Frames
 - ◆ RTS,CTS,ACK
- Management Frames
 - ◆ Authentication and De-Authentication
 - ◆ Association, Re-Association, and Disassociation
 - ◆ Beacon and Probe frames

802.11 frame



802.11 frame: addressing



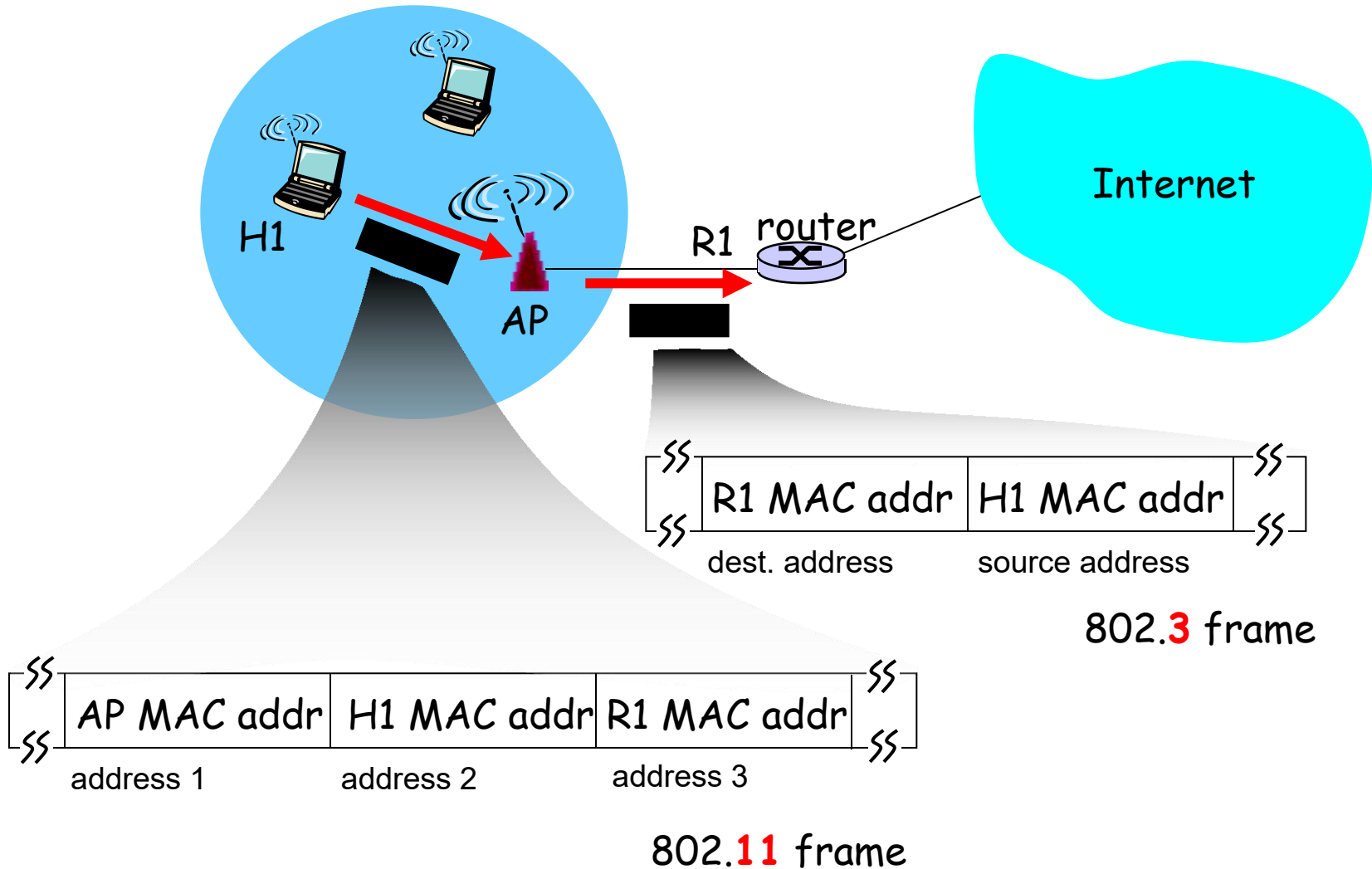
Address 1: MAC address of wireless host or AP to receive this frame

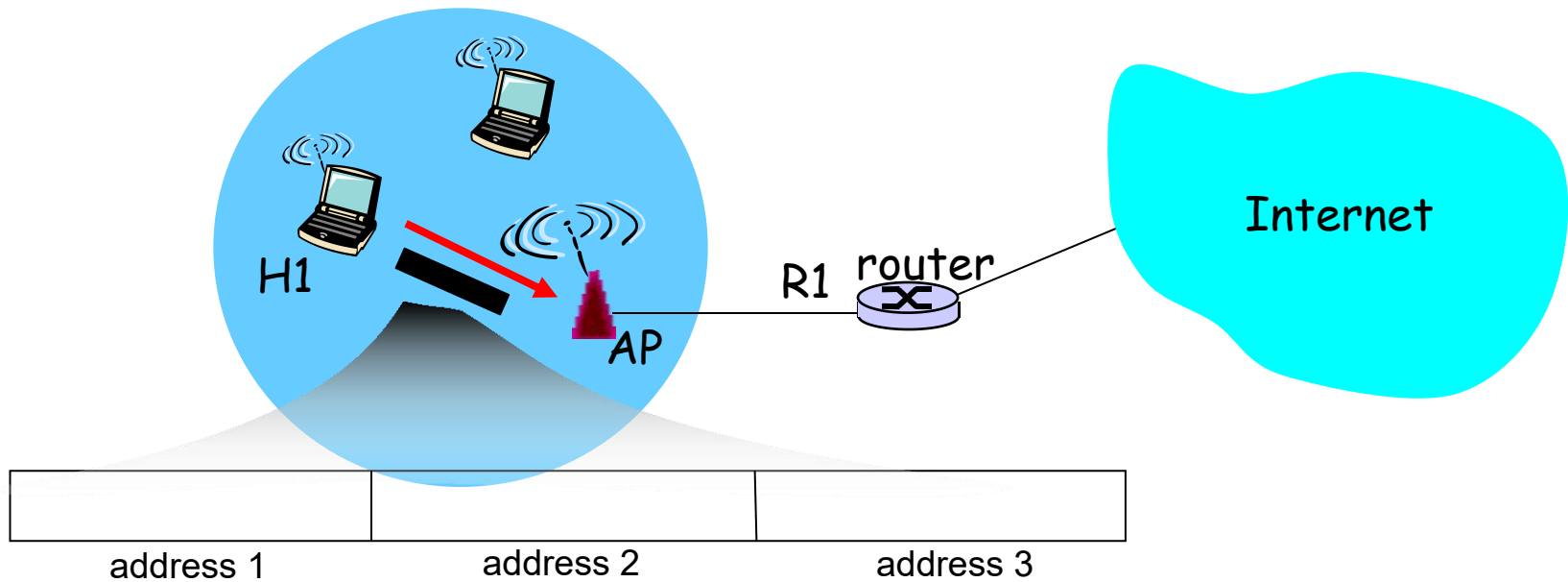
Address 2: MAC address of wireless host or AP transmitting this frame

Address 3: MAC address of router interface to which AP is attached

Address 4: used only in ad hoc mode

802.11 frame: addressing





AP MAC addr H1 MAC addr R1 MAC addr

SS
SS

SS
SS

802.11 frame

The Evolution of WiFi(补充)



Wi-Fi 0	IEEE802.11	The first 2.4GHz Wi-Fi standard
Wi-Fi 1	IEEE802.11b	It operated at 2.4GHz to achieve higher data rates.
Wi-Fi 2	IEEE802.11a	It supported 5GHz operation and its 20MHz bandwidth supported multiple data rates
Wi-Fi 3	IEEE802.11g	It allowed for faster data rates of up to 54Mbps in the same 2.4GHz frequency band as IEEE802.11b
Wi-Fi 4	IEEE802.11n	It supported the 2.4GHz and 5GHz frequency band with up to 600Mbps data rates
Wi-Fi 5	IEEE802.11ac	The first standard to enable the use of multiple input/multiple output(MIMO) technology. It supported the 5GHz frequency band with data rate up to 3.5Gbps.
Wi-Fi 6	IEEE802.11ax	It supports the 2.4GHz, 5GHz and 6GHz frequency band with up to 9.6Gbps data rate.
Wi-Fi 7	IEEE802.11be	It will support the 2.4GHz, 5GHz and 6GHz frequency band with up to 46Gbps data rate.

Outline

- 4.1 Channel Allocation Problem
- 4.2 Multiple Access Protocols
- (補)IEEE 802 Reference Model
- 4.3 Ethernet
- 4.4 Wireless LANs
- 4.8 Data Link Layer Switching

4.8 Data Link Layer Switching(交换)

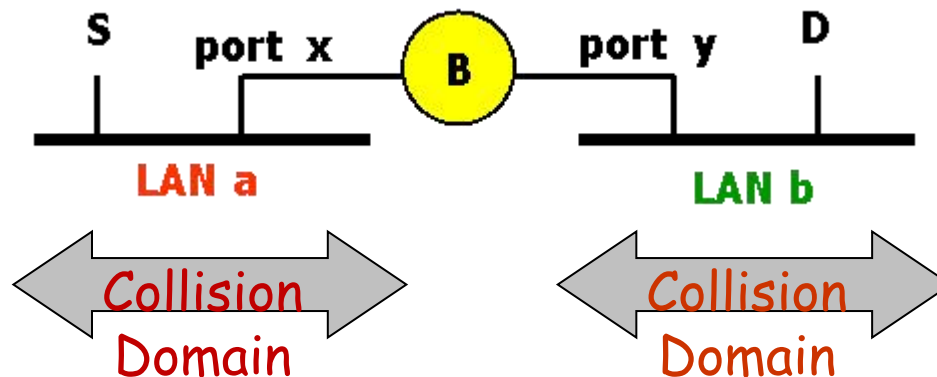
- *4.8.1 Uses of Bridges(网桥)*
- 4.8.2 Learning Bridges
- 4.8.4 Summary of connection devices
- 4.8.5 Virtual LANs(虚拟局域网)

(1) Why Interconnecting Multiple LANs?

- The need for interaction
- The organization may be geographically spread over several buildings
- Split a LAN into separate LANs to accommodate the load (multiple **collision domains**)
- Physical distance(2500m)
- **Reliability** (可靠性) --故障隔离
- **Security** (promiscuous mode, 混杂模式)

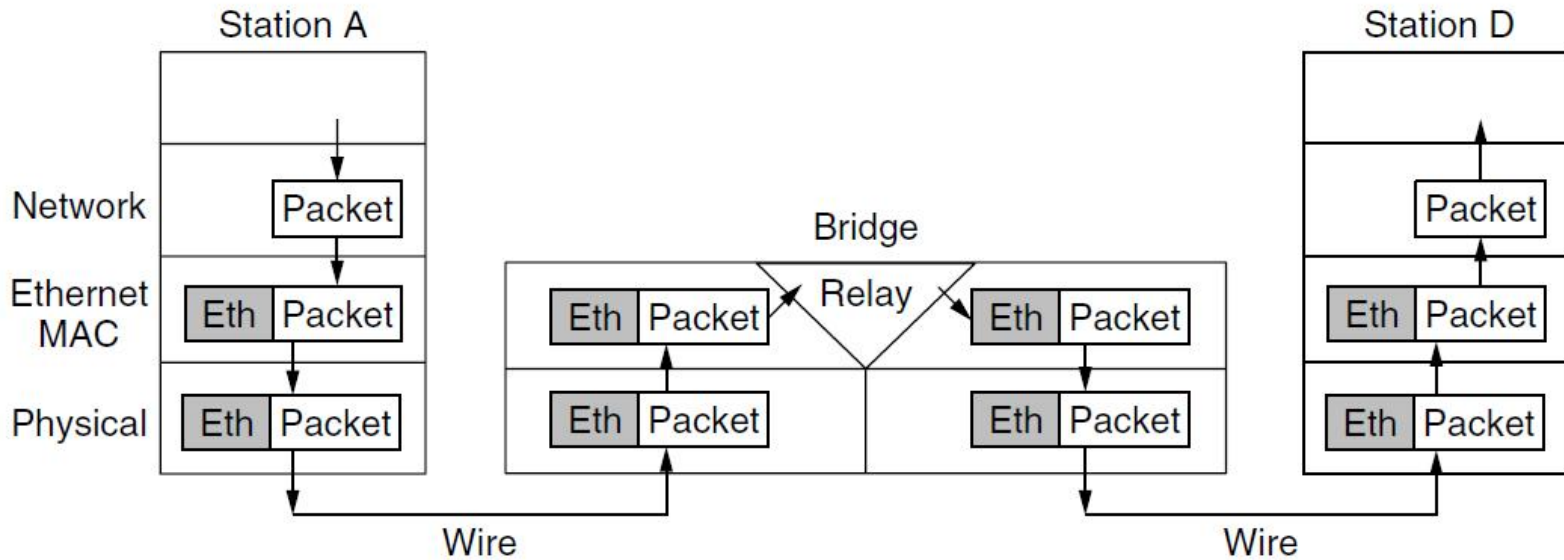
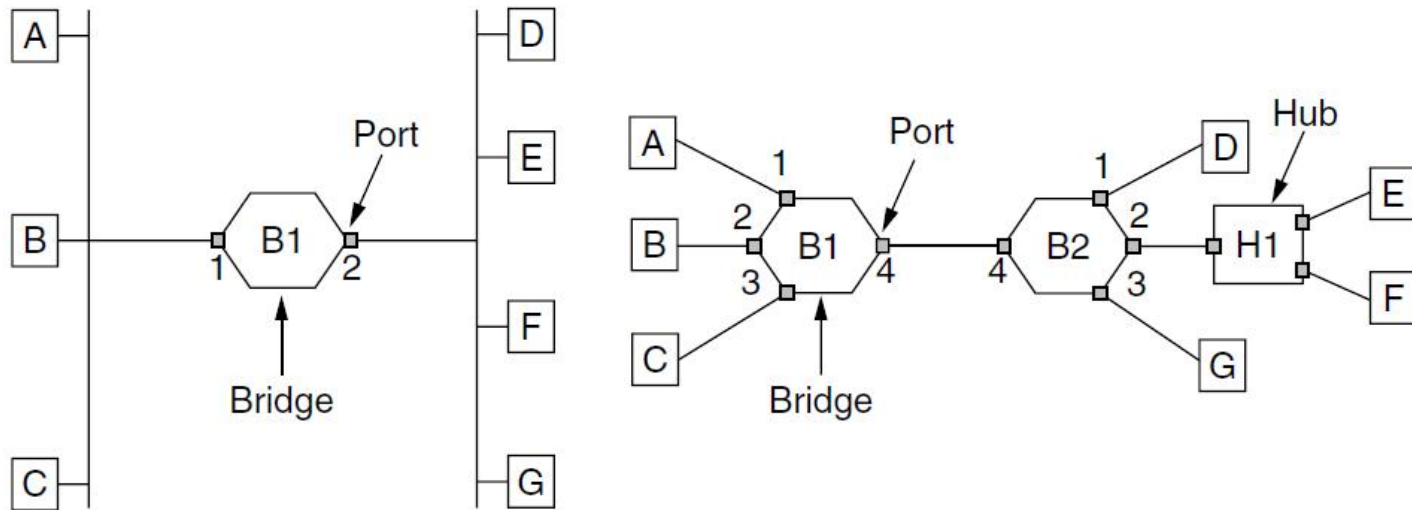
(2) What is Bridge(网桥)?

- Bridges are used to **interconnect LANs at data link layer**.
 - ◆ Forward (or discarded) a frame based on destination MAC addresses
- Plug and Play(即插即用)



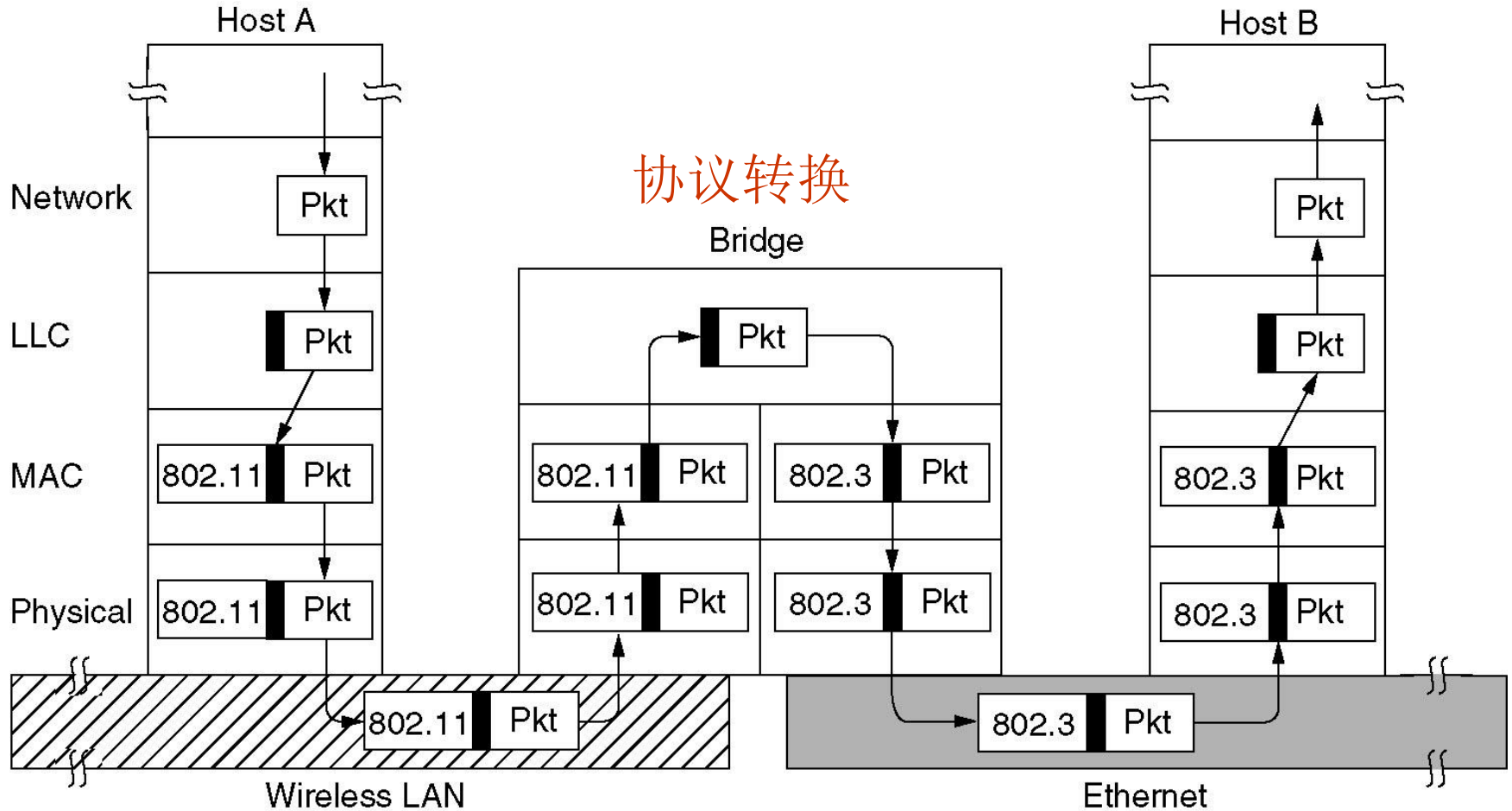
- Drawback
 - ◆ Delay increased(延迟增加)-Working in a **store-and-forward** way
 - ◆ **Broadcast storm(广播风暴)** can not be avoided

Bridge(网桥)



Protocol processing at a bridge

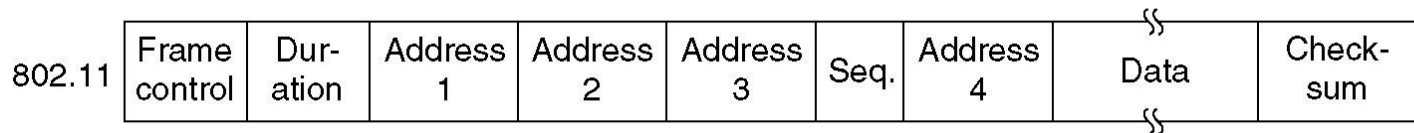
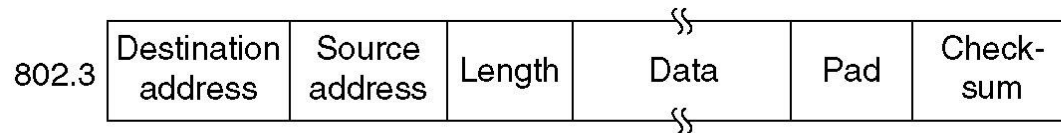
A LAN Bridge from 802.11 to 802.3



(3) Bridges(网桥) from 802.x to 802.y

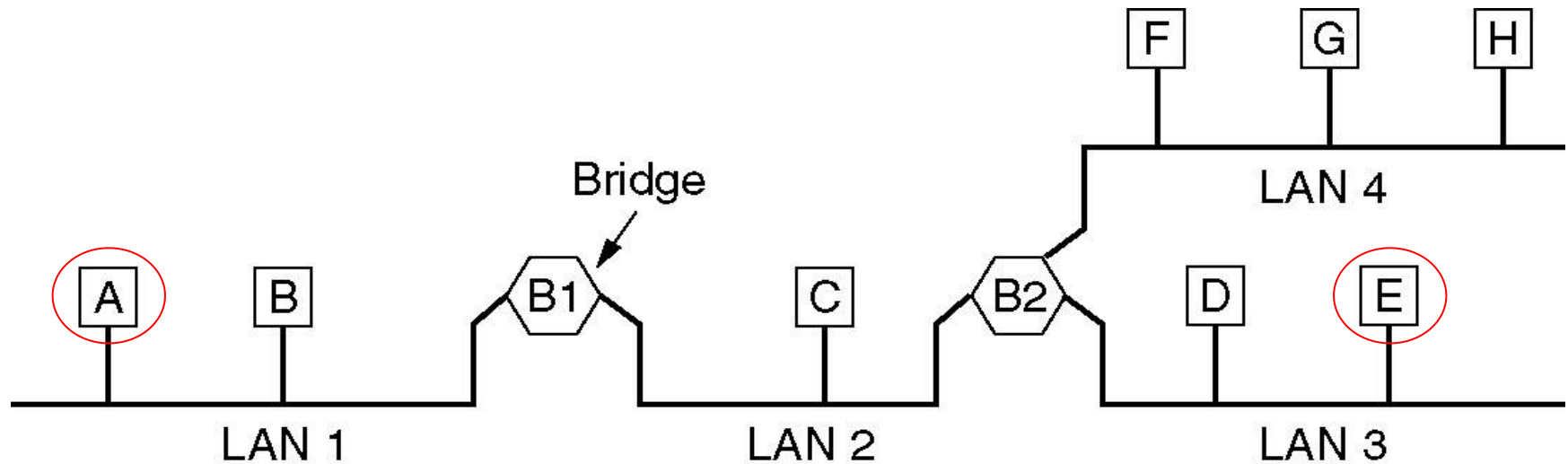
Problems

- ◆ Each of the LANs uses a **different frame 帧 format**
- ◆ LANs do **not** run at **the same data rate**
- ◆ Different LANs have **different maximum frame lengths**



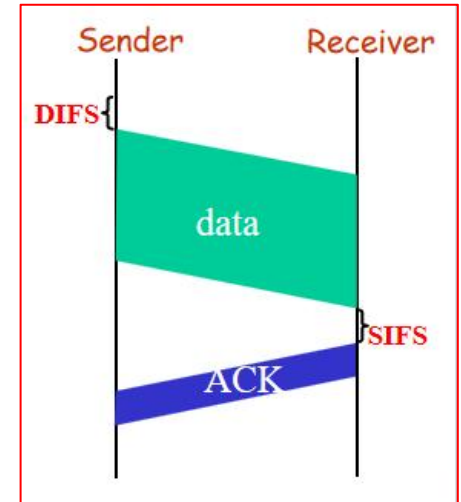
- ◆ **Security**: Wireless networks support encryption 加密 in the data link layer, Ethernet does not
- ◆ **QoS** 服务质量(Quality of service: 802.11、802.16)

A configuration with 4 LANs & 2 bridges

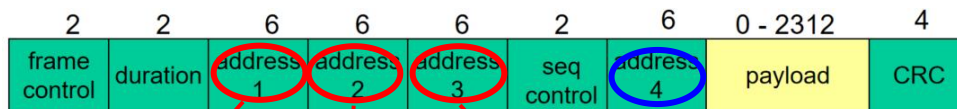


Review of IEEE802.11(WiFi)

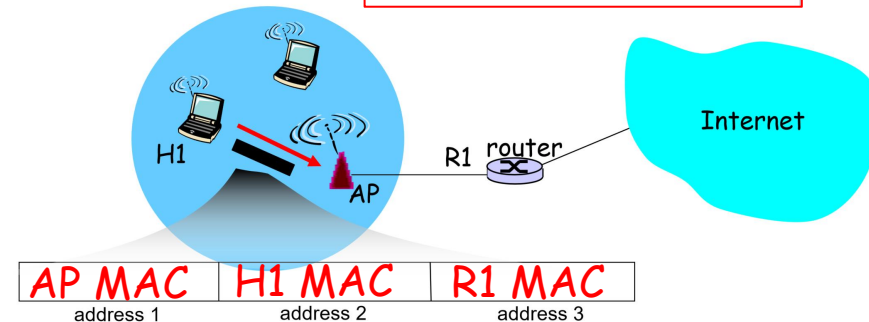
- WLAN (WiFi): IEEE802.11
- Acknowledged connectionless service
- MAC Sublayer: ($\neq$ MACA)
 - ◆ PCF(Point Coordination Function): 集中式, Polling
 - ◆ DCF(Distributed Coordination Function): 分布式, CSMA/CA
 - Physical Sensing & Virtual Sensing (NAV)
 - Random backoff
 - Data, Ack, RTS/CTS (optinal)
 - ◆ Reliability: fragment burst
 - ◆ Saving power: beacon frame, APSD
 - ◆ QoS
 - Interframe Spacing (DIFS, SIFS)
 - TXOP -- 会计算



Frame PDU

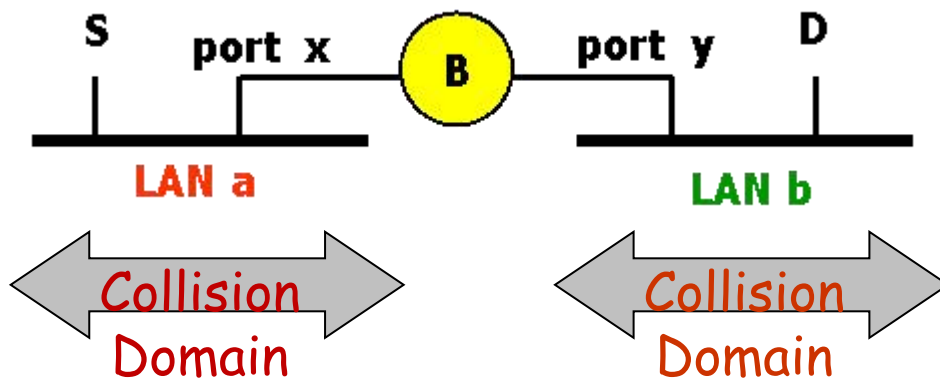


Receiver Addr
Transmitter Addr
Router Addr



Review of Bridge

- Bridges are used to **interconnect LANs** at **data link layer**.
- ◆ **Store-and-forward**: forward (or discarded) a frame based on destination MAC addresses



Station Address	Port/LAN Number
A	1
B	1
C	1
D	2
E	2

- ◆ **Broadcast storm**(广播风暴): broadcast MAC address->flooding

4.8 Data Link Layer Switching(交换)

- 4.8.1 Uses of Bridges
- *4.8.2 Learning Bridges*
- 4.8.3 Spanning Tree Bridges
- 4.8.4 Summary of connection devices
- 4.8.5 Virtual LANs

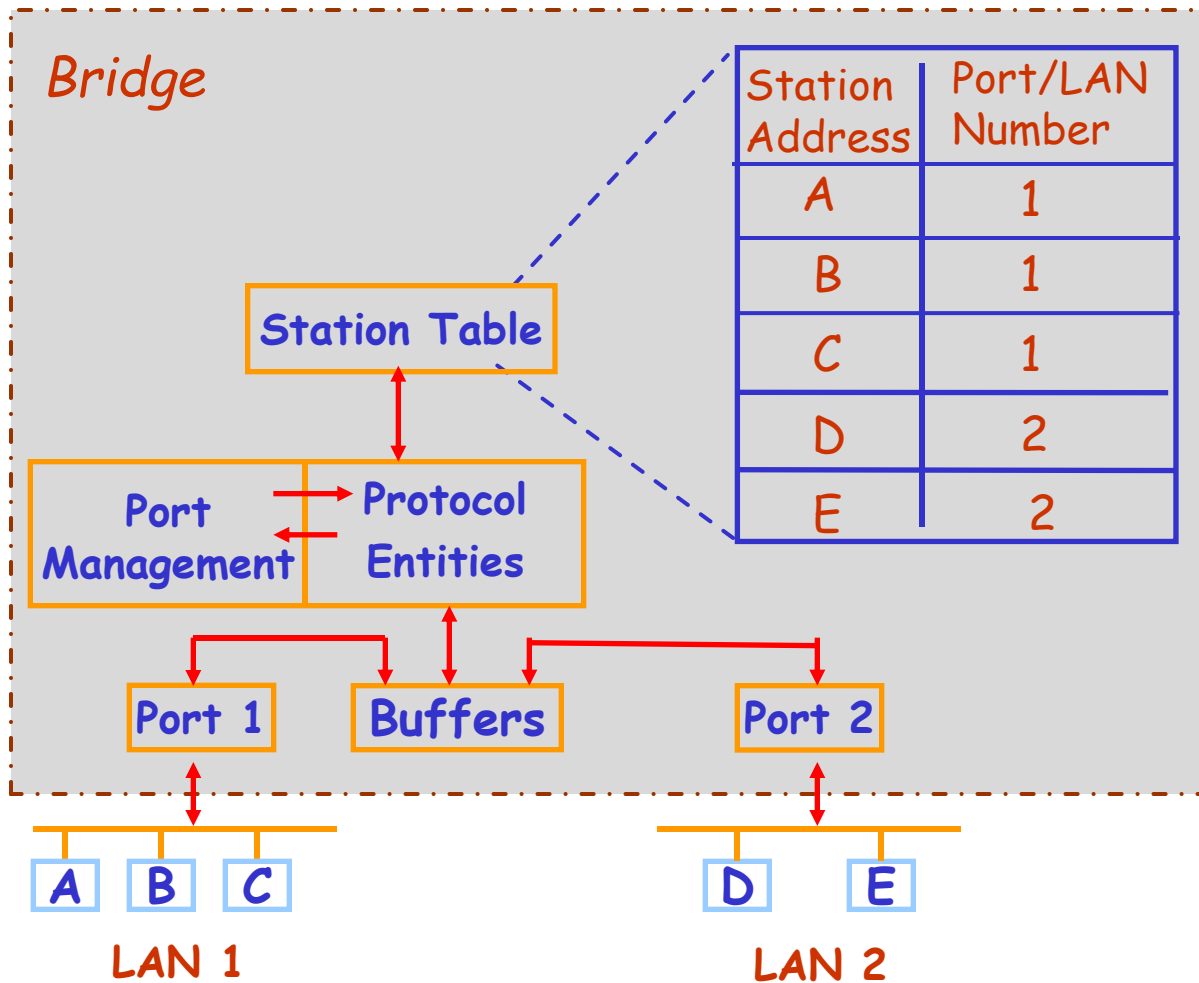
Bridge

- Related algorithms of Bridges
 - ◆ Backward learning(后向学习/逆向学习)
 - ◆ Spanning tree(生成树)

(1) Learning Bridges(学习网桥)

- Frame routing and forwarding is determined by bridges, which is transparent to hosts, also called **transparent bridges**(透明网桥)
- **Plug and play**(即插即用)
- Operates in **promiscuous mode**(混杂模式): Accepting every frame transmitted on all the LANs it attached
- For an incoming frame from **port-x**, checking **Station Table**(Forwarding Table/Hash Table, 站表/转发表)
 - ◆ If **destination port = port-x**, **discard**(丢弃) the frame
 - ◆ If **destination port $\neq$ port-x**, **forward**(转发) the frame
 - ◆ If **destination port** is **unknown** or for **broadcast address**, **flooding**(泛洪)
 - Outputting on all the ports connected except the one frame arrived on

Transparent Bridge(透明网桥): Operations



(2) How to build Station table

- Backward Learning(后向学习)

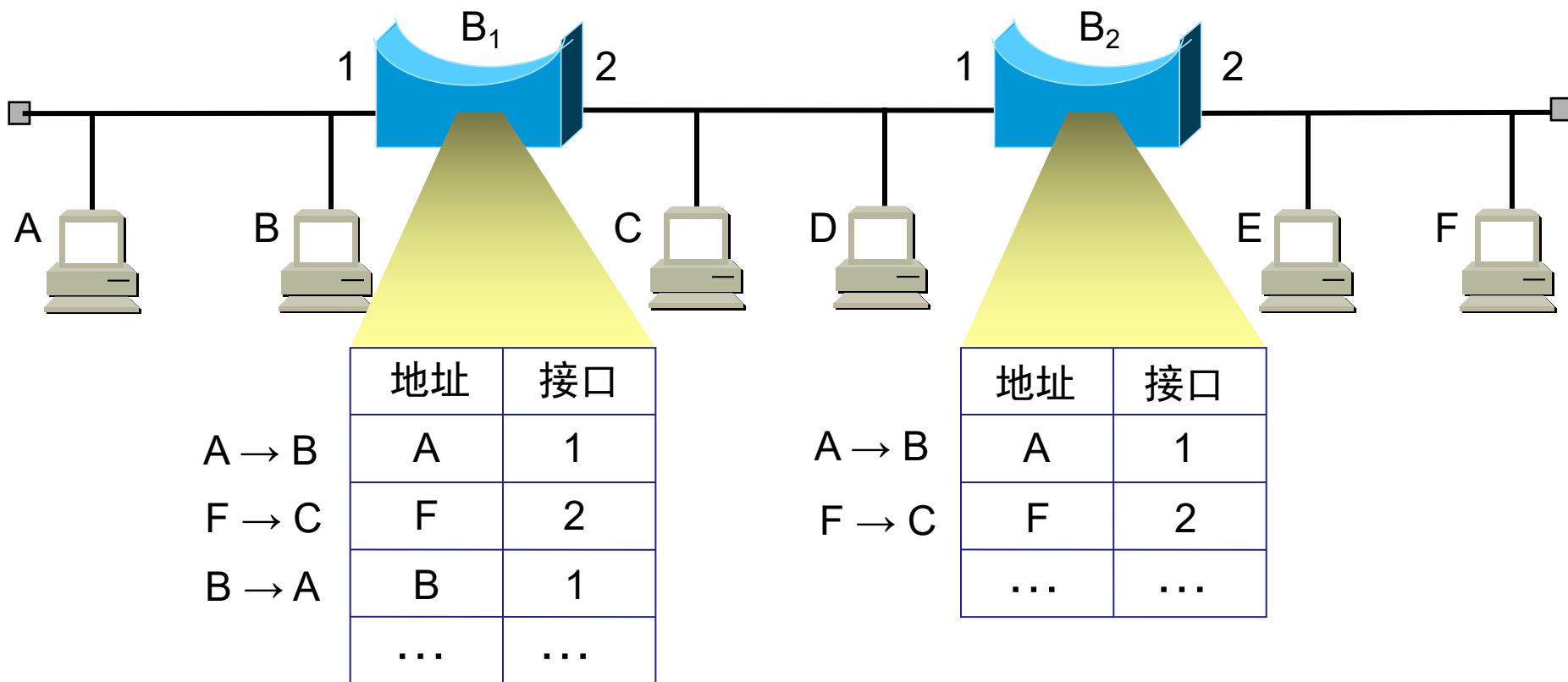
- Init

- ◆ When the bridges are first plugged in, all the hash tables are empty

- Adding

- ◆ Bridges operate in promiscuous mode混杂模式, so they see every frame sent on any of their LANs

Example of building station table

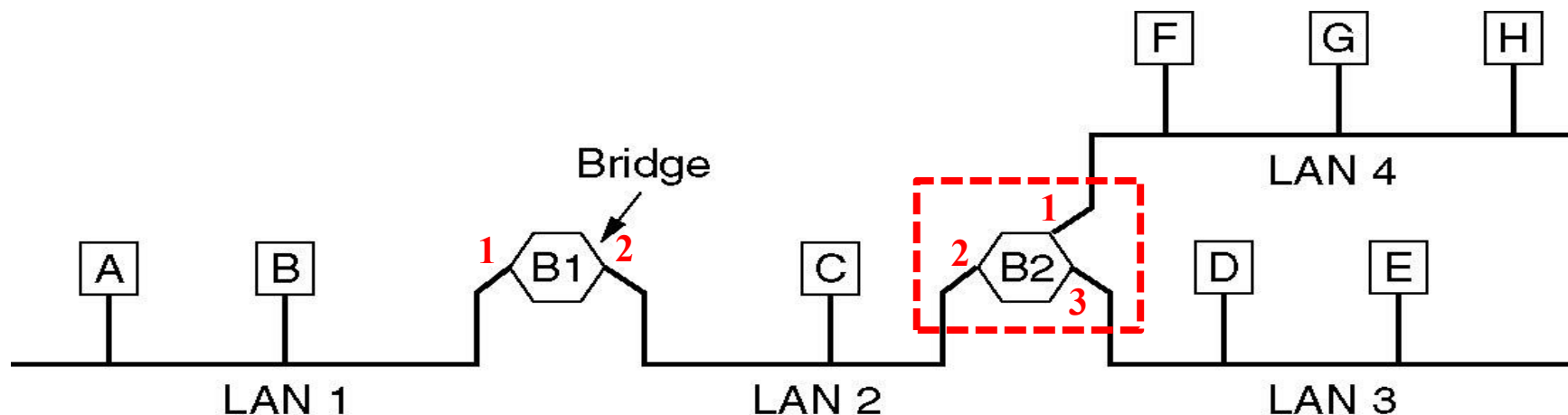


(1) flooding泛洪

(2) flooding

(3) discard丢弃

Example2

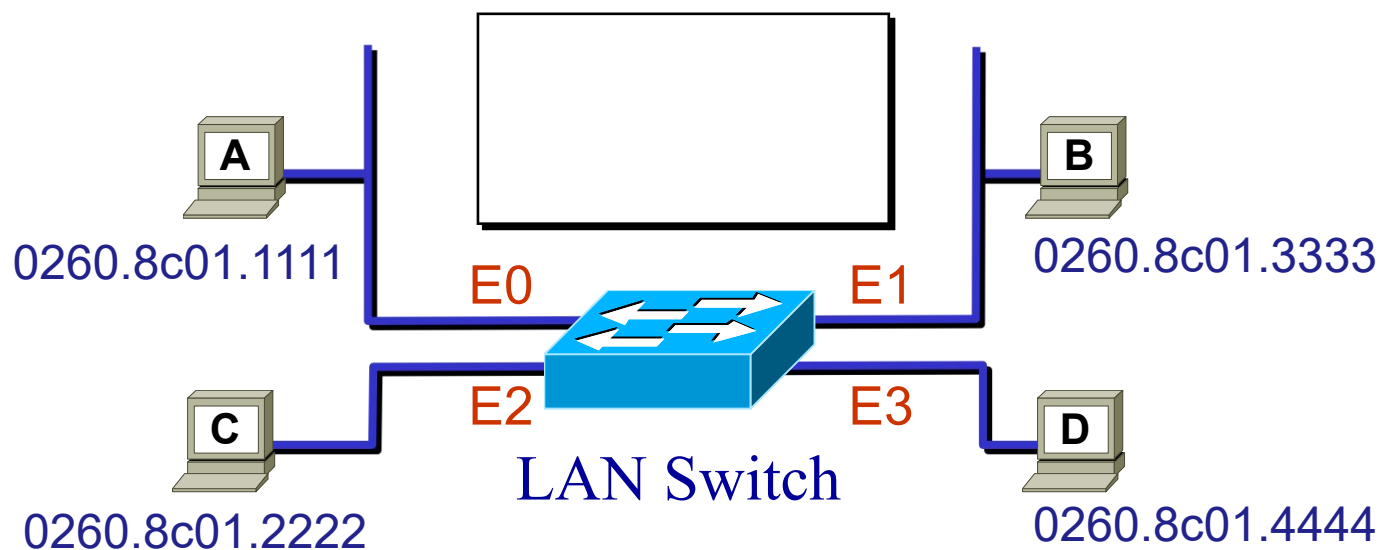


		B1	B2	
1	A send frame to C	<u>A 1</u>	<u>A 2</u>	flooding泛洪
2	E send frame to A	<u>E 2</u>	<u>E 3</u>	forward转发
3	D send frame to E	(未收到)	<u>D 3</u>	discard丢弃
4	E move to LAN2			
5	E send frame to A	<u>E 2</u>	<u>E 2</u>	discard

(补充)Switch交换机: How Switches Learn Host Addresses and Locations

交换机根据转发表将收到的帧转发到相应端口

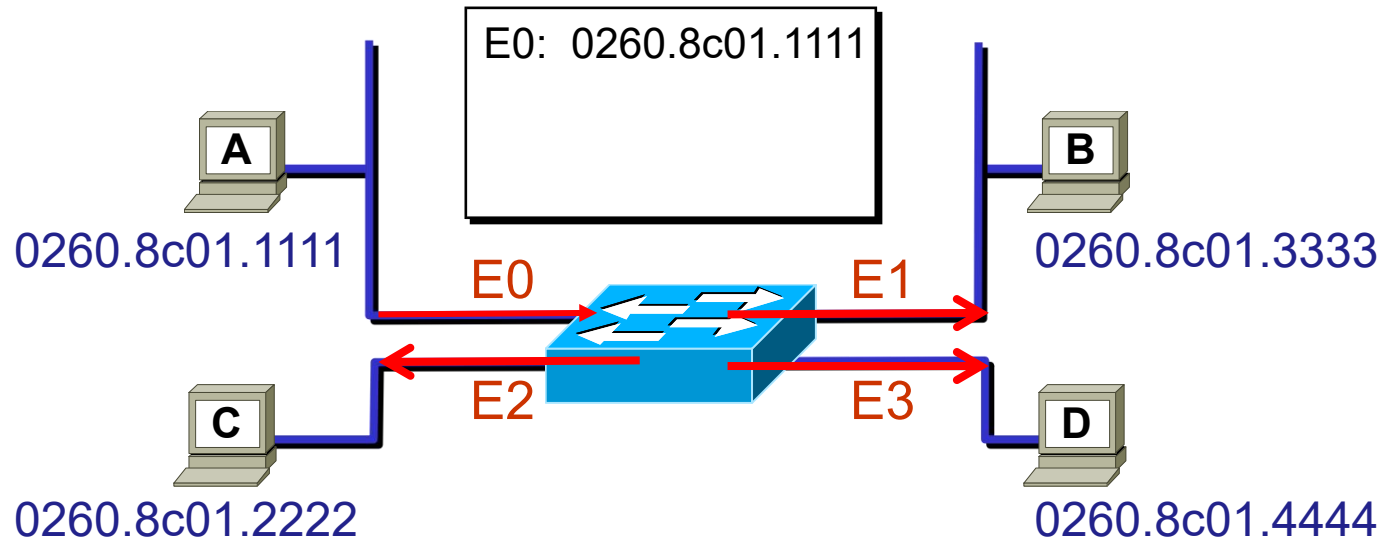
MAC forwarding table(转发表)



Initial MAC forwarding table is empty

Switch: How Switches Learn Host Addresses and Locations

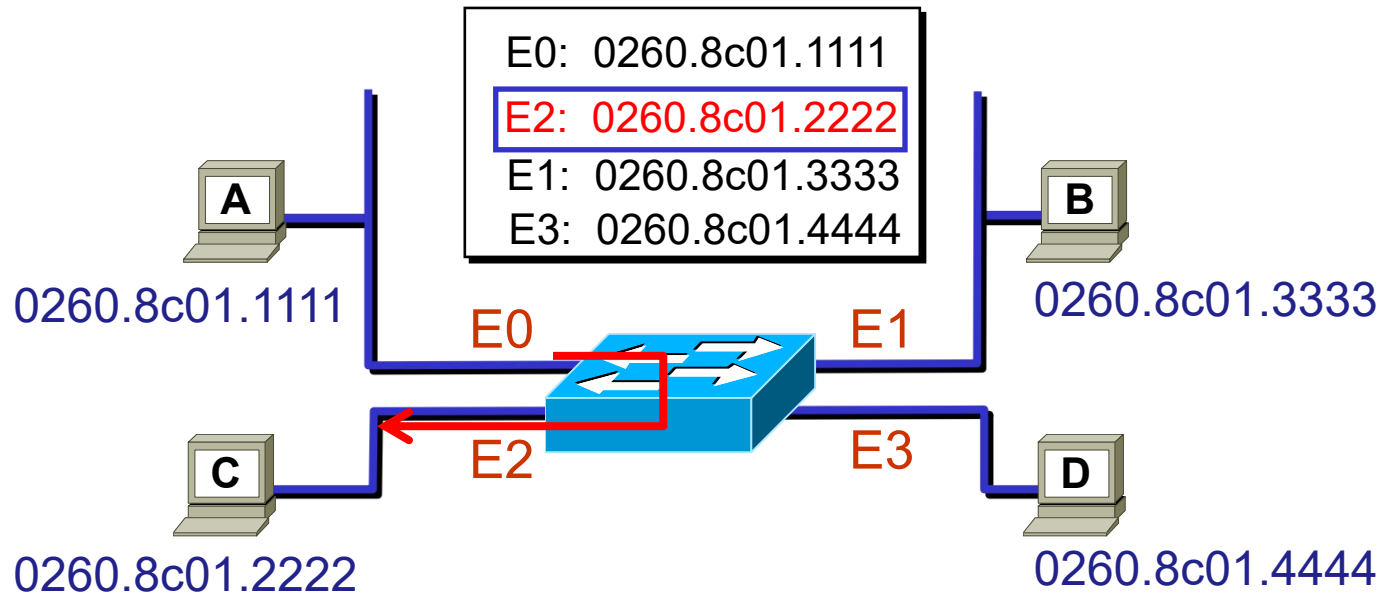
MAC forwarding table (转发表)



- Station **A** sends a frame to Station **C**
- The frame from station A to station C is **flooded** (洪泛转发) out to all ports except port E0 (unknown unicasts are flooded)
- Switch caches MAC address of station **A** to port **E0** by learning the source address of data frames (逆向学习)

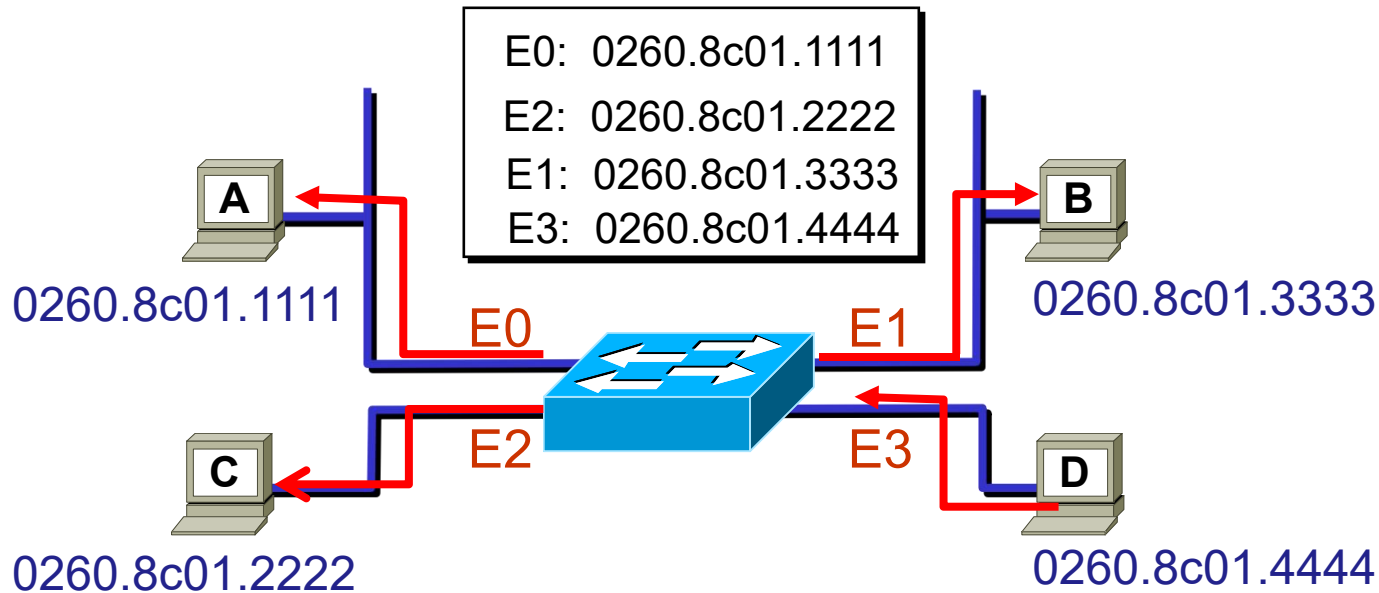
Switch:How Switches Filter(过滤) Frames

MAC forwarding table (转发表)



- Station **A** sends a frame to station **C**
- Destination is known, frame is **forwarded** to port E2

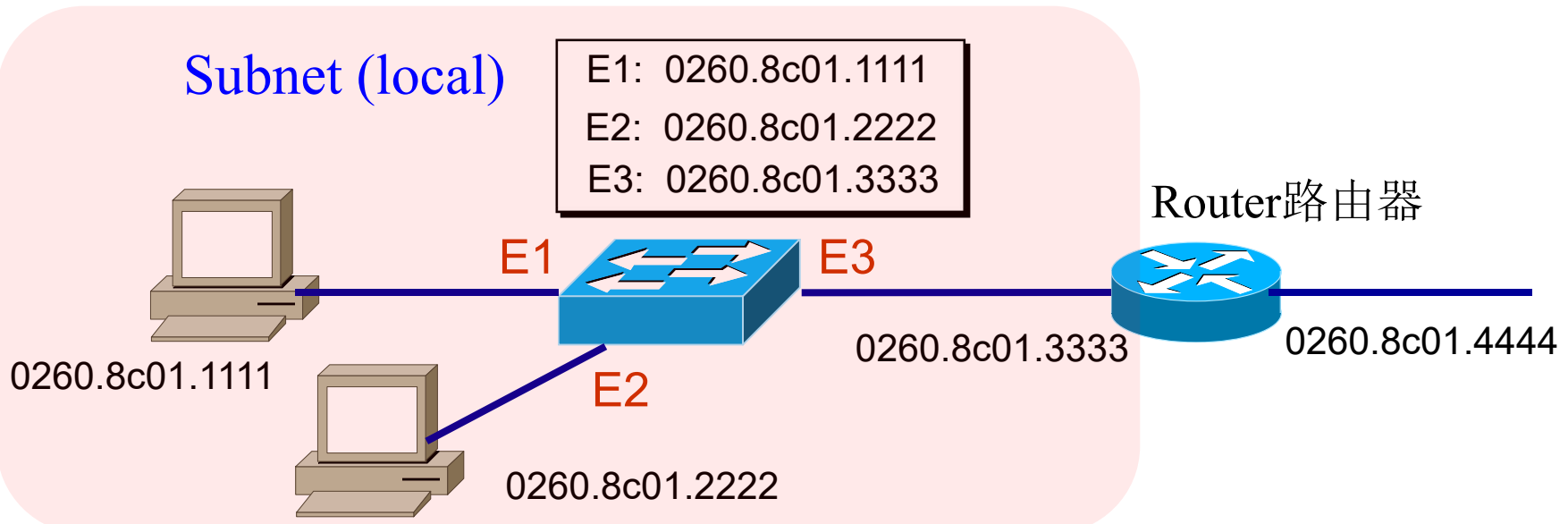
Switch: Broadcast Frames 广播帧



- Station **D** sends a **broadcast** frame (Dest MAC Address: FF-FF-FF-FF-FF-FF)
- Broadcast frames are **flooded** to all ports other than the originating port

Switch:MAC Forwarding Table

- Unlike IP address, MAC address is *local*(本地).
- MAC forwarding table is also *local*, within the *single broadcast domain* (which is an *IP subnet*子网).
- MAC address learning stops at the router, and a switch does not learn the MAC addresses at the other side of the router.
- ◆ The switch learns one MAC address of the router.



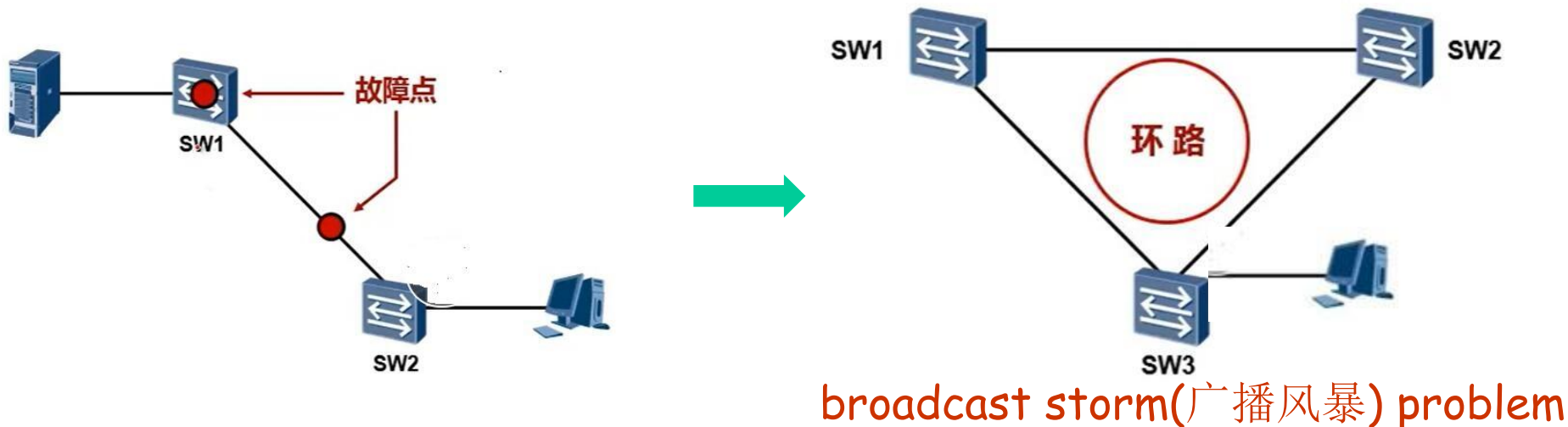
4.8 Data Link Layer Switching

- 4.8.1 Uses of Bridges
- 4.8.2 Learning Bridges
- *4.8.3 Spanning Tree Bridges*
- 4.8.4 Summary of connection devices
- 4.8.5 Virtual LANs

(1) Spanning Tree Bridges(生成树网桥): Motivation

■ Problem

- ◆ To increase reliability, **redundant links** can be used between bridges (**redundant topology**冗余拓扑)
- ◆ This redundancy introduces some problems because it creates **loops(环)** in the topology



Q: 如何在保证网络拓扑冗余的同时避免二层环路的发生?

[华为资料]

(2) Spanning Tree Bridges(生成树网桥): Ideas

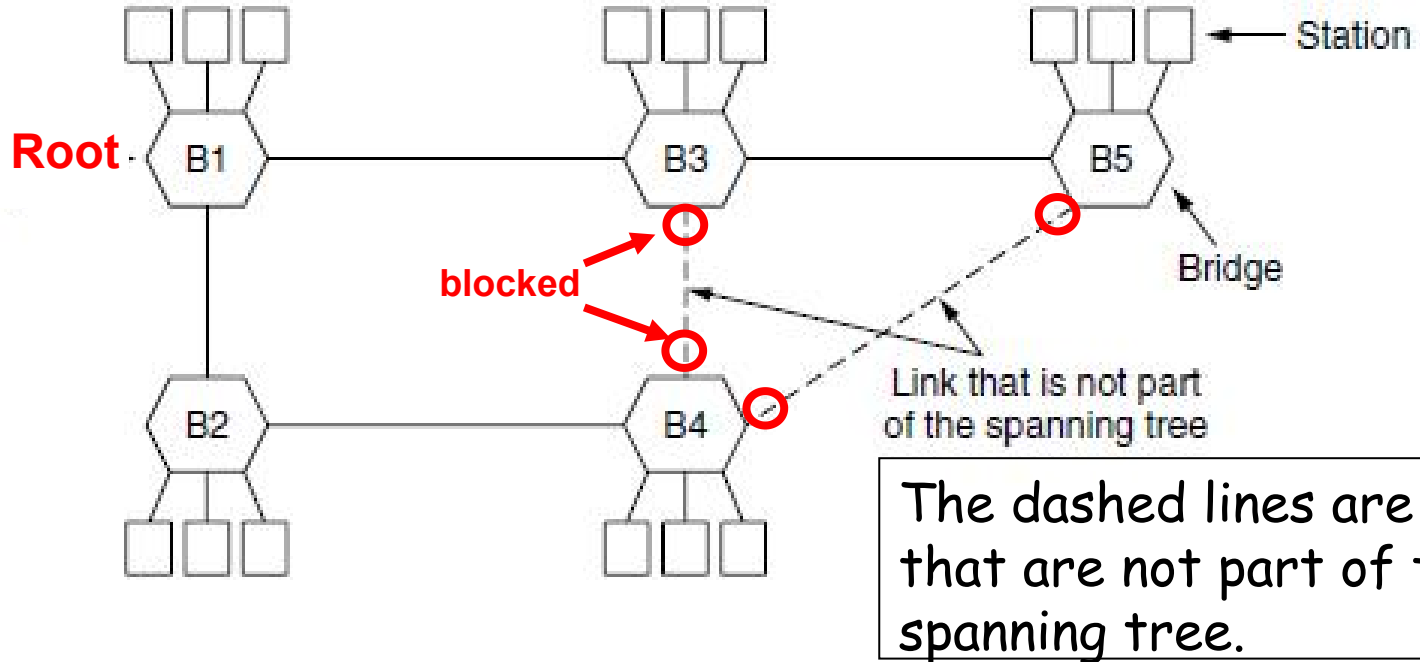
- Solution(Radia Perlman, 1983)
 - ◆ Bridges communicate with each other and overlay the actual topology with a **spanning tree** that reaches every bridges
 - ◆ Some potential connections between bridges are ignored in the interest of constructing a **fictitious loop-free topology(无环拓扑)** that is a **subset of the actual topology**
 - ◆ Using this spanning tree, there is exactly **one path** from every station to every other station, loops are impossible

断环，构成树形拓扑



Radia Perlman 2009
(born 1951)

(3) Spanning Tree Bridges: Examples



- (1) Spanning Tree **Disables** one of these connections.
- (2) It **keeps track** of each of these connections.
- (3) If the active connection becomes disconnected, it will **activate a disabled one to take over and make a new tree.**

(4) 802.1d: Constructing the Spanning Tree

生成树

- ◆ Each bridge periodically broadcasts a configuration message out all of its ports to its neighbors and processes the messages it receives from other bridges, this message is not forwarded, it includes:
 - an ID based on its MAC address
 - ID of the root it believe to be
 - the distance to root
- ◆ Chose the bridge with the lowest ID to be the root, after enough messages exchanged, all bridges will agree on the root
- ◆ Remembers the shortest path(最短路径) it finds to the root, if there are multiple equivalent paths, the path via bridge with lowest ID is chosen
- ◆ Turn off(block阻塞) ports that are not part of the shortest path
- ◆ Algorithm continues to run during normal operation to automatically detect topology changes and update the tree

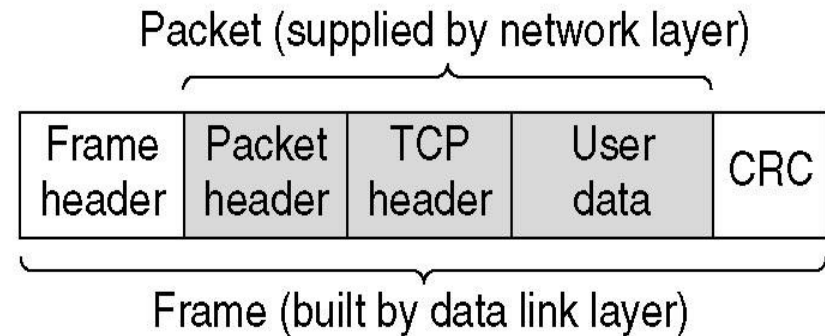
4.8 Data Link Layer Switching

- 4.8.1 Uses of Bridges
- 4.8.2 Learning Bridges
- 4.8.3 Spanning Tree Bridges
- *4.8.4 Summary of connection devices*
- 4.8.5 Virtual LANs

4.8.4 Interconnection(网络互联) Devices

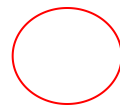
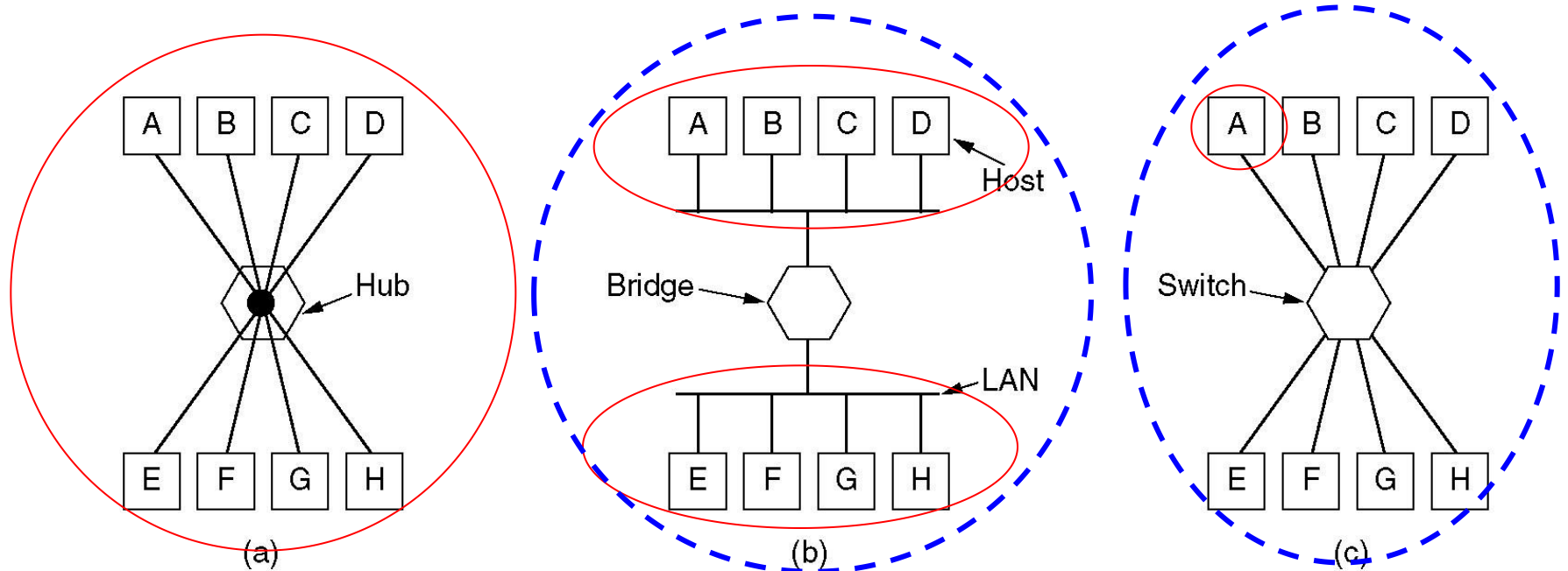
Application layer	Application gateway
Transport layer	Transport gateway
Network layer	Router
Data link layer	Bridge, switch
Physical layer	Repeater, hub

(a)

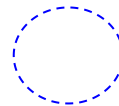


(b)

Hubs, Bridges, Switches



Collision Domain 冲突域



Broadcast Domain 广播域

Bridge(网桥) vs. Switch(交换机)

■ Bridge

- ◆ frame handling done in **software**
- ◆ analyzes and forwards **one frame at a time**
- ◆ uses **store-and-forward** operation

■ Switch

- ◆ performs address recognition and frame forwarding in **hardware**
- ◆ has **multiple parallel data paths**, can handle multiple frames at a time
- ◆ can have **cut-through(直通式)** operation

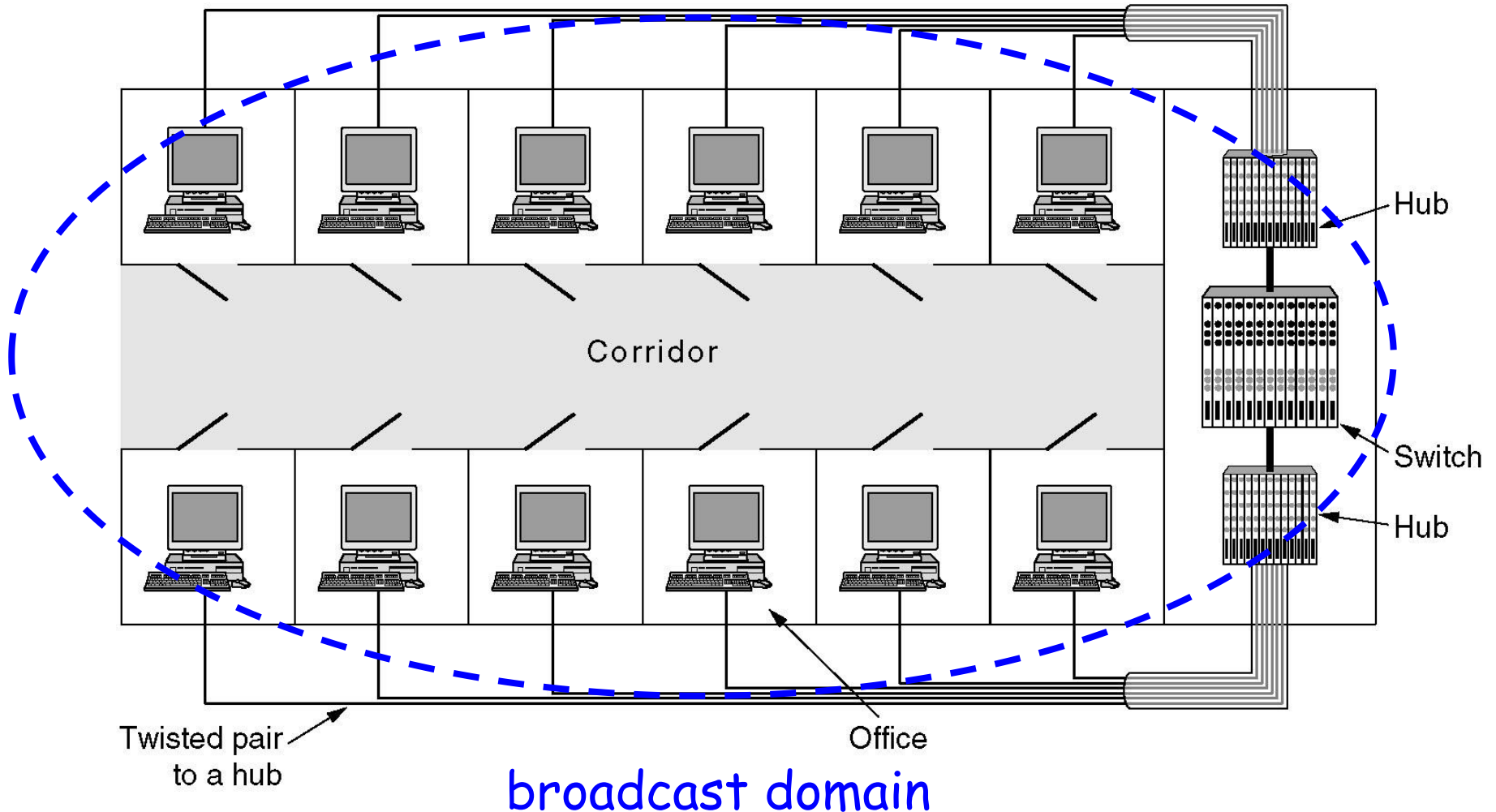
- New installations typically include layer 2 switches with bridge functionality rather than bridges

4.8 Data Link Layer Switching

- 4.8.1 Uses of Bridges
- 4.8.2 Learning Bridges
- 4.8.3 Spanning Tree Bridges
- 4.8.4 Summary of connection devices
- *4.8.5 Virtual LANs*

4.8.5 Virtual LANs(虚拟局域网)

A building with centralized wiring using hubs and a switch.



Reason for using Virtual LANs

■ Security

- ◆ Many departments have information that they do not want passed outside their department

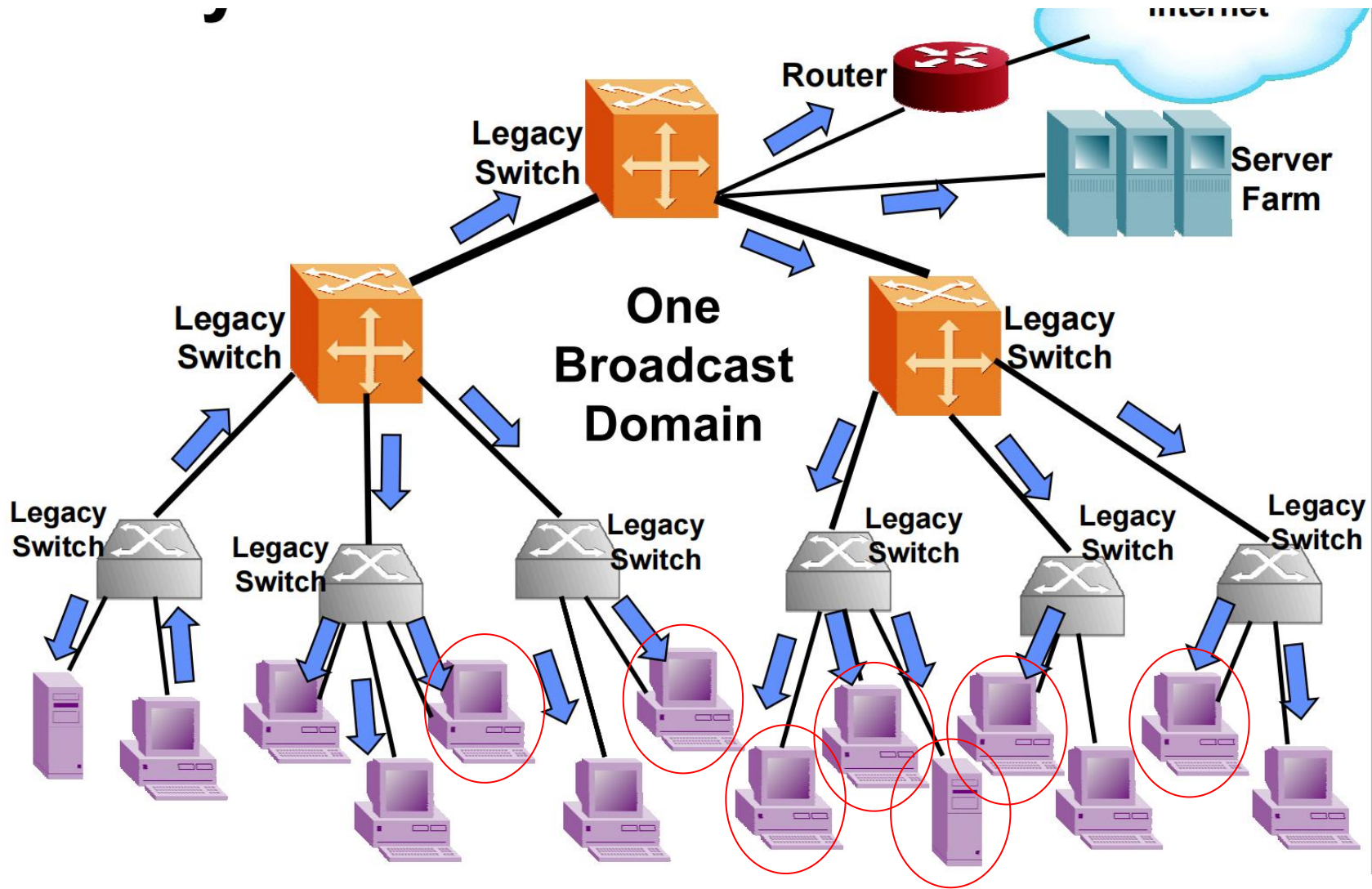
■ Load负载

- ◆ Some LANs are more heavily than others and it may be desirable to separate them

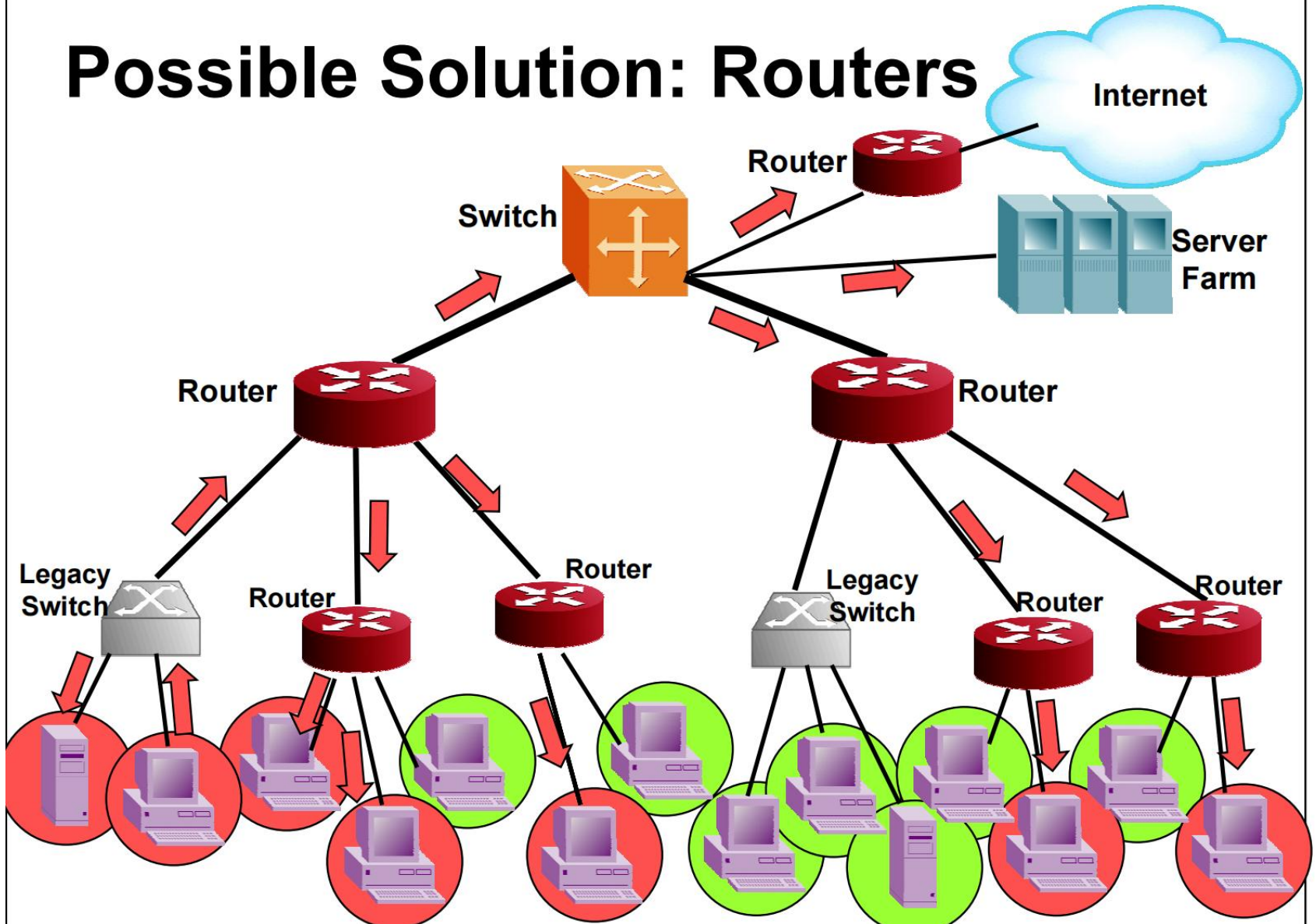
■ Broadcast traffic广播流量(Broadcast storm)

- ◆ Bridges **broadcast traffic** when the location of the destination is **unknown or broadcast address**.
- ◆ Broadcast consumes more of the LAN capacity than a regular frame
- ◆ **Broadcast storm**
 - LAN capacity is occupied by these frames
 - Machines are crippled just processing and discarding all the frames being broadcast

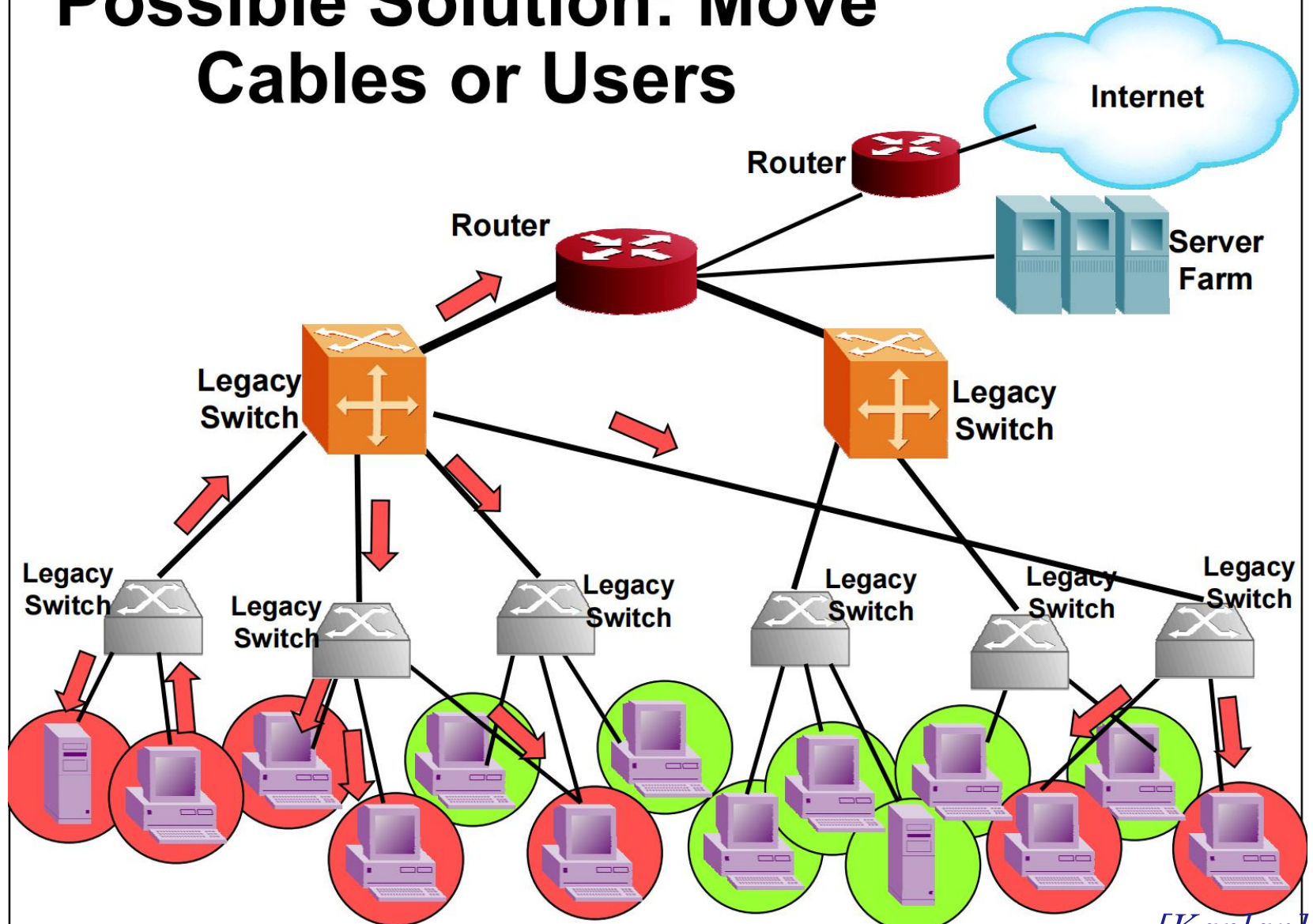
Why Are VLANs(虚拟局域网) Needed?



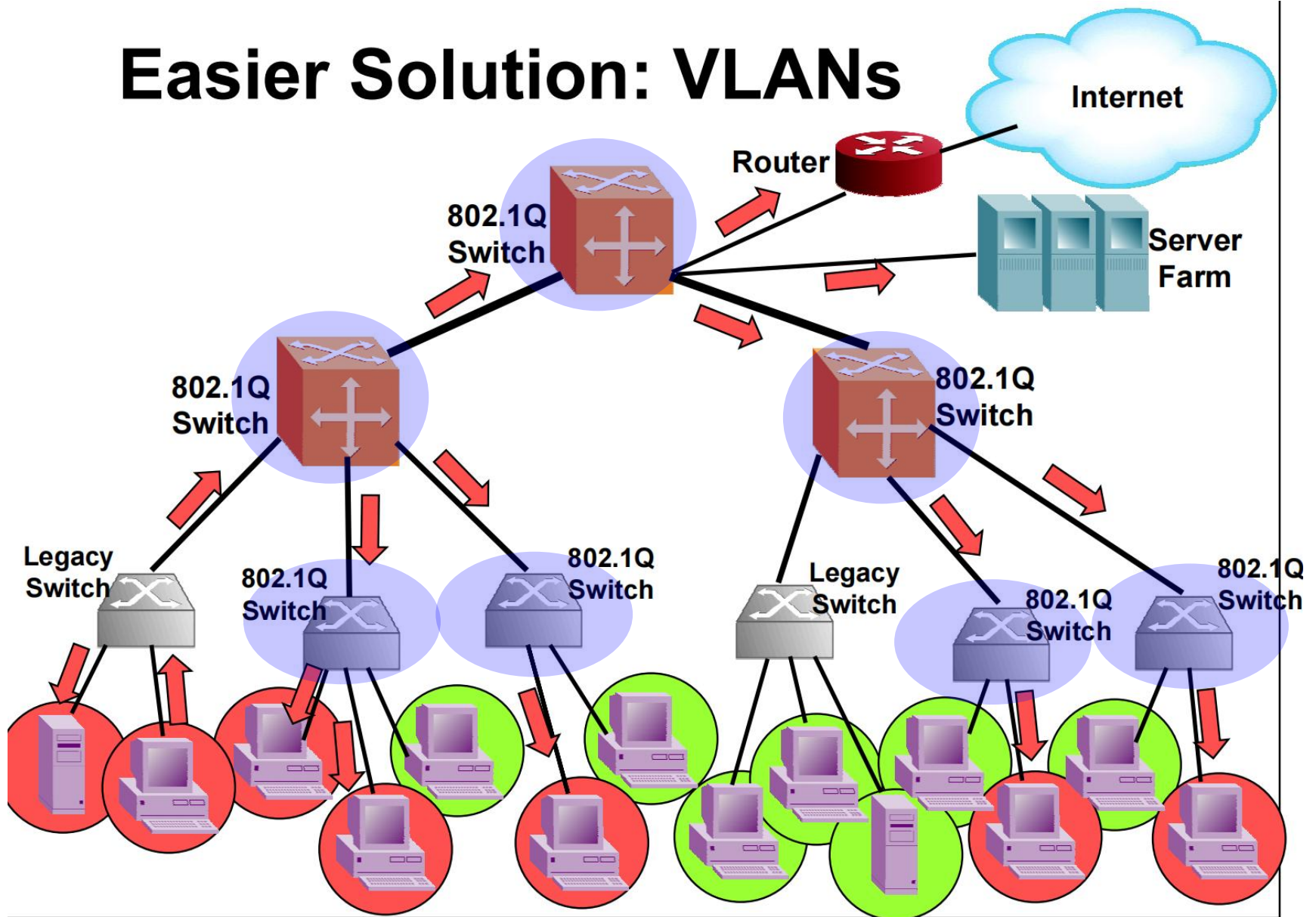
Possible Solution: Routers



Possible Solution: Move Cables or Users

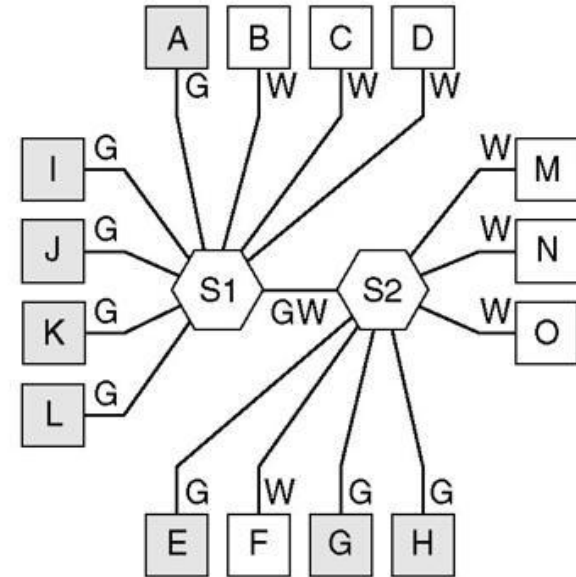


Easier Solution: VLANs

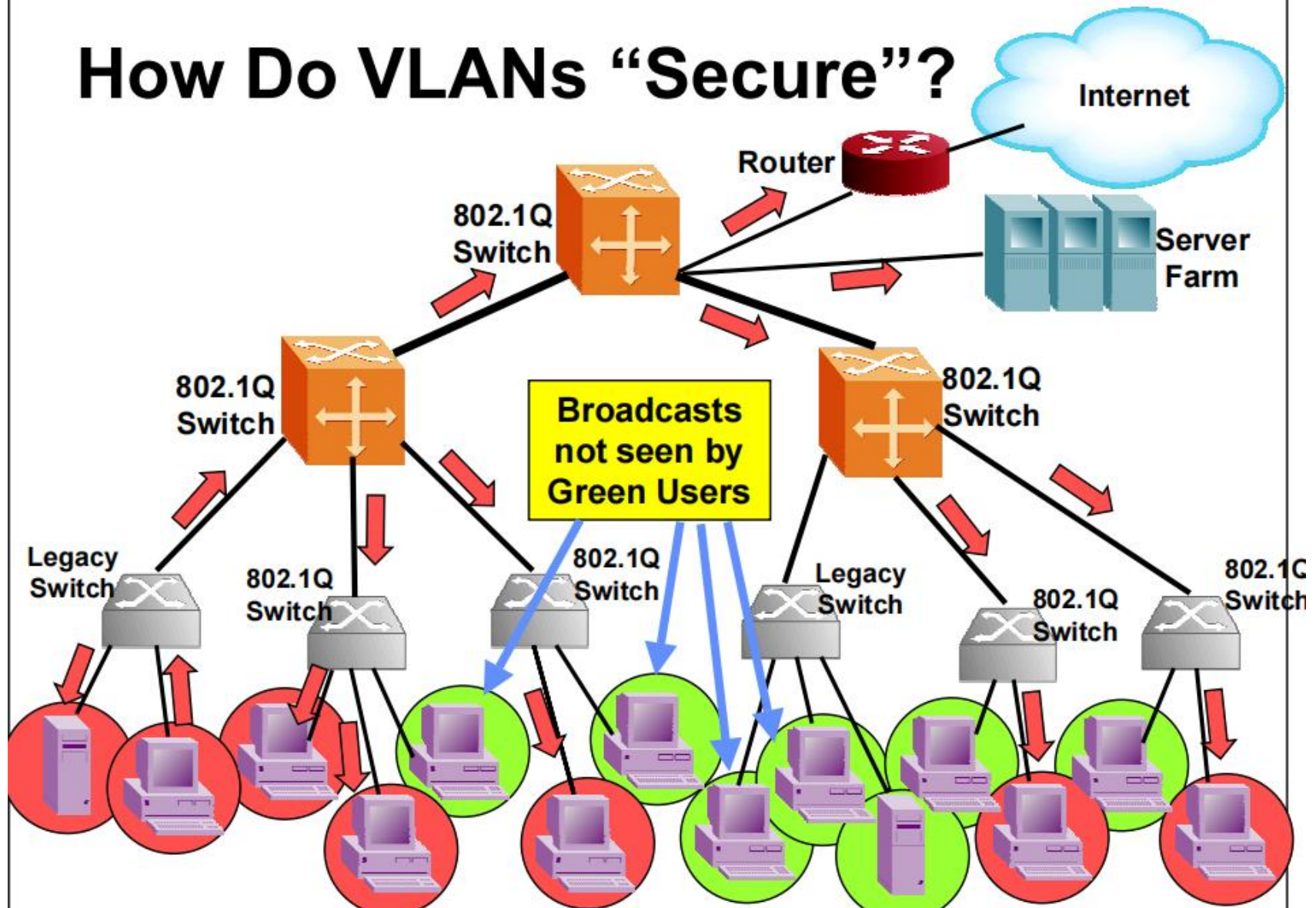


What are VLANs

- VLAN: Virtual LAN(虚拟局域网)
- Decoupling(解耦) the logical topology from the physical topology.
- ◆ Label each port with a "color"
- ◆ A port may be labeled with multiple VLAN colors
- Prohibit
 - ◆ packets from traveling to destinations on different VLANs
 - ◆ multicast and broadcast across colors.
- In the bridges or switches, configuration tables tell which VLANs are accessible via which ports



How Do VLANs “Secure”?



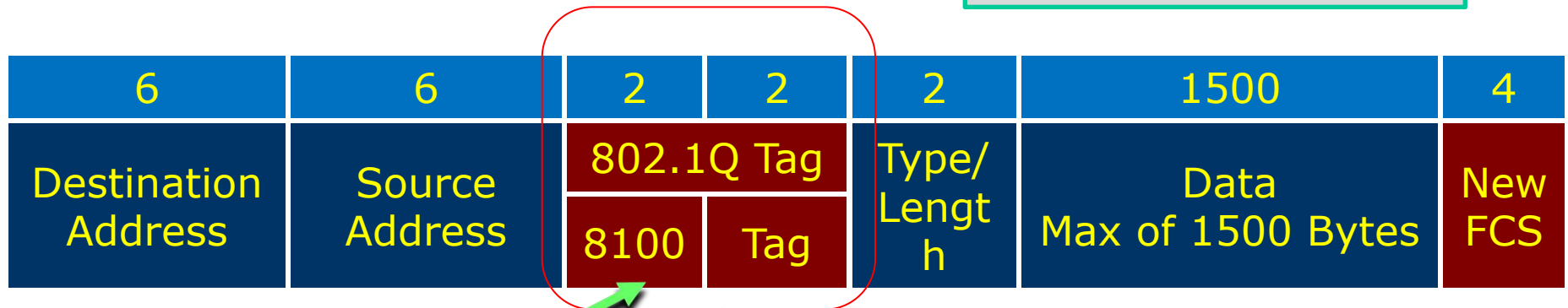
VLAN ID(自阅)

- The bridges and switches know what **VLAN ID** an incoming frame is
 - ◆ Every **port** is assigned a VLAN ID
 - ◆ This method only works if all machines on a port belong to the same VLAN
 - ◆ Every **MAC address** is assigned a VLAN ID
 - ◆ When a frame arrives, the bridge or switch extracts the MAC address and look it up in a table to see which VLAN the frame came from
 - ◆ Every **layer 3 protocol or IP address** is assigned a VLAN ID

IEEE 802.1Q Frame Tagging

■ IEEE802.1Q Frame

Length 1522 Bytes



VLAN
protocol
ID(0x8100)

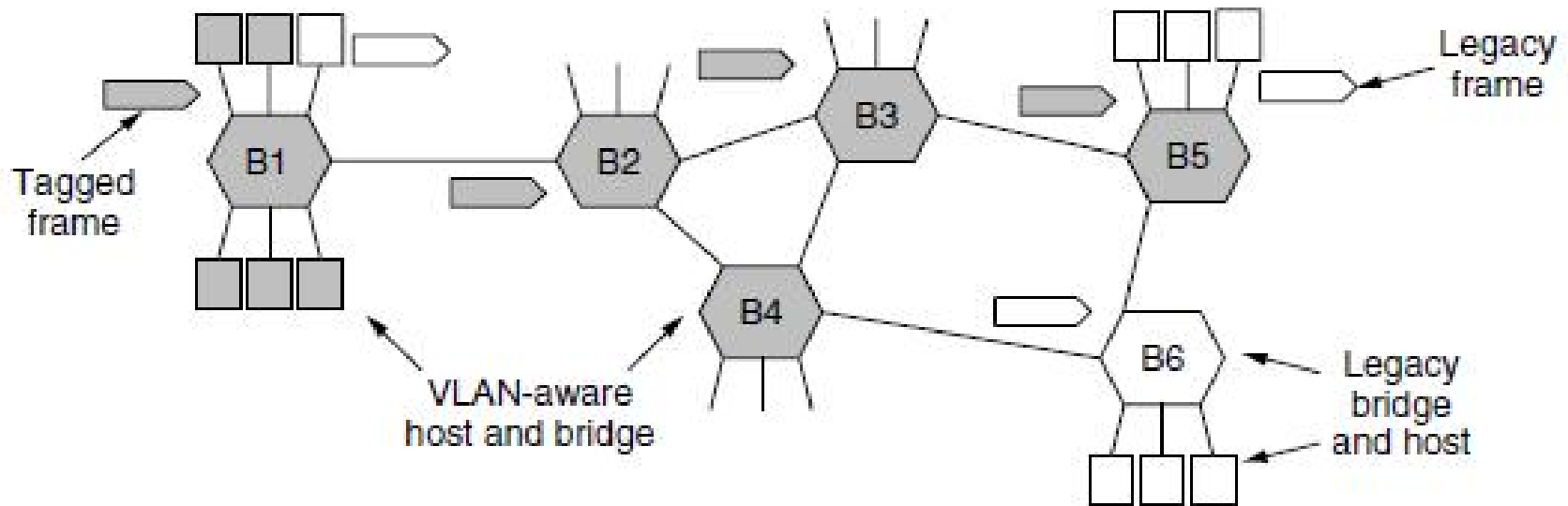


Canonical Format Identifier

CFI=1, VLAN ID有效

The IEEE 802.1Q Standard

- The shaded symbols are VLAN aware. The empty ones are not.
- The VLAN fields are only actually **used by the bridges and switches** and **not by the user machines**



First VLAN-aware bridge(switch) to touch a frame **adds** VLAN fields and the last one down the road **removes** them.

Key points of MAC sublayer (1)

- Static & Dynamic Channel Allocation
- Multiple Access Protocols
 - ◆ Carrier Sense Multiple Access (**CSMA/CD**)
 - ◆ carrier sense, collision detection, jam, binary exponential backoff
 - ◆ Wireless LAN Protocols (**MACA**)
- IEEE 802 Reference Model
 - ◆ Functions of LLC, MAC and physical layer
- Ethernet
 - ◆ Ethernet Cabling
 - ◆ **Manchester Encoding**
 - ◆ **Operations of Ethernet**
 - ◆ Fast Ethernet / Gigabit Ethernet
 - ◆ **LAN Switch**

Key points of MAC sublayer (2)

■ WLAN

- ◆ IEEE802.11: Usage, **MAC protocols**
- ◆ IEEE802.16: Usage
- ◆ IEEE802.15: Usage

■ Data Link Layer Switching

- ◆ **Learning Bridges**
- ◆ Spanning Tree Bridges
- ◆ **Repeaters, Hubs, Bridges, Switches, Routers, Gateways**
- ◆ **Layering and collision domain**
- ◆ **Virtual LANs**

Terms

缩写	英文全称	中文
	1-persisten CSMA	1-坚持CSMA
AP	Access Point	接入点
AIFS	Arbitration InterFrame Spacing	仲裁帧间隔
	auto-negotiation	自动协商
	backward learning	逆向学习
	beacon frame	信标帧
	binary exponential backoff	二进制指数退避
	binary countdown	二进制倒计时法
	bit-map protocol	位图协议
	bluetooth	蓝牙
	broadcast storm	广播风暴
	bridge	网桥
	broadcasting	广播
	carrier extension	载波扩展
	checksum	校验和/检查和

缩写	英文全称	中文
CRC	Cyclic Redundancy Check	循环冗余校验
	Collision Domain	冲突域
	collision free protocol	无冲突协议
	carrier sense	载波监听
CSMA	Carrier Sense Multiple Access	载波监听多点接入
	classic Ethernet	传统以太网
CAMA/CA	CSMA with Collision Avoidance	带冲突避免的载波监听多点接入
CSMA/CD	Carrier Sense Multiple Access/Collision Detection	带冲突检测的载波监听多点接入
CTS	Clear To Send	允许发送
	cut-through switching	直通交换
DCF	Distributed Coordination Function	分布协调功能
DIFS	DCF InterFrame Spacing	DCF帧间隔
EIFS	Extended InterFrame Spacing	扩展帧间隔
	Fast Ethernet	快速以太网

缩写	英文全称	中文
	Ethernet	以太网
	Exposed Terminal Problem	暴露站问题
	fragment	(帧)片段
	flow control	流量控制
FCS	Frame Check Sequence	帧校验序列
FDDI	Fiber Distributed Data Interface	光纤分布数据接口
	frame	帧
	frame bursting	帧突发
	gateway	网关
	Gigabit Ethernet	千兆以太网
	hidden terminal problem	隐蔽站问题
	HUB	集线器
IEEE	Institute of Electrical and Electronics Engineers	国际电气电子工程师协会
	jam	强化信号

缩写	英文全称	中文
LAN	Local Area Network	局域网
	Learning Bridge	学习网桥
	limited contention protocol	有限竞争协议
LLC	Logical Link Control	逻辑链路控制
MAC	Media Access Control	媒体(介质)访问控制
MACA	Multiple Access with Collision Avoidance	带冲突避免的多点接入
	multi-access channel	多接入信道
	Manchester Encoding	曼彻斯特编码
	multicasting	多播/组播
NAV	Network Allocation Vector	网络分配向量
NIC	Network Interface Card	网络接口卡
	non-persistent CSMA	非坚持CSMA
OUI	Organization Unique Identifier	组织唯一标识符
	packet	分组/包
	pad	填充

缩写	英文全称	中文
PCF	Point Coordination Function	点协调功能
	p-persistent CSMA	p坚持CSMA
PPP	Point-to-Point Protocol	点对点协议
	preamble	前导码
	promiscuous mode	混杂模式
	pure ALOHA	纯ALOHA
	random access channel	随机接入信道
	repeater	中继器
	router	路由器
	rate anomaly	速率异常
RTS	Request To Send	请求发送
SIFS	Short InterFrame Spacing	短帧间隔
	store-and-forward	存储-转发
	slot	时隙/时槽
	slotted ALOHA	时隙ALOHA
	Spanning Tree Bridge	生成树网桥
	station	站点

缩写	英文全称	中文
	subnet	子网
	switch	交换机
	switching	交换
	switched Ethernet	交换式以太网
	thick Ethernet	粗缆以太网
	thin Ethernet	细缆以太网
	token	令牌
	token bus	令牌总线
	token ring	令牌环
TXOP	Transmission Opportunity	发送机会
VLAN	Virtual LAN	虚拟局域网
	Vulnerable Period	脆弱期
WAN	Wide Area Network	广域网
WiFi	Wireless Fidelity	无线保真
WiMAX	Worldwide Interoperability for Microwave Access	全球微波互联接入
WLAN	Wireless Local Area Network	无线局域网

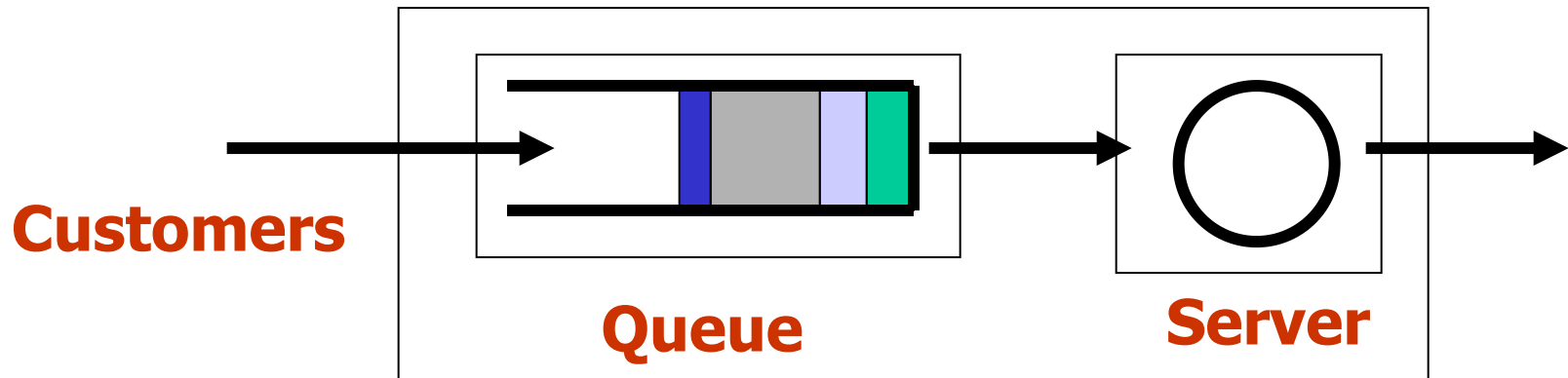
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 - ◆ Nen-Fu Huang, Lecture notes "IEEE 802.3 Carrier Sense Multiple Access with Collision Detection (CSMA/CD)", Natinal Tsinghua University, which are marked with [Huang]
 - ◆ Hadriel Kaplan & Robert Noseworthy, "The Ethernet Evolution From 10 Meg to 10 Gig How it all Works!" which are marked with [Kaplan]

Self Study

M/M/1 Model

- First M: Arrivals are a “Markovian / Poisson” process (泊松过程)
- Second M: **exponential** distribution for service time (指数分布)
- 1: **1** server



Queuing System

M/M/1 Model

- Probability that k customers arrived during period T

$$P[k] = \frac{(\lambda T)^k e^{-\lambda T}}{k!}$$

- We can get

$$T = \frac{1}{\mu - \lambda}$$

μ : *AverageServiceRate*

λ : *AverageArrivalRate*

T : *AverageWaitingTime(QueuingTime + ServiceTime)*

Static Channel Allocation

■ Single channel

- ◆ Channel capacity(data transmission rate): $C(\text{bps})$
- ◆ Mean delay: T
- ◆ Frame arrival rate: $\lambda \text{ frames/sec}$
- ◆ mean frame length: $1/\mu (\text{bits})$

$$\text{Mean delay } T = 1/(\mu C - \lambda)$$

■ Divide the channel into $N(\text{FDM})$

$$\text{Mean delay } T_{\text{FDM}} = 1/[\mu(C/N) - (\lambda/N)] = N/(\mu C - \lambda) = NT$$

Static Channel Allocation: Example

■ Single channel

- ◆ data transmission rate: 100 Mbps, mean frame length 10,000 bits, Frame arrival rate is 5000 frames/sec

$\mu = 1/10000$, $C = 100\text{Mbps}$, $\mu C = 10000$ frames/sec

Mean delay $T = 1/(\mu C - \lambda) = 0.2$ ms

■ Divide the channel into N

- ◆ Replace the 100-Mbps network with 10 networks of 10 Mbps each and statically allocate each user to one of them

Mean delay $T = NT = 10 * 0.2$ ms = 2ms

Multiple Access Protocols

- *ALOHA: Pure ALOHA & Slotted ALOHA*
- Carrier Sense Multiple Access Protocols
- Collision-Free Protocols
- Wireless LAN Protocols

Pure ALOHA: Retransmission

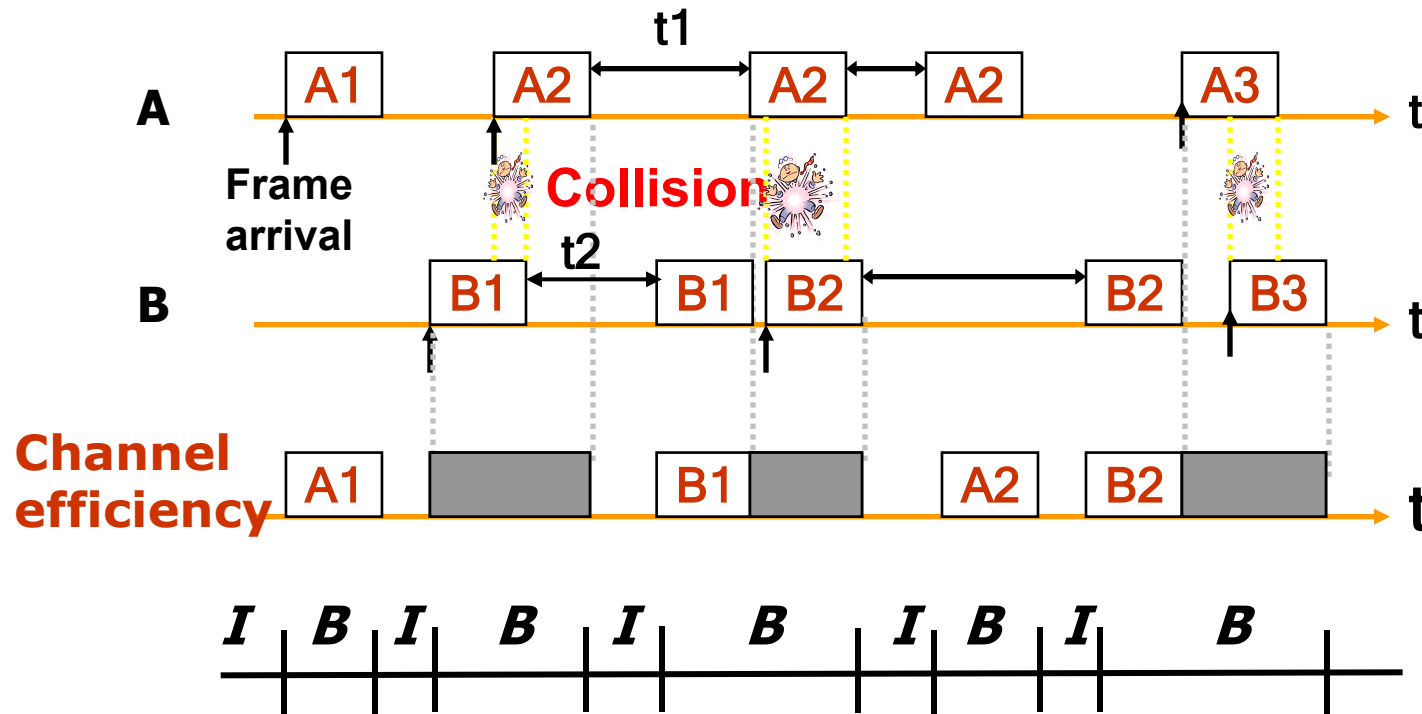
■ Retransmission

- ◆ waiting for a random time before retransmission.
- ◆ Retransmission again after waiting another random time on collision. Repeating until successful transmission

■ Simple but low efficiency

- ◆ Max Utilization: 18.4%

Illustration of Pure ALOHA



Frames are transmitted at completely arbitrary times.

Review: M/M/1 Model

- Probability that k customers arrived during period T

$$P[k] = \frac{(\lambda T)^k e^{-\lambda T}}{k!}$$

- We can get

$$T = \frac{1}{\mu - \lambda}$$

μ : *AverageServiceRate*

λ : *AverageArrivalRate*

T : *AverageWaitingTime(QueuingTime + ServiceTime)*

Pure ALOHA: Performance Analysis(1)

Assumptions

- ◆ Fixed frame length, frame time is t
- ◆ Frames generated according to Poisson distribution
- ◆ New frames and retransmitting frames also accord to Poisson distribution

Parameters

- ◆ Throughput S : mean successful frames per frame time
 $0 \leq S \leq 1$
- ◆ Network load G : mean frames per frame time $G \geq S$
- ◆ probability P_0 : successful transmission

$$S = G \times P_0$$

Pure ALOHA: Performance Analysis(2)

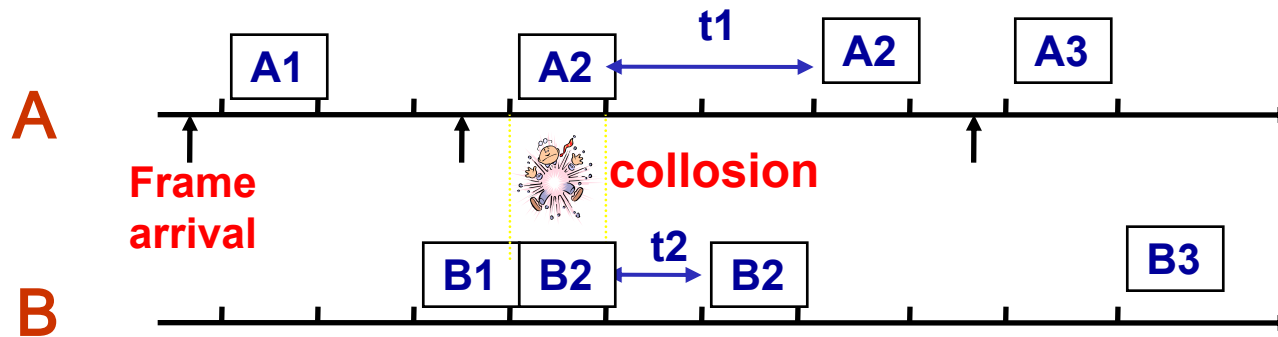
$$P_r(k) = \frac{G^k e^{-G}}{k!}$$

$$P_r(0) = P_0 = e^{-G} \text{ (within frame time } t\text{)}$$

- $P_0 = P$ [no other traffic being initiated during the entire vulnerable period] = e^{-2G}
- Throughput: $S = G \times P_0 = G e^{-2G}$
- Maximum throughput occurs when $G=0.5$,
 $S=1/2e \approx 0.184$

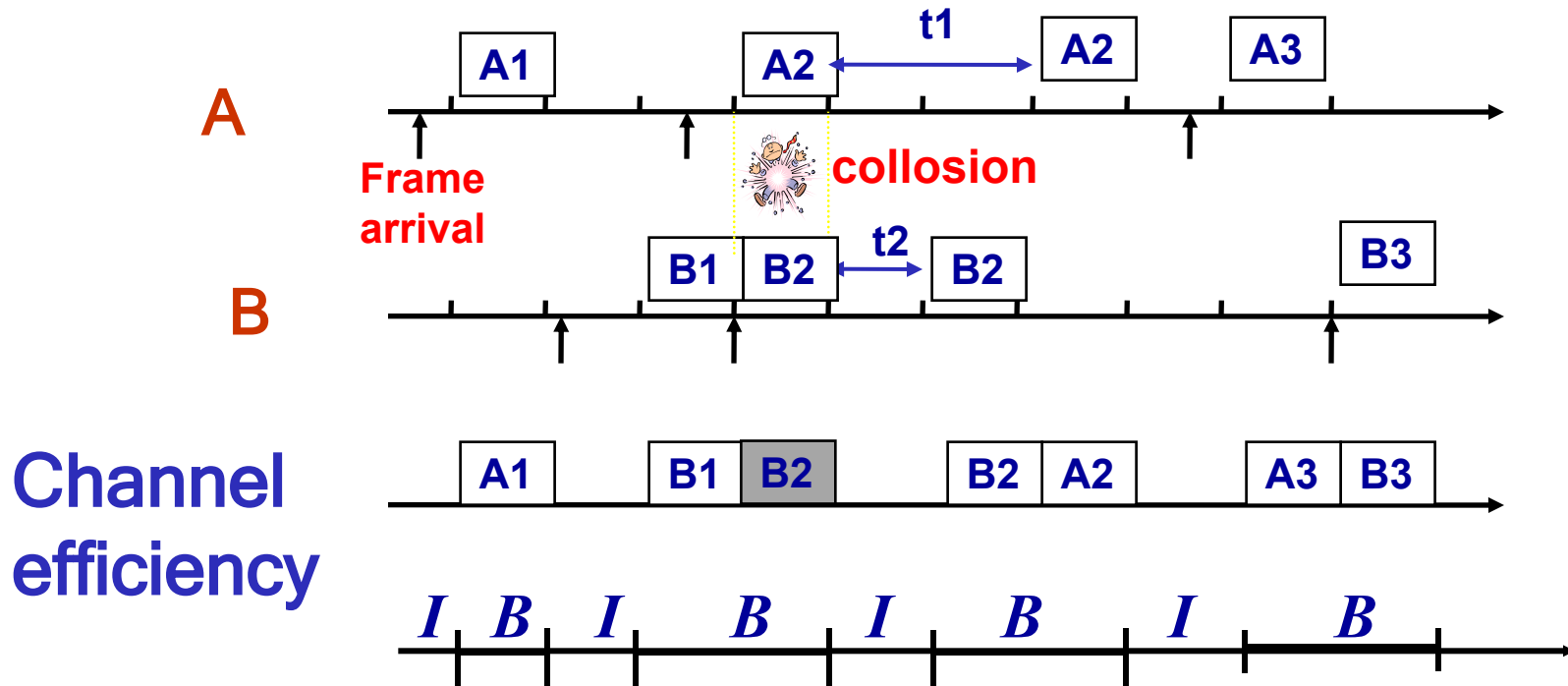
(2) Slotted ALOHA(时隙/分槽ALOHA)

- Time in uniform slots equal T_0
- Need central clock (or other sync mechanism)
- Transmission begins at slot boundary



- Frames either miss or overlap totally
- Vulnerable Period: t (frame time帧时, 发送1帧的时延)
- Max utilization(S): 36.8%

Illustration of Slotted ALOHA



Vulnerable Period: t

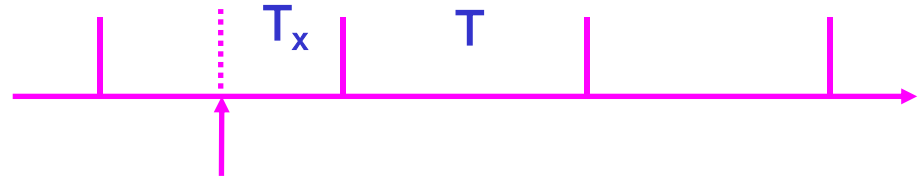
Slotted ALOHA: Performance Analysis

■ Collision probability

- ◆ No frame sent between t_0 and t_0+t

$$P_r(0) = P_0 = e^{-G} \text{ (within frame time } t\text{)}$$

$$S = GP_0 = Ge^{-G}$$



- ◆ Maximum $S=0.37$ at $G=1$

37%: Slots empty

37%: Successes

26%: Collisions

Expected number of transmissions

■ Pure ALOHA

成功传输概率: $P_0 = e^{-2G}$

尝试 k 次才能成功的概率: $P_k = e^{-2G}(1 - e^{-2G})^{k-1}$

平均传输次数: $E = \sum k P_k = e^{2G}$

■ Slotted ALOHA

$P_0 = e^{-G}$

$P_k = e^{-G}(1 - e^{-G})^{k-1}$

$E = \sum k P_k = e^G$

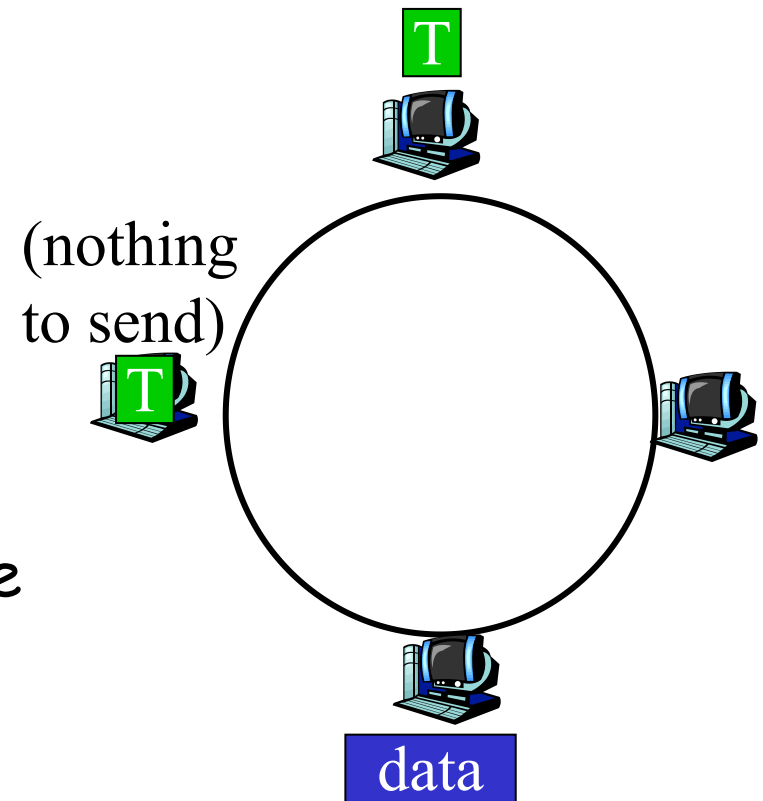
Higher values of G reduces number of empties but
increases number of collisions **exponentially**

Multiple Access Protocols

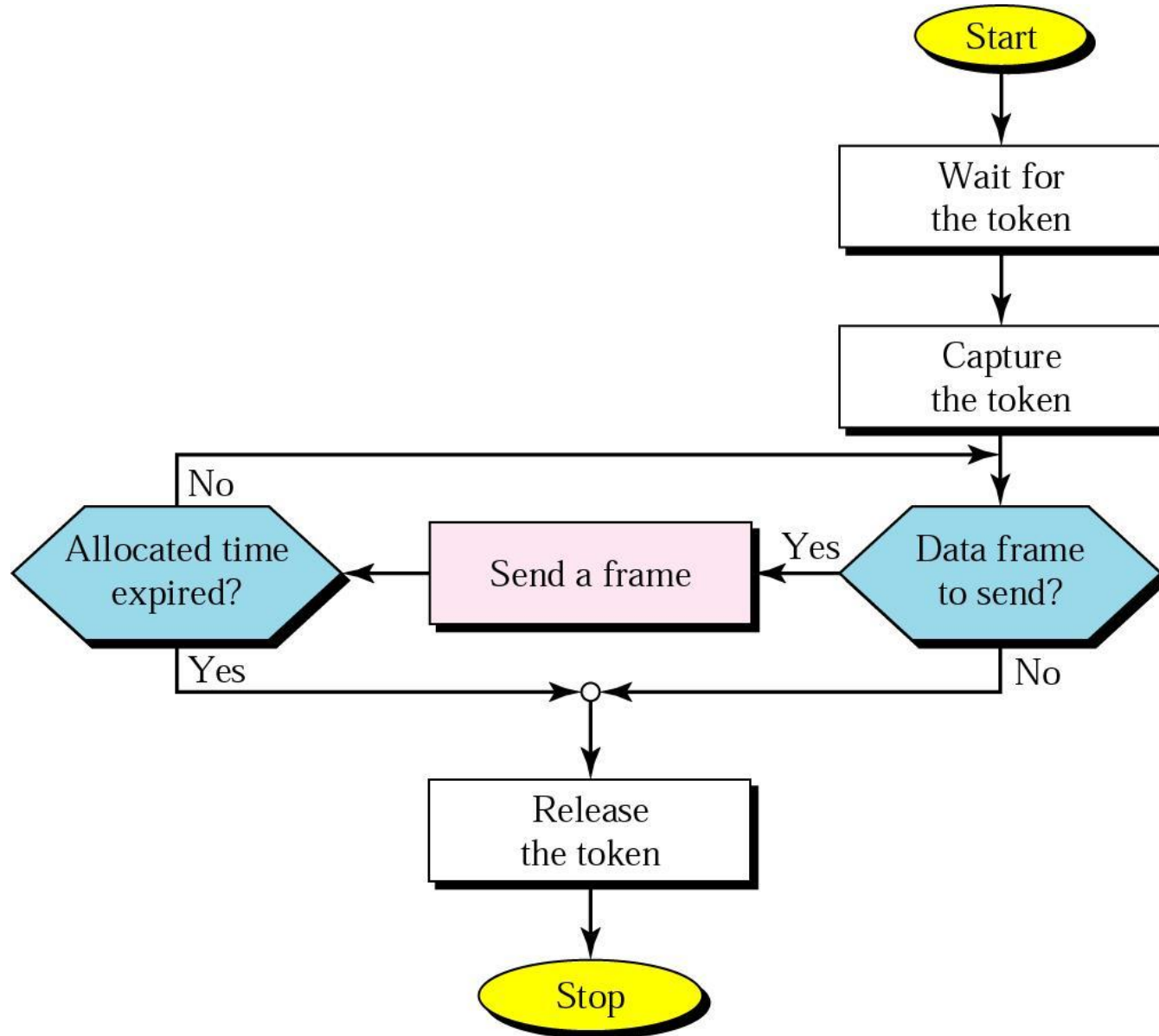
- ALOHA: Pure ALOHA & Slotted ALOHA
- Carrier Sense Multiple Access Protocols
- *Collision-Free Protocols* 无冲突协议
- Wireless LAN Protocols

Token Passing (令牌传递)

- Token: a small message representing permission to send.
- Token ring
 - ◆ All stations connects to a ring
 - ◆ Token passed from one node to next sequentially
 - ◆ Station can transmit only when seizing token
 - ◆ Collision free
 - ◆ Protocols: 802.4, 802.5



Procedure of Token Passing



[Forouzan]

4.3.3 Performance of IEEE 802.3

- Assumed
 - frame length F
 - network bandwidth B
 - cable length L
 - speed of signal propagation c , for optimal case $A=1/e$

$$\text{Channel efficiency} = \frac{1}{1+2BLc/cF}$$

$$2BLc/cF=2BDc/F$$

Increasing network bandwidth or distance reduces efficiency for a given frame size

IEEE802.3定义的传输介质与编码体系

	传输介质
5	最长为 500m 的粗同轴电缆
2	最长为 200m 的细同轴电缆
T	Twisted Pair （双绞线）
F	Fiber （光纤）
B	Bi-directional (1 芯单模光缆)
S	Short Reach (100m)(2 芯多模光缆)
L	Long Reach (10km)(2 芯单模或多模光缆)
E	Extented Long Reach (40km)(2 芯单模光缆)
Z	Long Reach Simple Mode (70km)(2 芯单模光缆)
C	Co-axial （ 2 芯屏蔽同轴电缆）
P	PON (1 芯单模光缆,单点到多点)

	编码体系
X	快速以太网时使用 4B/5B 编码
	千兆以太网时使用 8B/10B 编码
R	使用 64B/66B 编码



About ALOHA

■ Norman Abramson先后毕业于
Harvard大学、UCLA（加州大学
Los Angeles分校）、Stanford

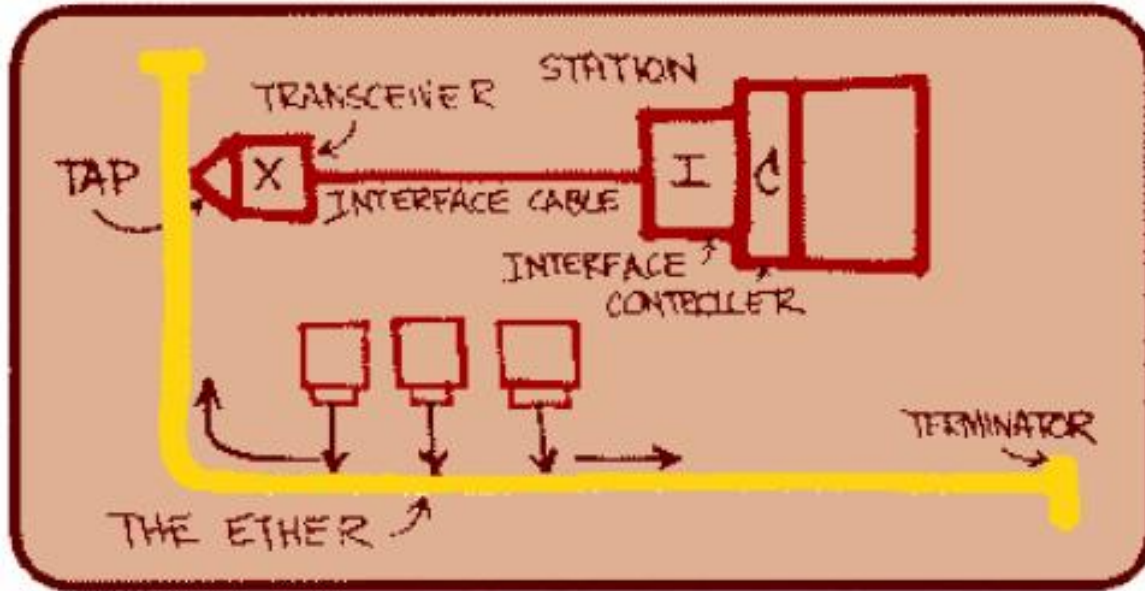


大学，分别获学士、硕士、博士学位，毕业后在Stanford大学电子工程系任教。1967年开始任Hawaii大学信息与计算机科学系教授，并从事通信理论研究，领导ALOHA系统研究项目，提出了著名的**ALOHA多址通信协议**和**CRC（循环冗余校验）**专利技术，并成功建立了世界上第一个数据通信网络ALOHANET，开创了计算机局域网和现代数据通信网，同时也首先实现了无线分组数据接入方法。

Ethernet History

- Invented by Metcalf at Xerox in 1973 and patented in 1976, IEEE standard in 1985
- Why is it called Ethernet?
 - ◆ In late 1972, Metcalfe and his Xerox PARC colleagues developed the first experimental Ethernet system(called the Alto Aloha Network) with data transmission rate of 2.94 Mbps.
 - ◆ In 1973 Metcalfe changed the name to "Ethernet," to make it clear that the system could support any computer and to point out that his new network mechanisms had evolved well beyond the Aloha system.
 - ◆ The word "ether" describes an essential feature of the system: the physical medium carries bits to all stations, much the same way that the old "luminiferous ether" was once thought to propagate electromagnetic waves through space. Thus, Ethernet was born.

Bob Metcalfe's Vision



Bob Metcalfe's original drawing for Ethernet

(INTEL had just developed 8080, running at 4.77M)

■ Robert Metcalfe (1946~)

- ◆ 1969 Graduated from MIT
- ◆ 1973 Harvard (Ph.D)
- ◆ 1972~1979 Xerox PARC (Palo Alto Research Center)
 - 1973 Ethernet, 3Mbps
- ◆ 1979~1990 3Com
- ◆ 1990~2000 Column author, 2011~ UT Austin